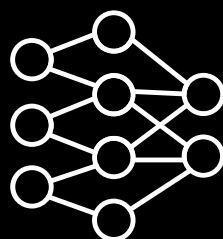


AI

IN 100 IMAGES

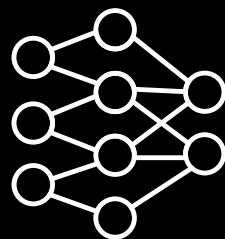


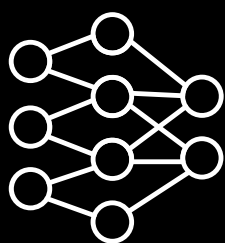
@ASHISH_BAMANIA



AI IN 100 IMAGES

ASHISH BAMANIA





HEY, FRIEND! IT'S ASHISH HERE.

I'M THRILLED THAT YOU'VE PICKED UP THIS BOOK.

INSIDE, YOU'LL FIND A VISUAL SUMMARY OF 100 MUST-KNOW CONCEPTS IN AI TODAY.

AS AN EMERGENCY PHYSICIAN AND A SELF-TAUGHT SOFTWARE DEVELOPER, I'M ALWAYS FEELING PASSIONATE ABOUT LEARNING AND SHARING FASCINATING IDEAS TO THE WORLD.

AND, THIS BOOK IS A CONTINUATION OF THAT PASSION.

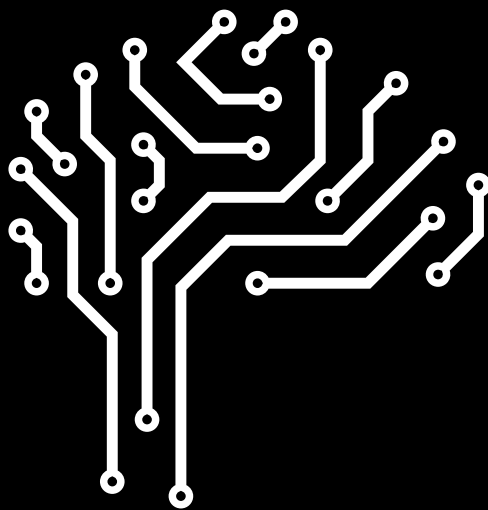
I HOPE YOU ENJOY READING IT AS MUCH AS I ENJOYED CREATING IT.

LET'S DIVE IN!

ARTIFICIAL INTELLIGENCE

IT IS THE CAPABILITY OF MACHINES TO PERFORM TASKS THAT USUALLY REQUIRE HUMAN INTELLIGENCE, SUCH AS LEARNING, REASONING, AND PROBLEM-SOLVING.

ARTIFICIAL INTELLIGENCE INCLUDES RULE-BASED SYSTEMS, EVOLUTIONARY ALGORITHMS, MACHINE LEARNING ALGORITHMS AND MORE.

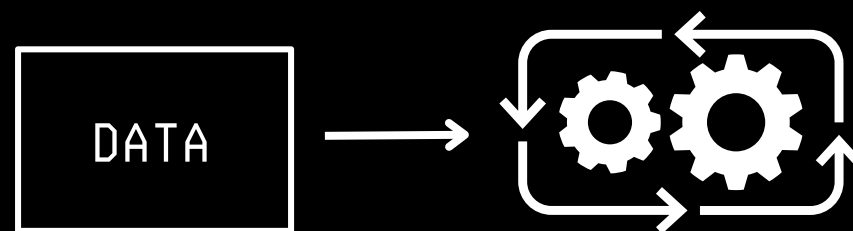


MACHINE LEARNING

A BRANCH OF ARTIFICIAL INTELLIGENCE THAT EMPOWERS COMPUTERS TO LEARN FROM DATA AND IMPROVE PERFORMANCE OVER TIME WITHOUT EXPLICIT PROGRAMMING.

ITS 3 SUBTYPES ARE:

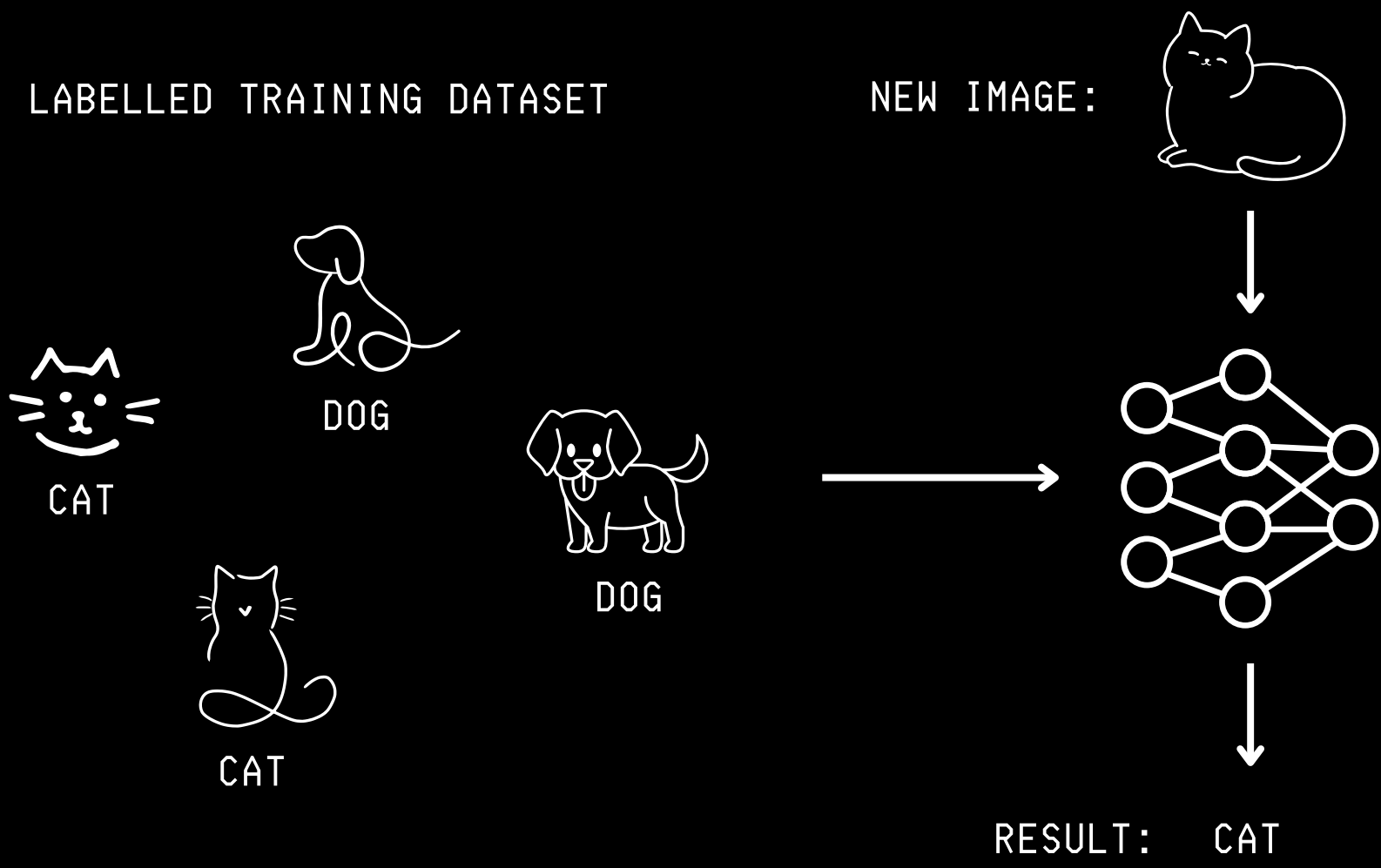
1. SUPERVISED LEARNING
2. UNSUPERVISED LEARNING
3. REINFORCEMENT LEARNING



IMPROVING
ML MODEL

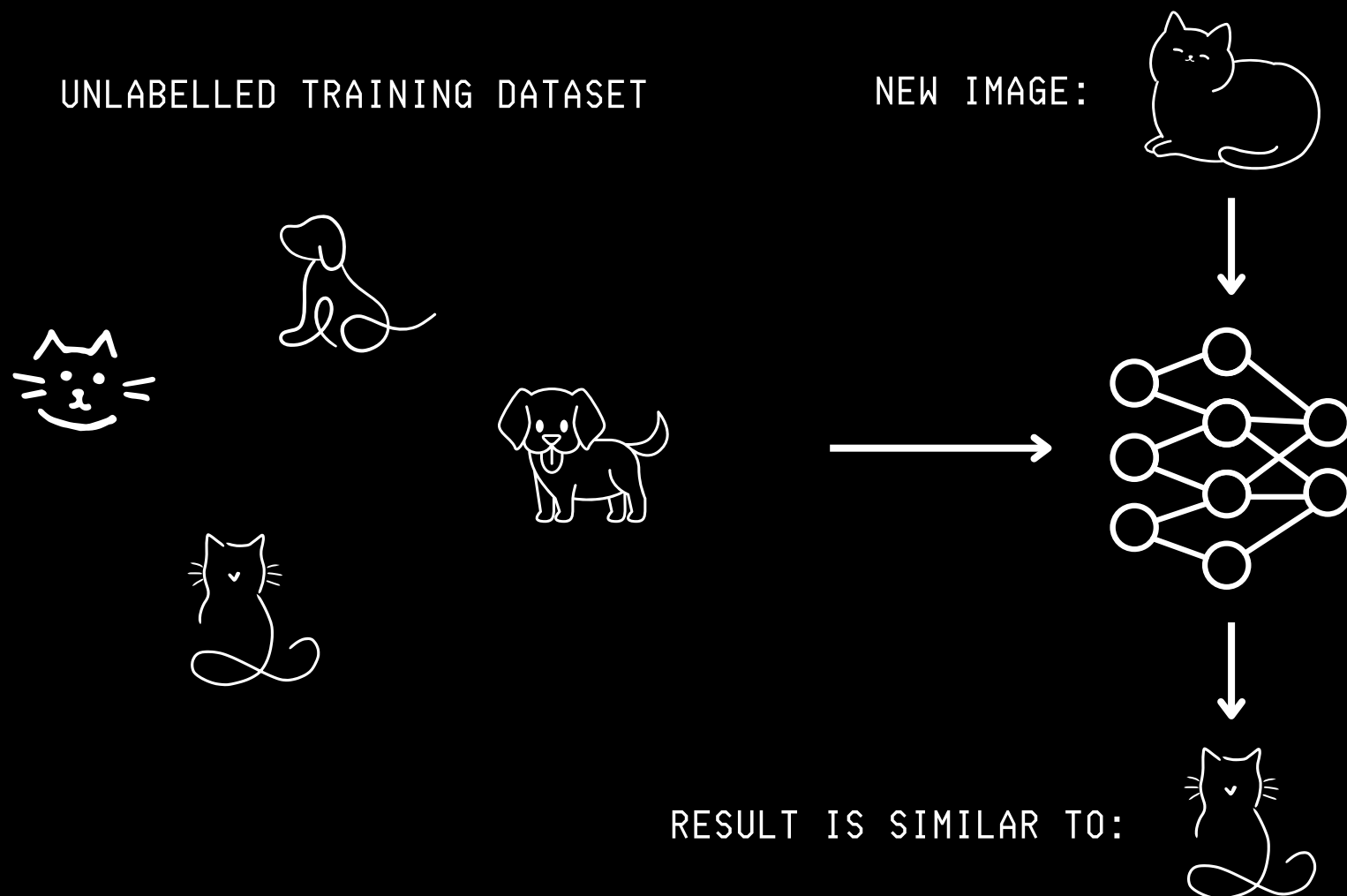
SUPERVISED LEARNING

A TYPE OF MACHINE LEARNING WHERE THE MODEL IS TRAINED ON LABELLED DATA TO MAKE PREDICTIONS OR CLASSIFY NEW DATA.



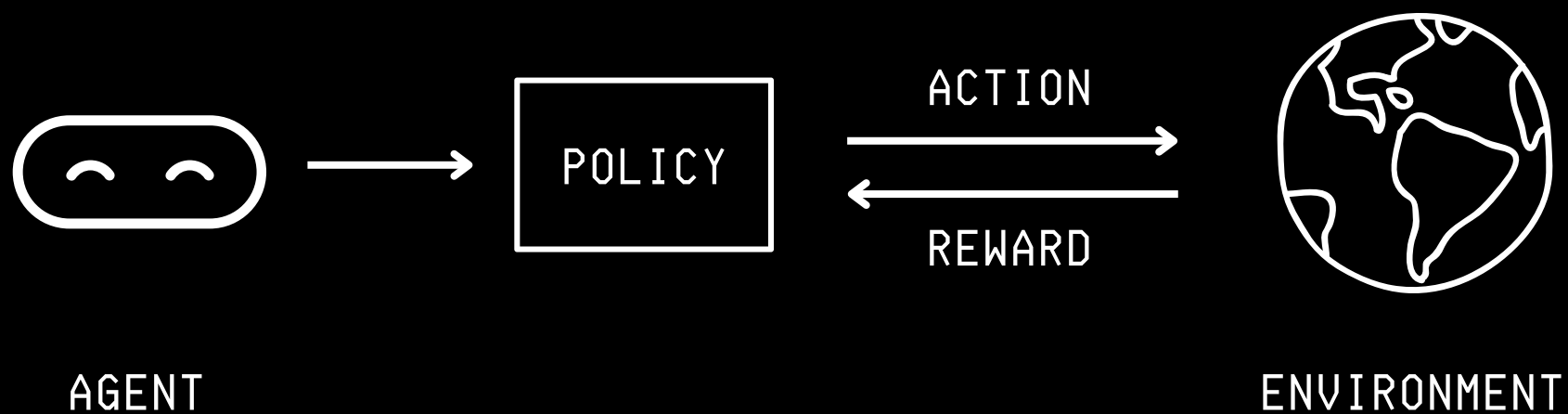
UNSUPERVISED LEARNING

A TYPE OF MACHINE LEARNING WHERE THE MODEL FINDS PATTERNS AND RELATIONSHIPS IN DATA WITHOUT USING LABELLED OUTPUTS.



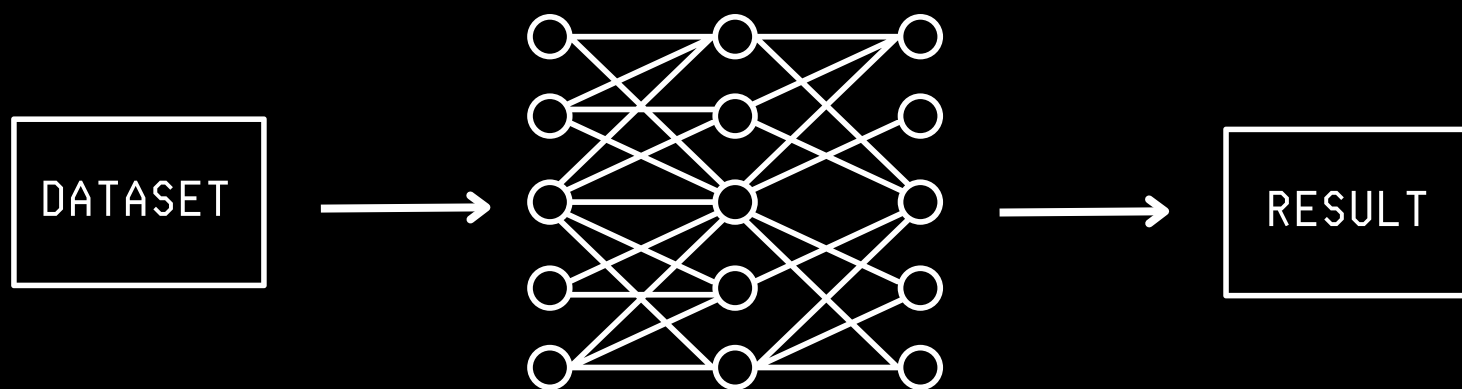
REINFORCEMENT LEARNING

A TYPE OF MACHINE LEARNING WHERE AN AGENT LEARNS TO MAKE DECISIONS BY TAKING ACTIONS IN AN ENVIRONMENT TO MAXIMIZE CUMULATIVE REWARDS, BASED ON STATES, ACTIONS, REWARDS, AND POLICIES.



DEEP LEARNING

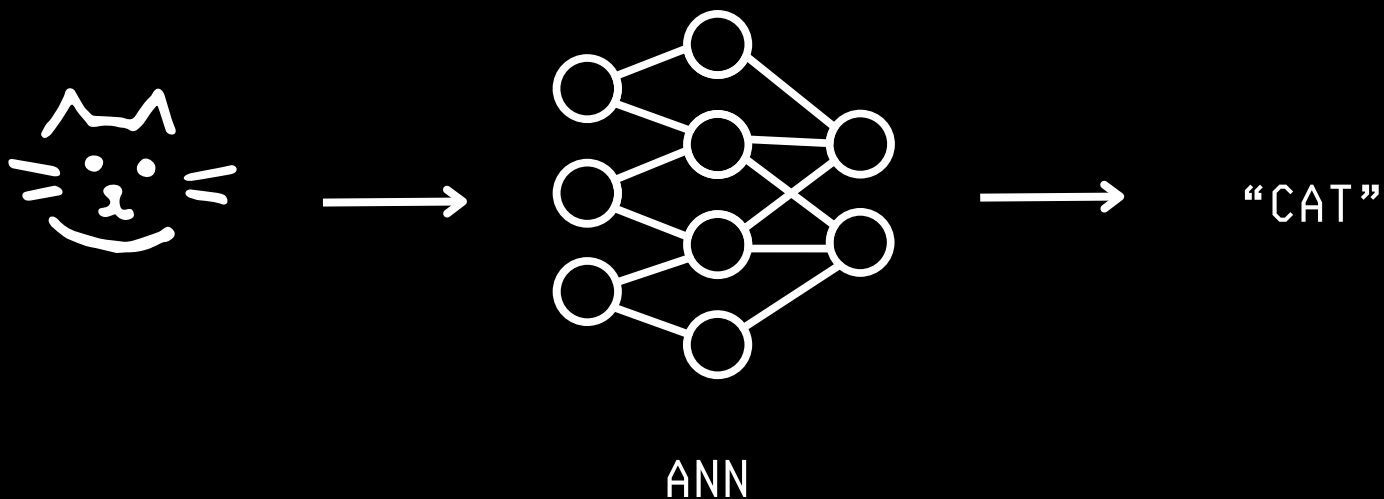
A SUBSET OF MACHINE LEARNING THAT USES NEURAL NETWORKS WITH MULTIPLE LAYERS TO MODEL AND UNDERSTAND COMPLEX PATTERNS IN LARGE DATASETS.



ANN

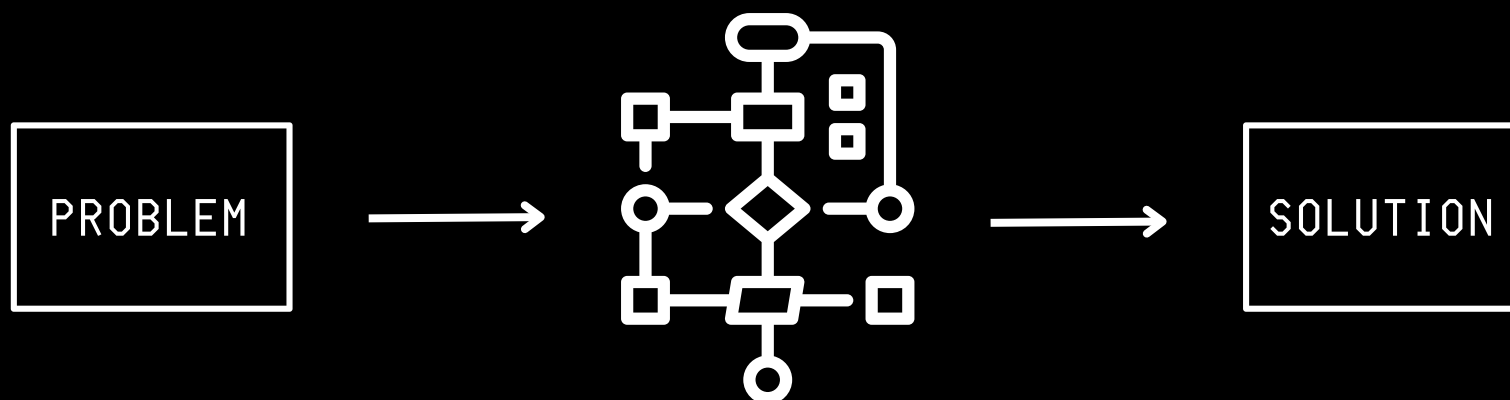
ARTIFICIAL NEURAL NETWORK

A COMPUTATIONAL MODEL INSPIRED BY THE HUMAN BRAIN,
CONSISTING OF INTERCONNECTED NODES CAPABLE OF LEARNING
AND PROCESSING COMPLEX PATTERNS FROM DATA.



ALGORITHM

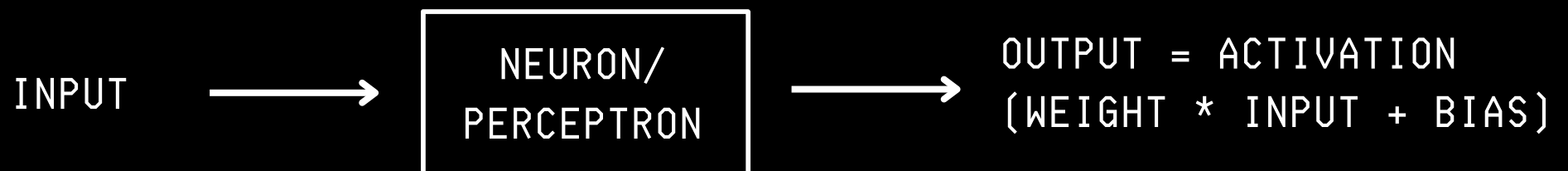
A STEP-BY-STEP PROCEDURE OR SET OF RULES DESIGNED TO PERFORM A SPECIFIC TASK OR SOLVE A PROBLEM.



PARAMETERS

VARIABLES IN A MACHINE LEARNING MODEL THAT IS LEARNED FROM THE TRAINING DATA.

AN ARTIFICIAL NEURAL NETWORK HAS WEIGHTS AND BIASES AS ITS PARAMETERS.



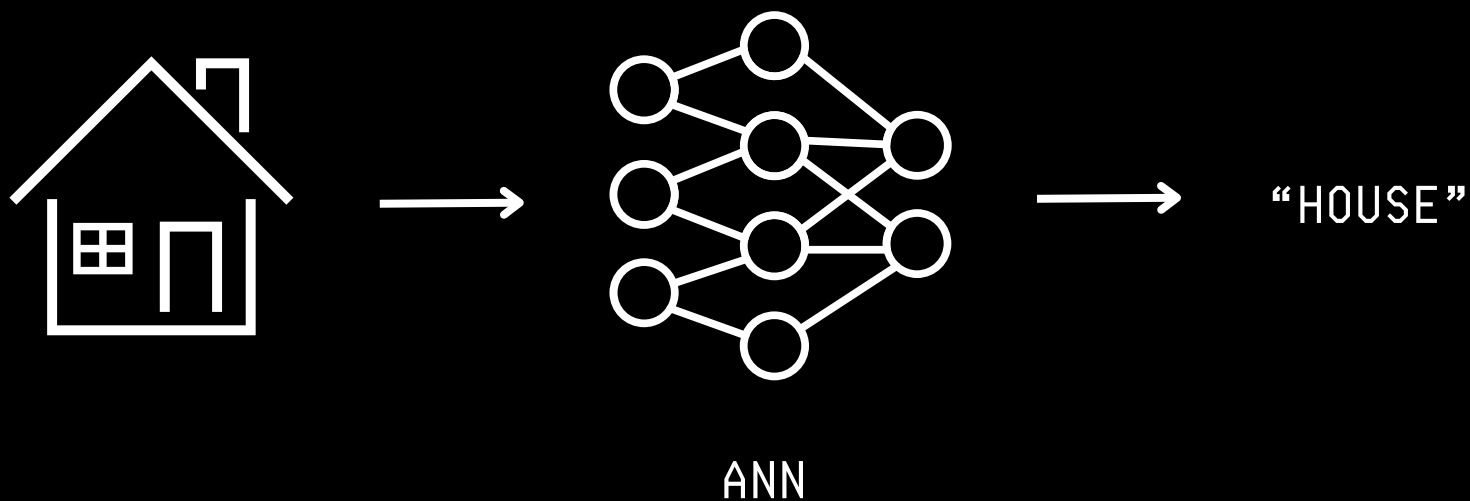
HYPERPARAMETERS

A CONFIGURATION VALUE SET BEFORE THE AI MODEL TRAINING PROCESS BEGINS, WHICH DETERMINES THE MODEL'S STRUCTURE OR HOW IT IS TRAINED.

- | | |
|----------------------|------------------------|
| 1. LEARNING RATE | 6. ACTIVATION FUNCTION |
| 2. BATCH SIZE | 7. DROPOUT RATE |
| 3. NUMBER OF EPOCHS | 8. MOMENTUM |
| 4. NUMBER OF LAYERS | 9. DECAY RATE |
| 5. NUMBER OF NEURONS | 10. OPTIMIZER |

CLASSIFICATION

A SUPERVISED LEARNING TASK WHERE A MODEL IS TRAINED TO PREDICT THE CATEGORY OR CLASS OF AN INPUT FROM A SET OF PREDEFINED CLASSES BASED ON ITS FEATURES.



REGRESSION

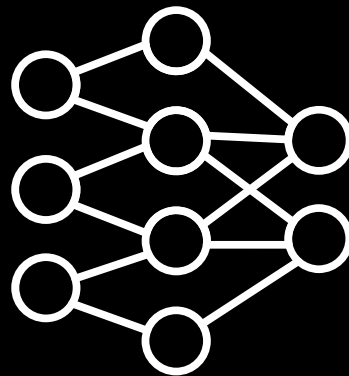
A SUPERVISED LEARNING TASK WHERE A MODEL IS TRAINED TO PREDICT A CONTINUOUS NUMERICAL VALUE BASED ON INPUT FEATURES.

HOUSE DETAILS

4 BEDROOMS

LOCATED IN LA

HAS A GARDEN



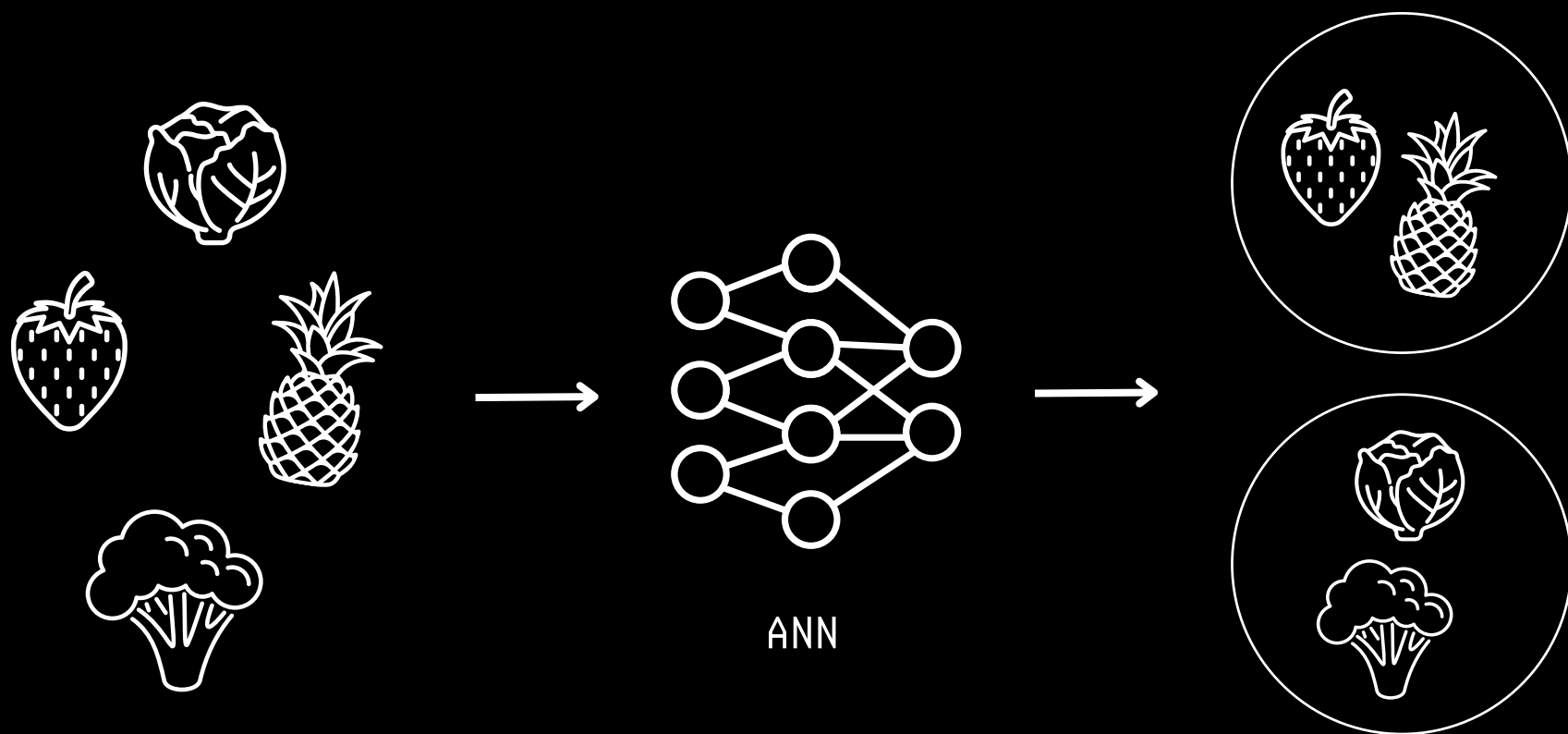
ANN



110,000 USD

CLUSTERING

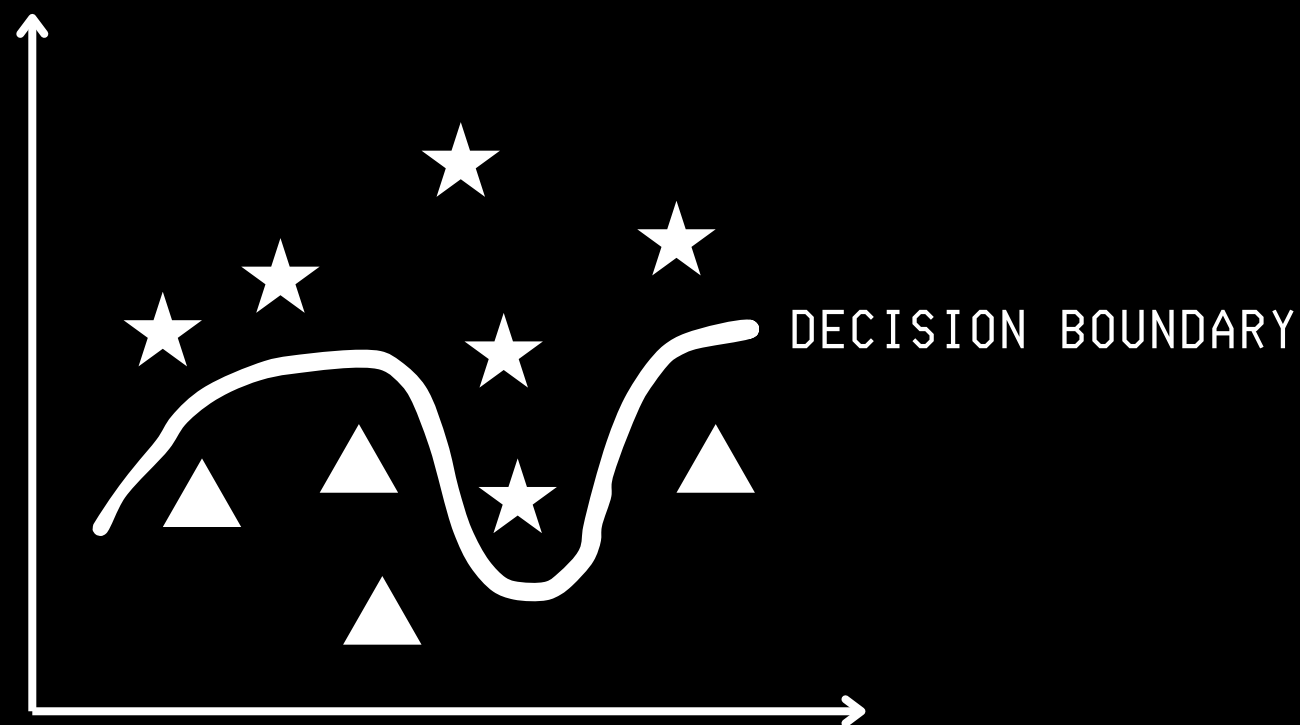
AN UNSUPERVISED LEARNING TECHNIQUE USED TO GROUP SIMILAR DATA POINTS TOGETHER INTO CLUSTERS BASED ON THEIR CHARACTERISTICS, WITHOUT USING PREDEFINED LABELS.



OVERFITTING

A MODELLING ERROR THAT IS SEEN WHEN A MACHINE LEARNING MODEL LEARNS THE TRAINING DATA TOO WELL RESULTING IN POOR GENERALIZATION TO NEW AND UNSEEN DATA.

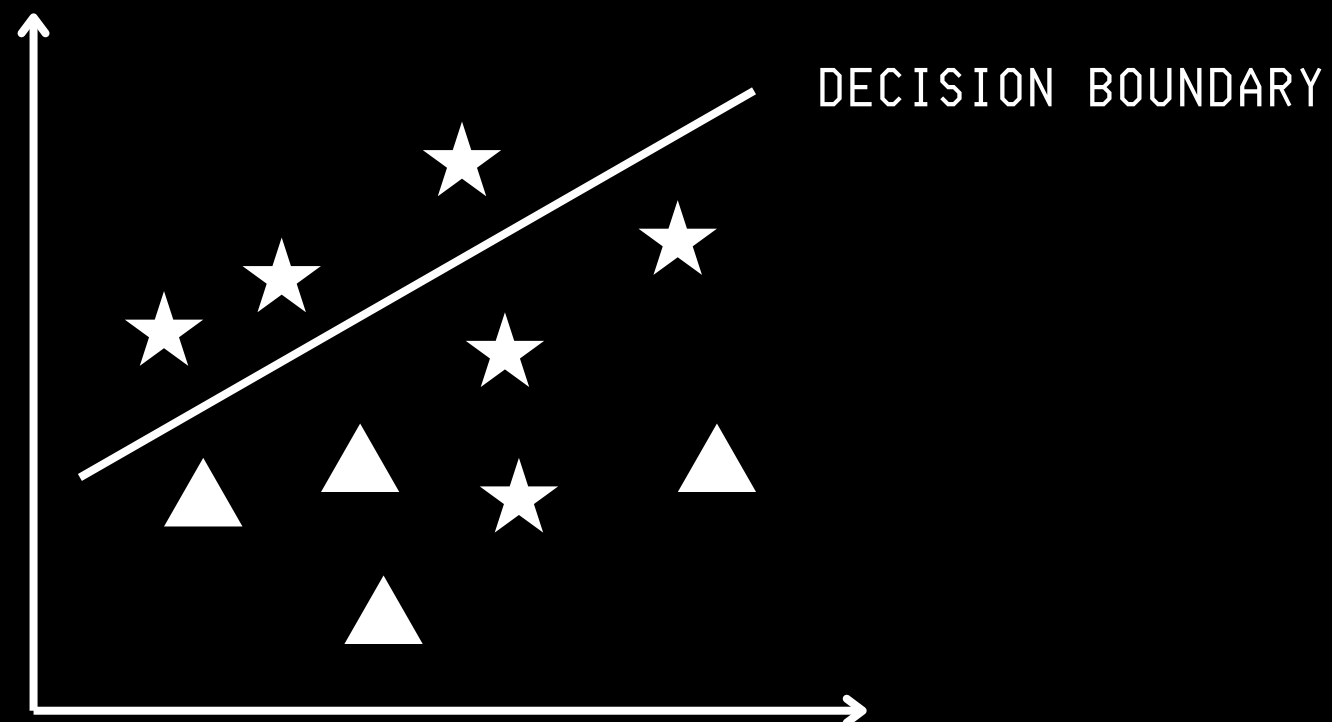
AN OVERFITTED MODEL PERFORMS BETTER ON TRAINING DATA THAN THE TESTING DATA.



UNDERFITTING

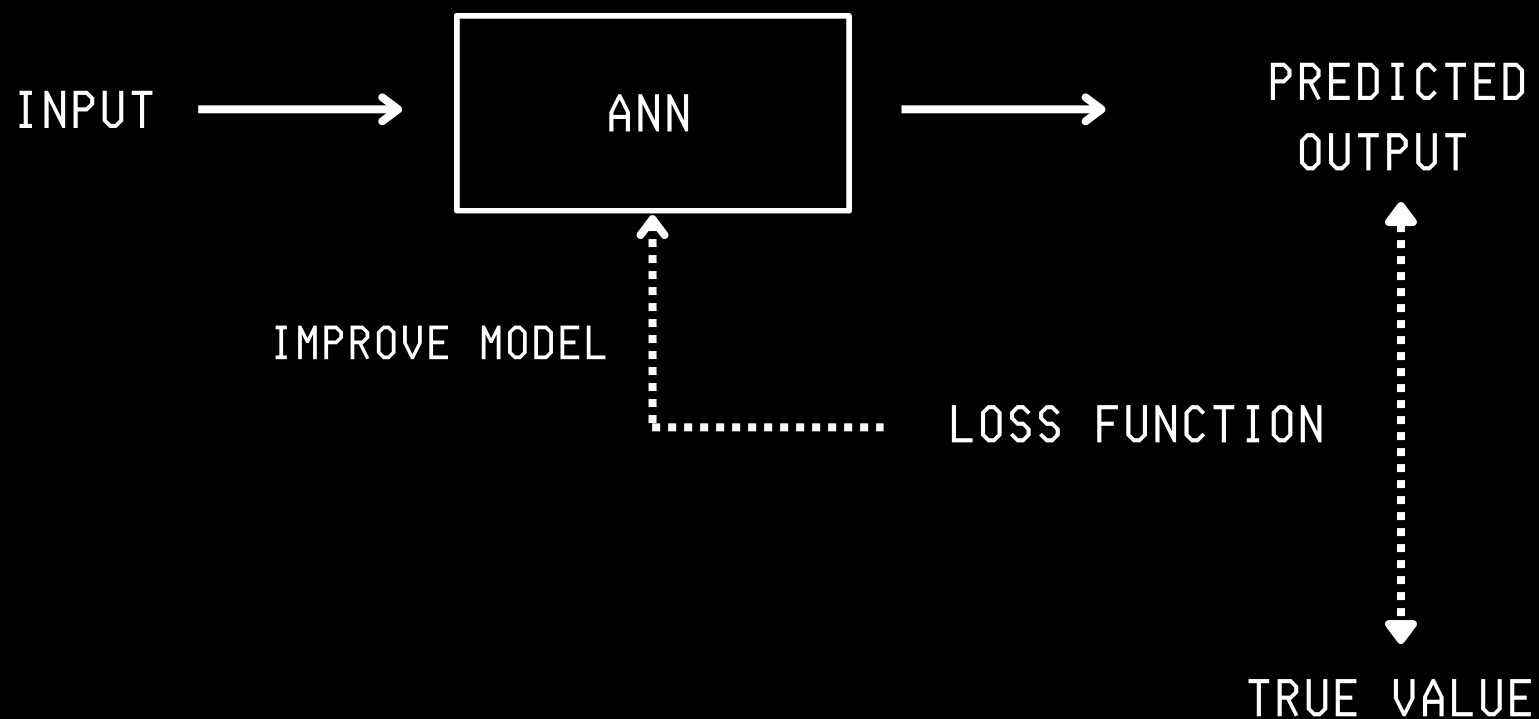
A MODELLING ERROR THAT IS SEEN WHEN A MACHINE LEARNING MODEL IS TOO SIMPLE TO CAPTURE THE UNDERLYING PATTERNS IN THE TRAINING DATA.

AN UNDERFITTED MODEL PERFORMS BADLY ON BOTH TRAINING DATA AND TESTING DATA.



LOSS FUNCTION

A MATHEMATICAL FUNCTION THAT MEASURES THE ERROR BETWEEN PREDICTED OUTPUTS AND TRUE VALUES, GUIDING AN AI MODEL'S TRAINING PROCESS BY MINIMIZING THIS ERROR.



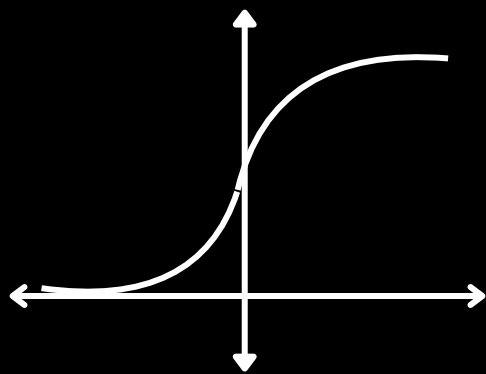
COST FUNCTION

THE AGGREGATE OF THE LOSS FUNCTIONS OVER THE ENTIRE TRAINING DATASET THAT IS USED TO GUIDE THE OPTIMIZATION ALGORITHM IN TRAINING AN AI MODEL.

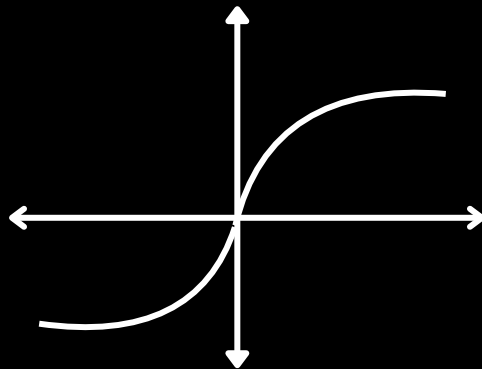
$$\text{COST FUNCTION} = \sum \text{LOSS FUNCTION FOR EACH TRAINING EXAMPLE}$$

ACTIVATION FUNCTION

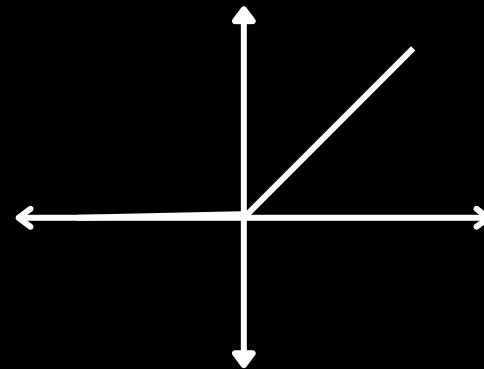
A MATHEMATICAL FUNCTION APPLIED TO A NEURON'S OUTPUT TO INTRODUCE NON-LINEARITY INTO A NEURAL NETWORK, ENABLING IT TO LEARN COMPLEX PATTERNS.



SIGMOID



TANH

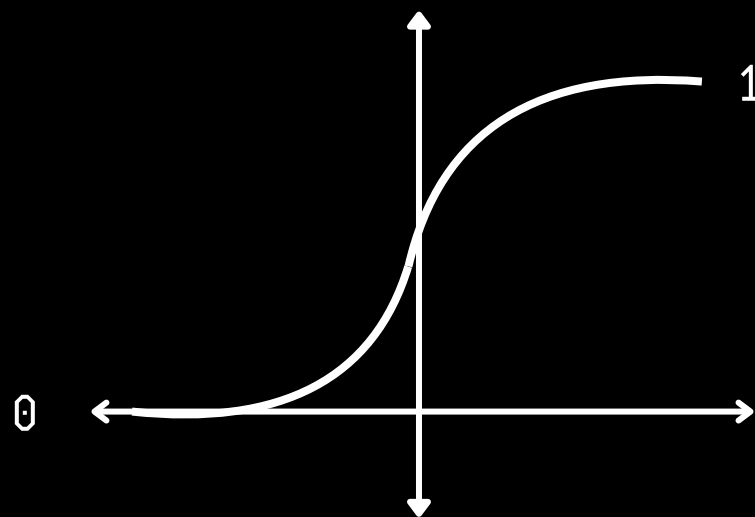


RELU

SIGMOID FUNCTION

AN ACTIVATION FUNCTION THAT MAPS ANY REAL-VALUED NUMBER TO A VALUE BETWEEN 0 AND 1.

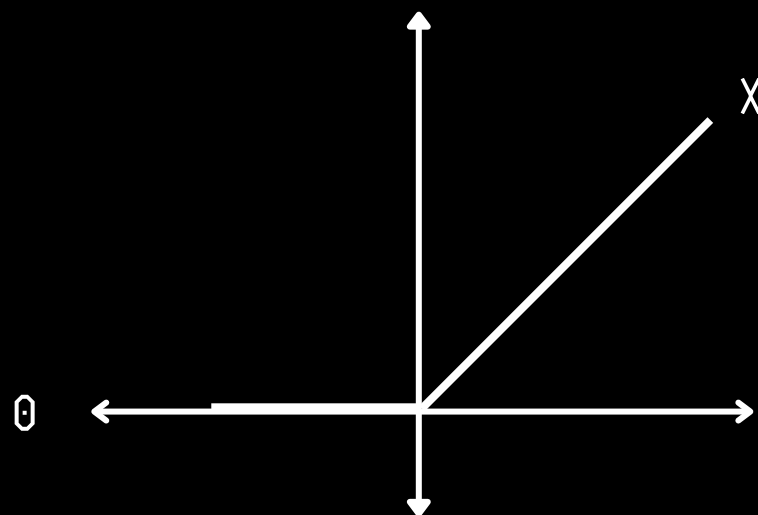
$$\sigma(x) = 1 / (1 + e^{-x})$$



RELU FUNCTION

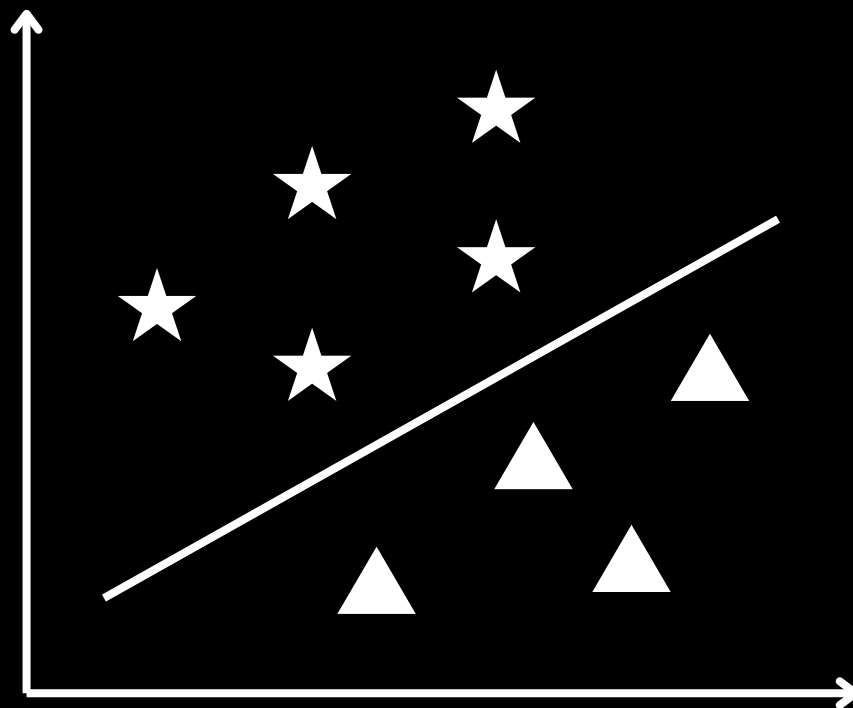
RELU (RECTIFIED LINEAR UNIT) IS AN ACTIVATION FUNCTION THAT OUTPUTS THE INPUT DIRECTLY IF IT IS POSITIVE, OTHERWISE, IT OUTPUTS 0.

$$\text{RELU}(x) = \max(0, x)$$



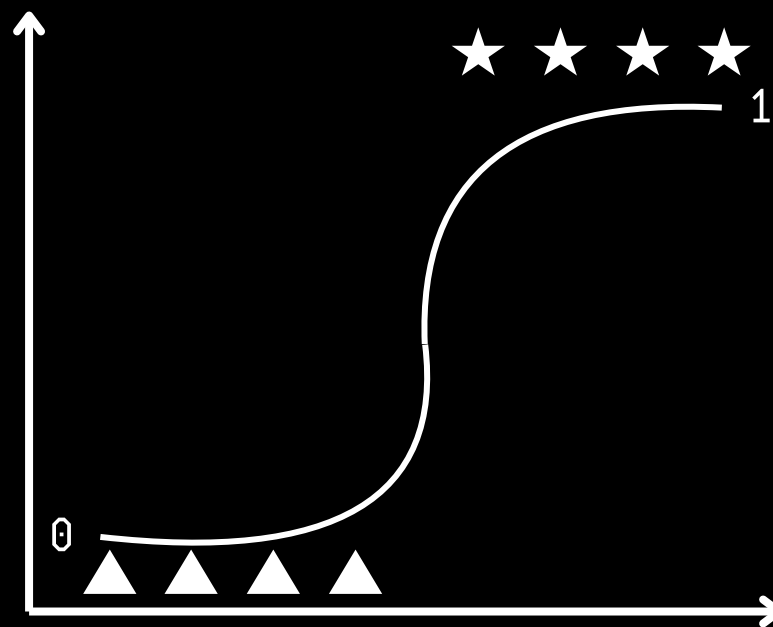
LINEAR REGRESSION

A SUPERVISED LEARNING ALGORITHM THAT PREDICTS A CONTINUOUS TARGET VARIABLE BY LEARNING THE LINEAR RELATIONSHIP BETWEEN INPUT FEATURES AND THE TARGET VARIABLE.



LOGISTIC REGRESSION

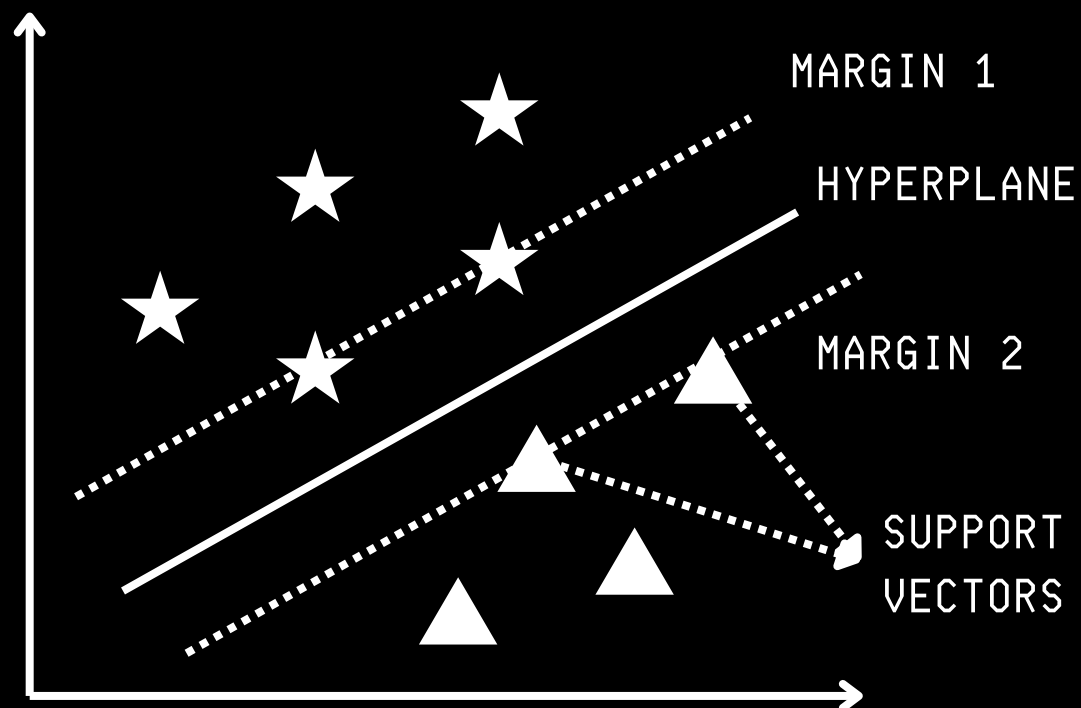
A SUPERVISED LEARNING ALGORITHM USED FOR CLASSIFICATION TASKS THAT PREDICTS THE PROBABILITY OF A CATEGORICAL OUTCOME BY MODELING THE RELATIONSHIP BETWEEN INPUT FEATURES AND THE LOG-ODDS OF THE TARGET VARIABLE USING A LOGISTIC FUNCTION.



SVM

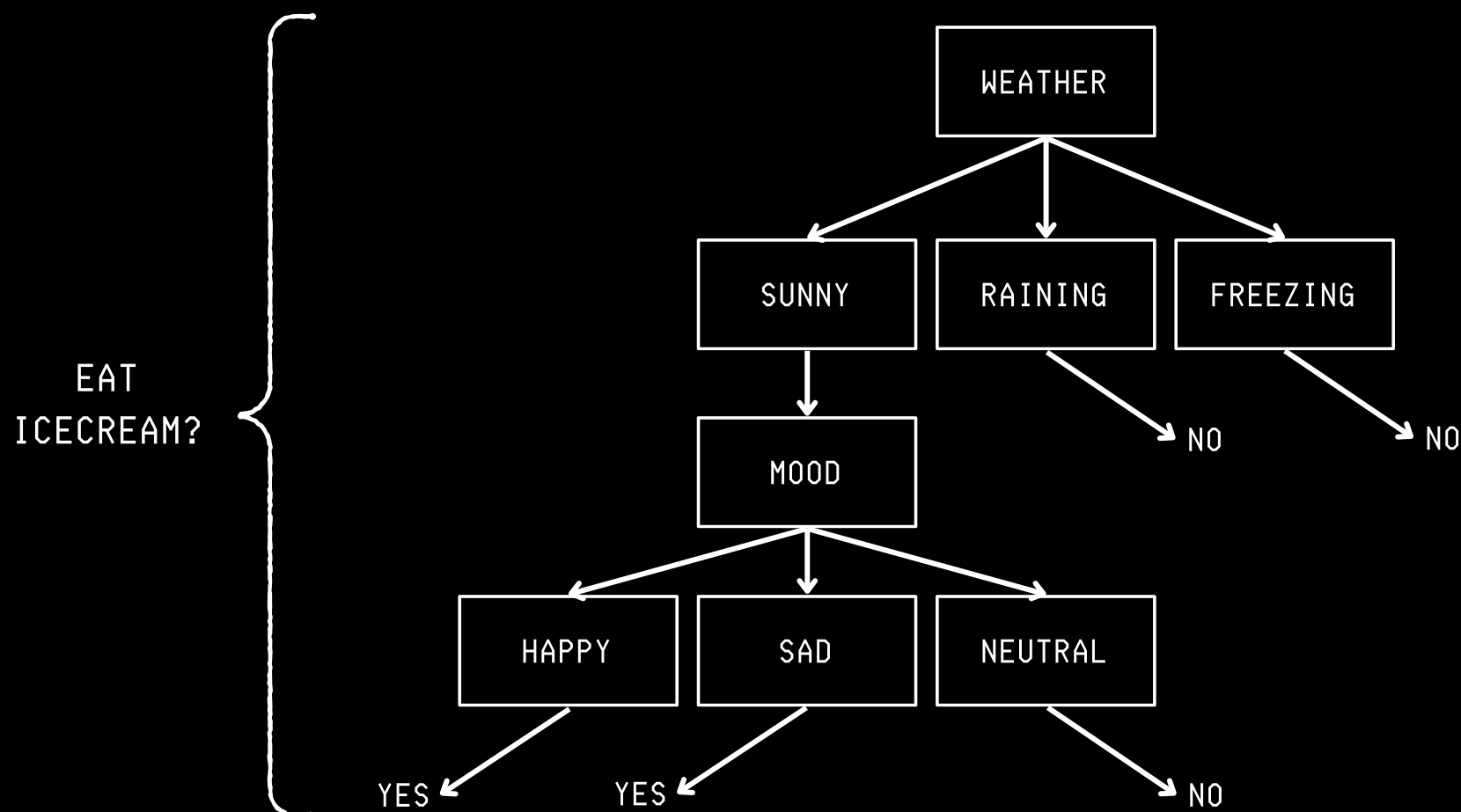
SUPPORT VECTOR MACHINE

A SUPERVISED LEARNING ALGORITHM THAT FINDS THE OPTIMAL HYPERPLANE TO SEPARATE DATA POINTS OF DIFFERENT CLASSES IN A HIGH-DIMENSIONAL SPACE.



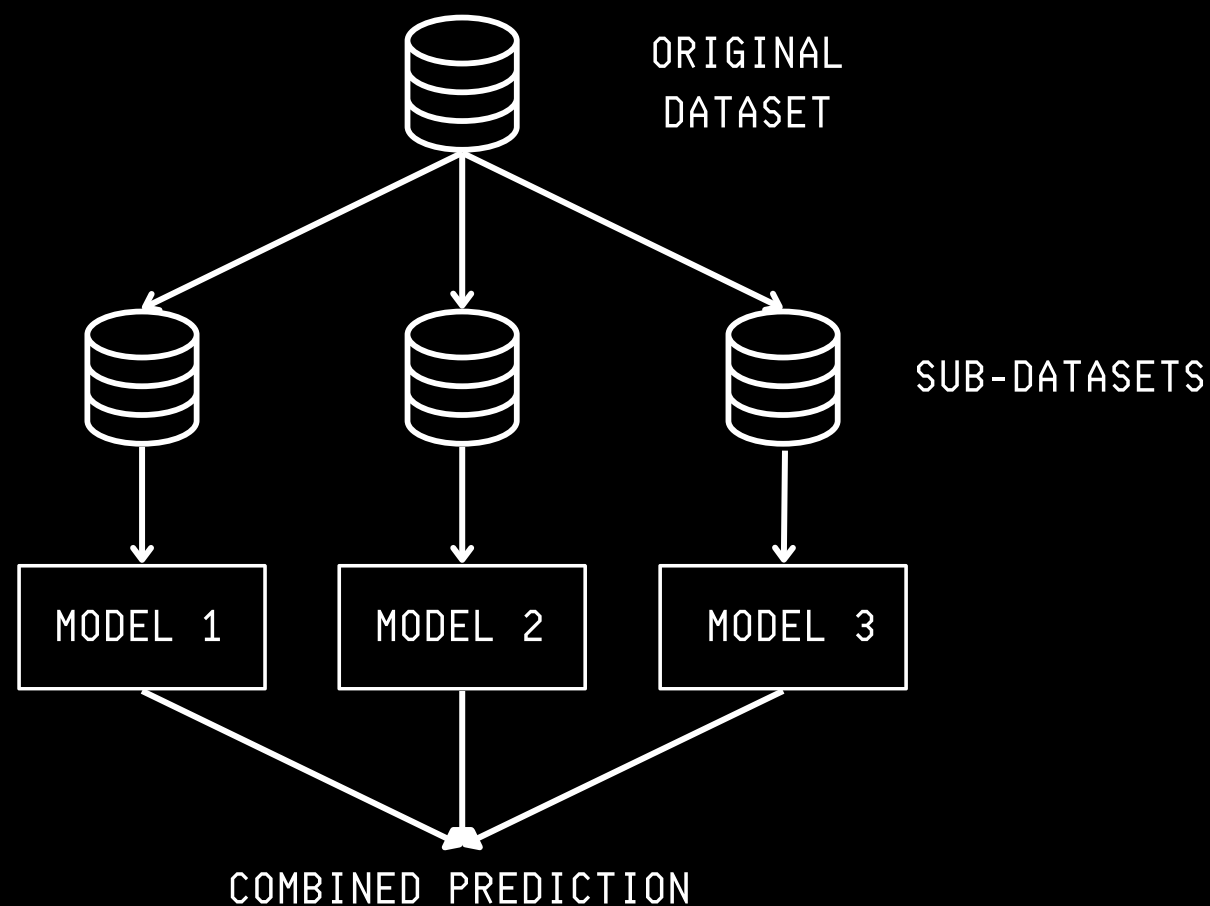
DECISION TREE

A SUPERVISED LEARNING ALGORITHM THAT MAKES PREDICTIONS BY RECURSIVELY SPLITTING DATA INTO SUBSETS BASED ON FEATURE VALUES, FORMING A TREE-LIKE STRUCTURE OF DECISIONS.



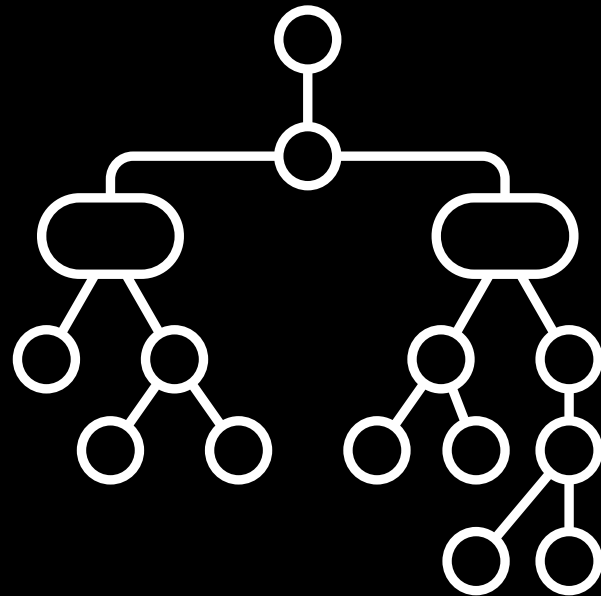
ENSEMBLE LEARNING

A TECHNIQUE THAT COMBINES THE PREDICTIONS OF MULTIPLE MODELS TO IMPROVE OVERALL PERFORMANCE AND ROBUSTNESS COMPARED TO INDIVIDUAL MODELS.



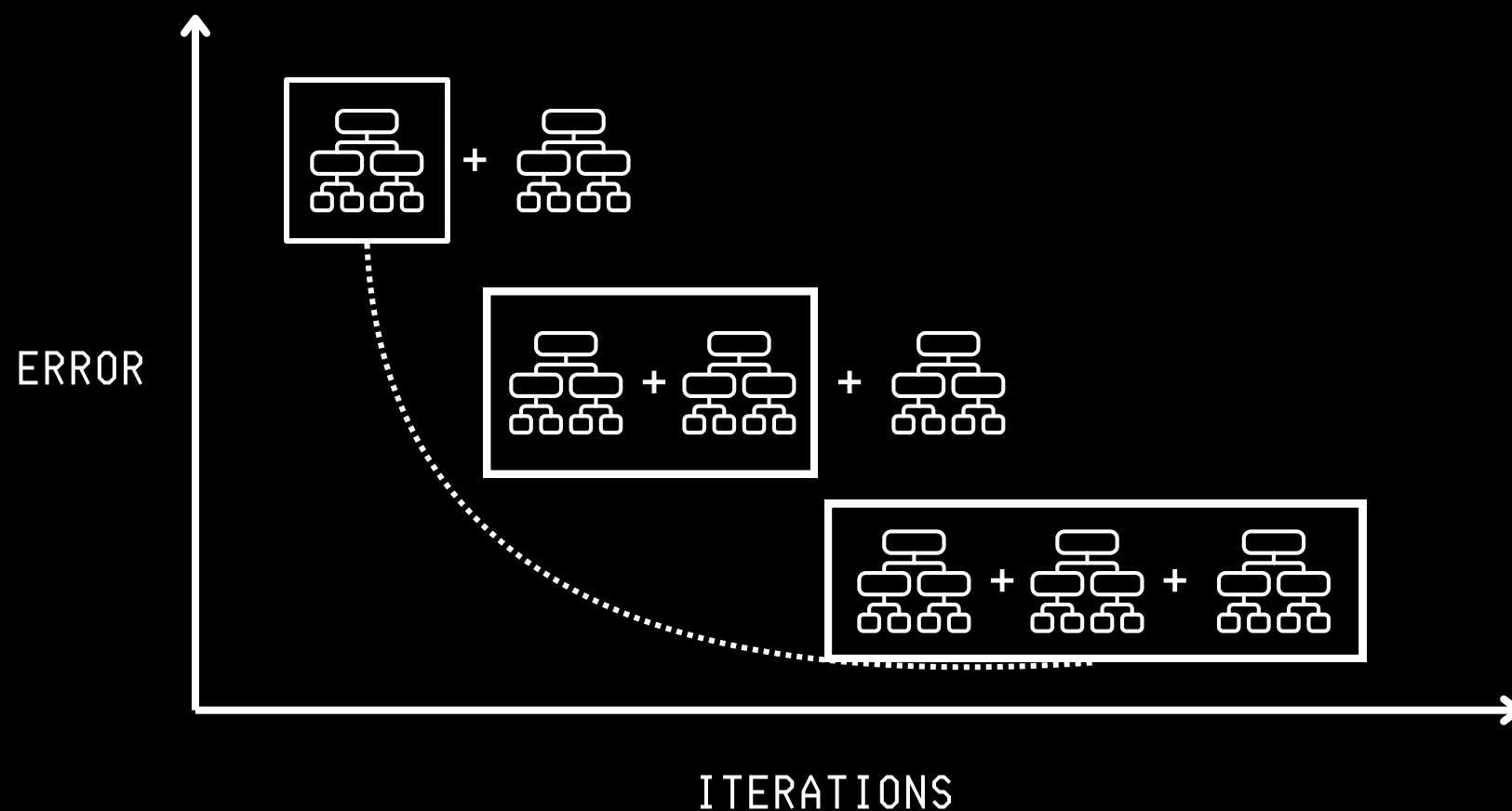
RANDOM FOREST

AN ENSEMBLE LEARNING METHOD THAT IMPROVES CLASSIFICATION AND REGRESSION ACCURACY BY CONSTRUCTING MULTIPLE DECISION TREES WITH RANDOM SUBSETS OF DATA AND FEATURES, AND COMBINING THEIR PREDICTIONS.



GRADIENT BOOSTING

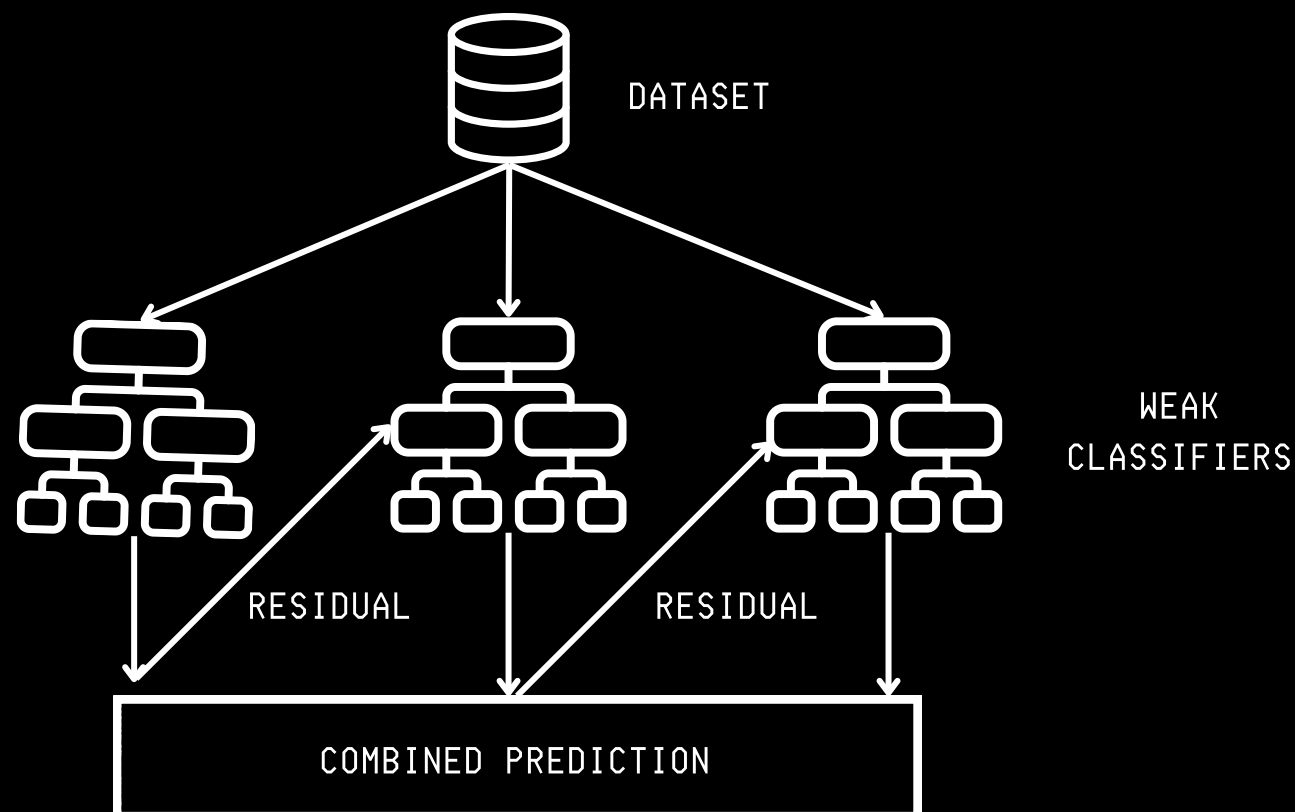
AN ENSEMBLE LEARNING TECHNIQUE THAT ITERATIVELY BUILDS A MODEL OF WEAK LEARNERS, USUALLY DECISION TREES, WHERE EACH SUBSEQUENT MODEL CORRECTS THE ERRORS OF THE PREVIOUS ONES BY OPTIMIZING A LOSS FUNCTION USING GRADIENT DESCENT.



XGBOOST

EXTREME GRADIENT BOOSTING

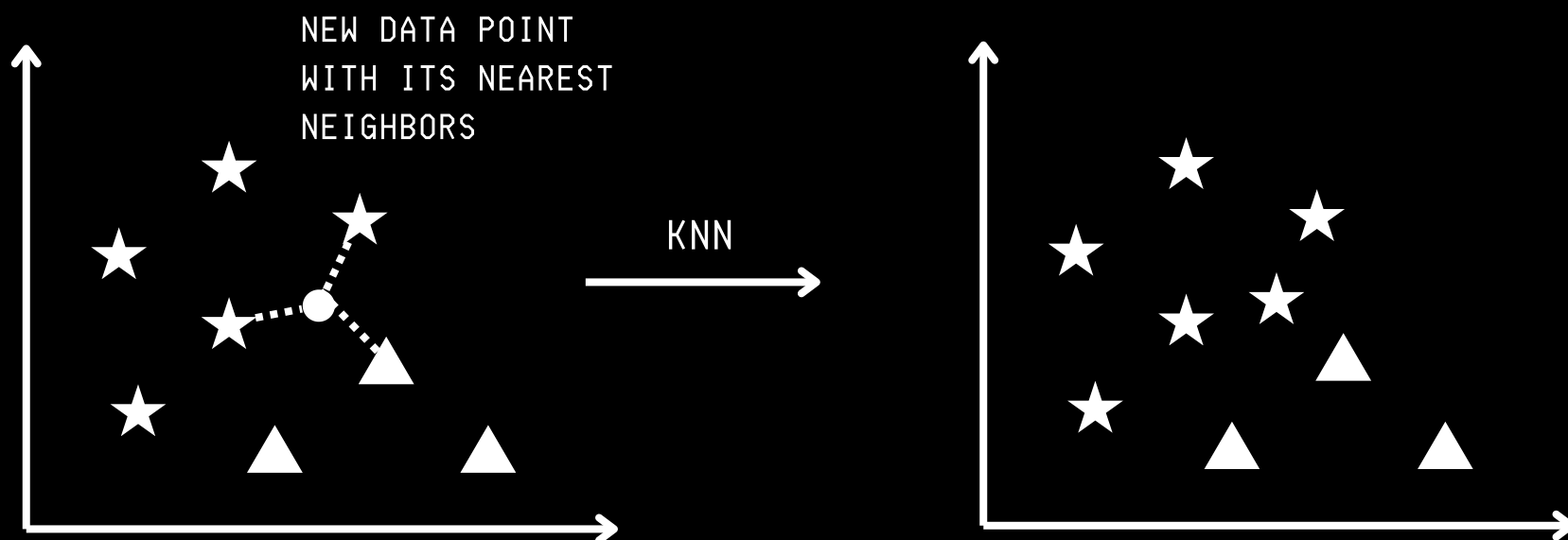
AN OPTIMIZED GRADIENT BOOSTING ALGORITHM THAT ENHANCES PERFORMANCE AND EFFICIENCY THROUGH ADVANCED TECHNIQUES LIKE REGULARIZATION, PARALLEL PROCESSING, AND HANDLING MISSING VALUES, MAKING IT HIGHLY EFFECTIVE FOR CLASSIFICATION AND REGRESSION TASKS IN LARGE DATASETS.



KNN

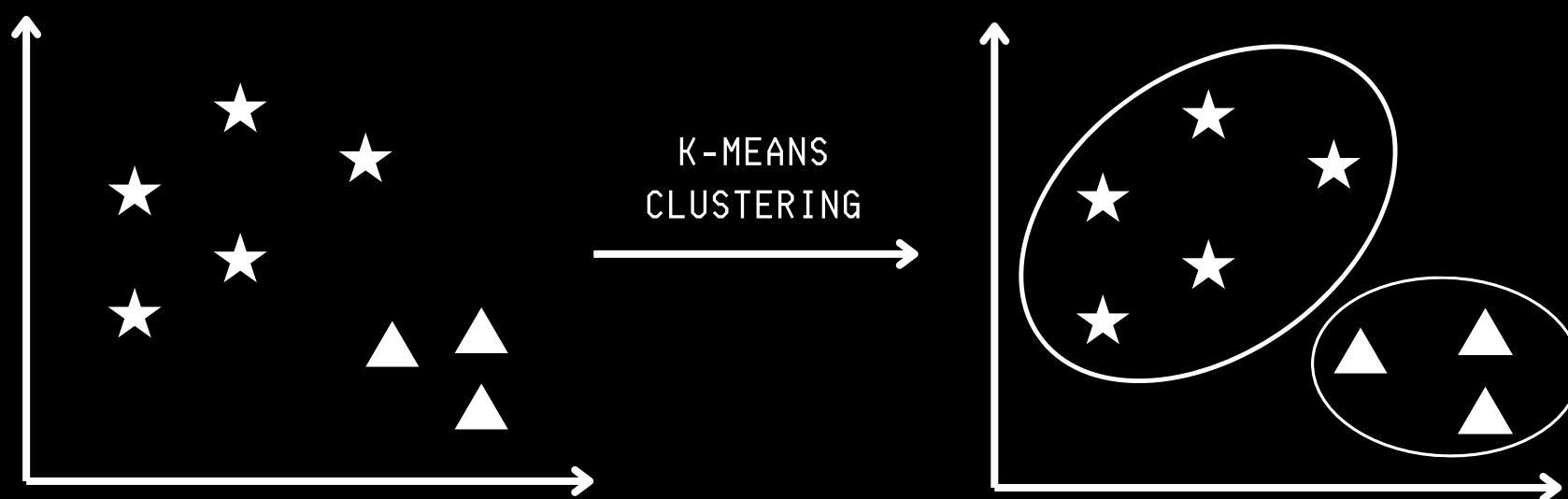
K-NEAREST NEIGHBORS

A SUPERVISED LEARNING ALGORITHM USED FOR CLASSIFICATION AND REGRESSION TASKS THAT PREDICTS THE LABEL OF A DATA POINT BASED ON THE MAJORITY LABEL (CLASSIFICATION) OR AVERAGE VALUE (REGRESSION) OF ITS K NEAREST NEIGHBORS IN THE FEATURE SPACE.



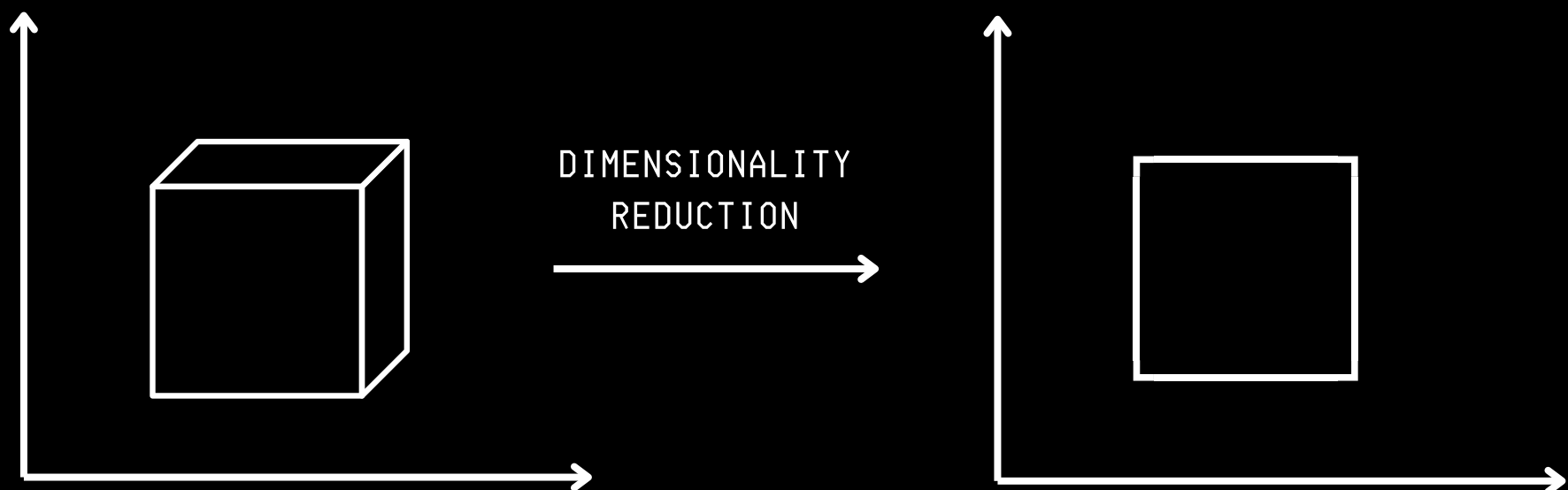
K-MEANS CLUSTERING

AN UNSUPERVISED LEARNING ALGORITHM THAT PARTITIONS A DATASET INTO K DISTINCT CLUSTERS BY MINIMIZING THE VARIANCE WITHIN EACH CLUSTER.



DIMENSIONALITY REDUCTION

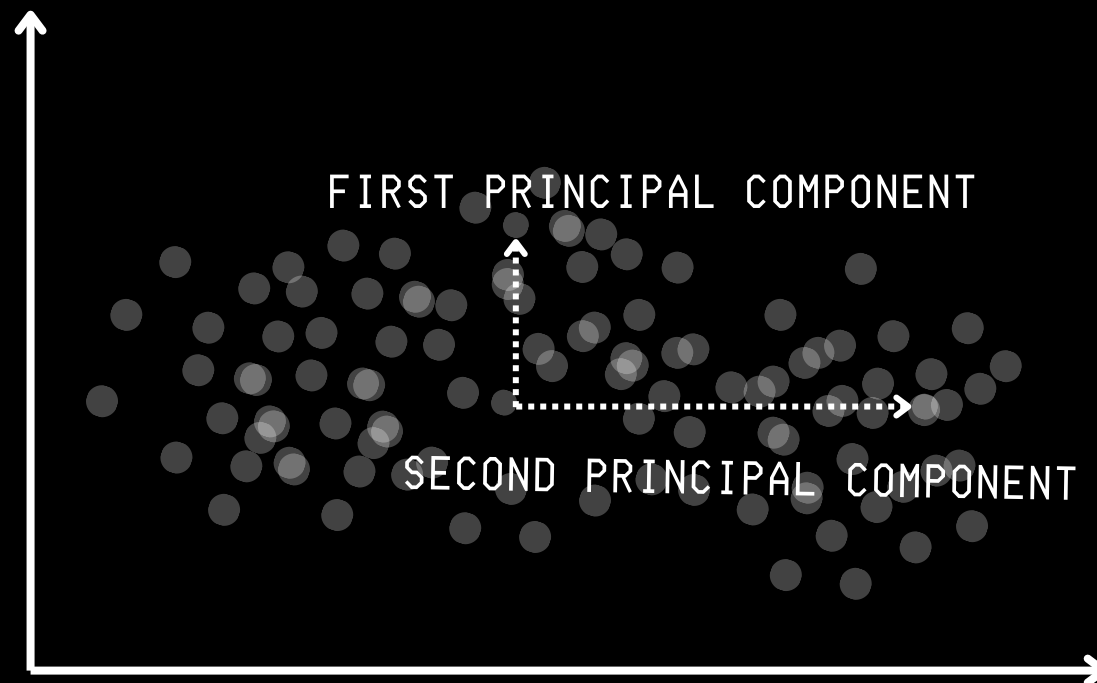
A PROCESS THAT REDUCES THE NUMBER OF INPUT VARIABLES OR FEATURES IN A DATASET WHILE PRESERVING AS MUCH INFORMATION AS POSSIBLE, TO SIMPLIFY AI MODELS, ENHANCE VISUALIZATION, AND IMPROVE COMPUTATIONAL EFFICIENCY.



PCA

PRINCIPAL COMPONENT ANALYSIS

AN UNSUPERVISED LEARNING TECHNIQUE USED FOR DIMENSIONALITY REDUCTION, WHICH TRANSFORMS THE ORIGINAL FEATURES INTO A NEW SET OF ORTHOGONAL FEATURES (PRINCIPAL COMPONENTS) THAT CAPTURE THE MAXIMUM VARIANCE IN THE DATA.



PRECISION

A METRIC USED TO EVALUATE THE ACCURACY OF A CLASSIFICATION MODEL.

IT IS DEFINED AS THE RATIO OF TRUE POSITIVE PREDICTIONS TO THE TOTAL NUMBER OF POSITIVE PREDICTIONS MADE BY THE AI MODEL (TRUE POSITIVES PLUS FALSE POSITIVES).

$$\text{PRECISION} = \frac{\text{TRUE POSITIVES (TP)}}{\text{TRUE POSITIVES (TP) + FALSE POSITIVES (FP)}}$$

RECALL

{CALLED SENSITIVITY OR TRUE POSITIVE RATE}

A METRIC USED TO EVALUATE THE EFFECTIVENESS OF A CLASSIFICATION MODEL.

IT IS DEFINED AS THE RATIO OF TRUE POSITIVE PREDICTIONS TO THE TOTAL NUMBER OF ACTUAL POSITIVES {TRUE POSITIVES PLUS FALSE NEGATIVES}.

$$\text{RECALL} = \frac{\text{TRUE POSITIVES (TP)}}{\text{TRUE POSITIVES (TP) + FALSE NEGATIVES (FN)}}$$

F1-SCORE

A METRIC USED TO EVALUATE THE PERFORMANCE OF A CLASSIFICATION MODEL.

IT IS DEFINED AS THE HARMONIC MEAN OF PRECISION AND RECALL, PROVIDING A BALANCE BETWEEN THE TWO METRICS.

$$\text{F1-SCORE} = 2 \times \frac{\text{PRECISION} \times \text{RECALL}}{\text{PRECISION} + \text{RECALL}}$$

SPECIFICITY

{CALLED TRUE NEGATIVE RATE}

A METRIC USED TO EVALUATE THE PERFORMANCE OF A CLASSIFICATION MODEL.

IT IS DEFINED AS THE RATIO OF TRUE NEGATIVE PREDICTIONS TO THE TOTAL NUMBER OF ACTUAL NEGATIVES {TRUE NEGATIVES PLUS FALSE POSITIVES}.

$$\text{SPECIFICITY} = \frac{\text{TRUE NEGATIVES (TN)}}{\text{TRUE NEGATIVES (TN) + FALSE POSITIVES (FP)}}$$

CONFUSION MATRIX

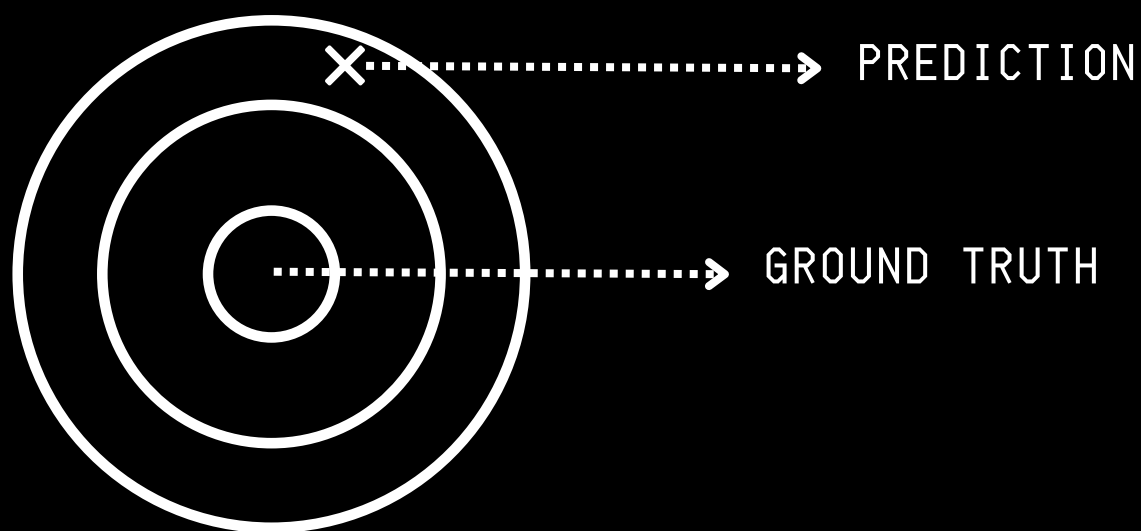
A TABLE USED TO EVALUATE THE PERFORMANCE OF A CLASSIFICATION MODEL BY COMPARING PREDICTED AND ACTUAL VALUES, SHOWING TRUE POSITIVES, FALSE POSITIVES, TRUE NEGATIVES, AND FALSE NEGATIVES.

ACTUAL VALUES	
PREDICTED VALUES	TRUE POSITIVES
	FALSE POSITIVES
PREDICTED VALUES	FALSE NEGATIVES
	TRUE NEGATIVES

BIAS

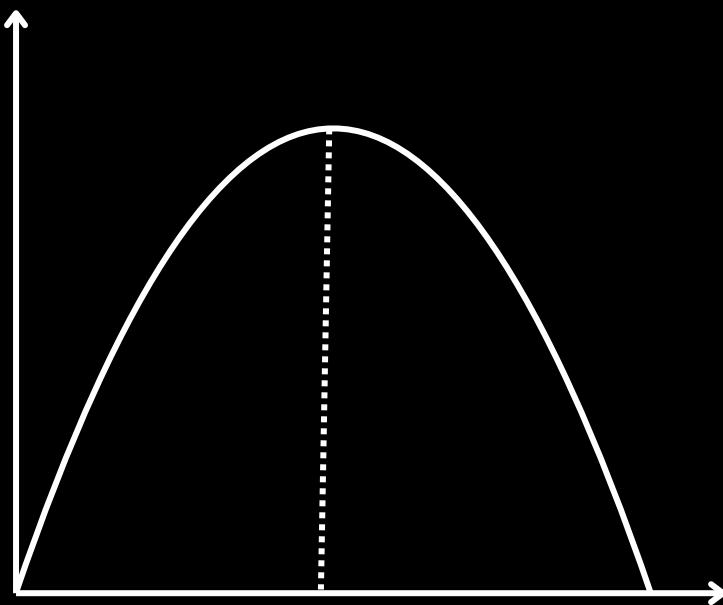
AN ERROR INTRODUCED BY APPROXIMATING A COMPLEX REAL-WORLD PROBLEM WITH A SIMPLIFIED MODEL, THAT LEADS TO SYSTEMATIC DEVIATIONS IN PREDICTIONS FROM THE ACTUAL VALUES.

WHEN HIGH BIAS IS A CAUSE, AND UNDERFITTING IS AN OUTCOME.

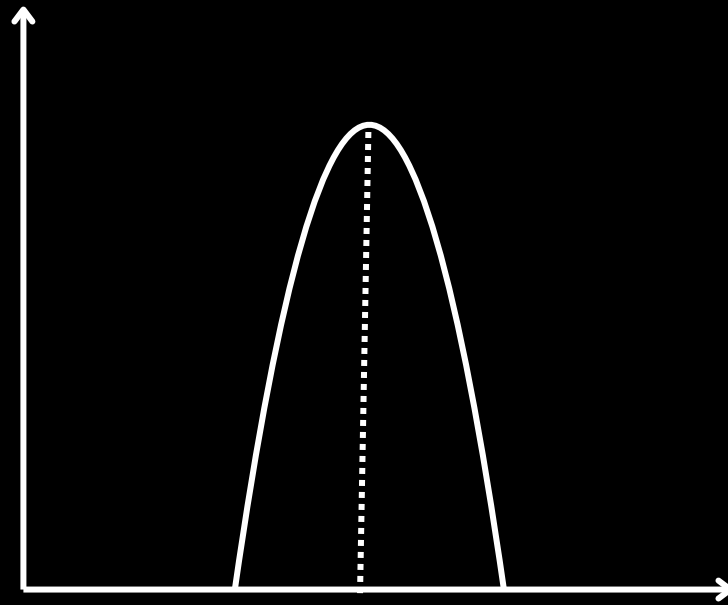


VARIANCE

A STATISTICAL MEASURE THAT QUANTIFIES THE DISPERSION OR SPREAD OF A SET OF DATA POINTS AROUND THEIR MEAN.



HIGH VARIANCE



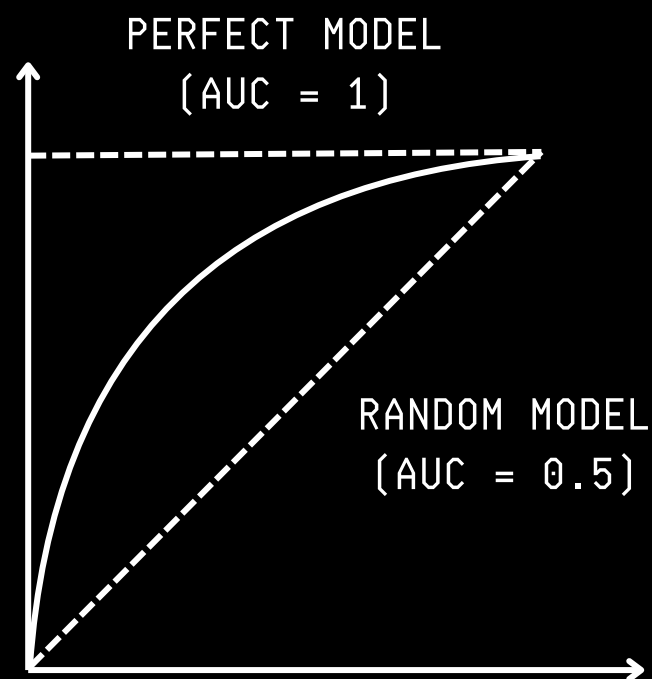
LOW VARIANCE

ROC CURVE

RECEIVER OPERATING CHARACTERISTIC CURVE

A GRAPHICAL REPRESENTATION USED TO EVALUATE THE PERFORMANCE OF A BINARY CLASSIFICATION MODEL BY PLOTTING THE TRUE POSITIVE RATE (TPR) AGAINST THE FALSE POSITIVE RATE (FPR) AT VARIOUS THRESHOLD SETTINGS.

THE AREA UNDER THE ROC CURVE (AUC) IS A VALUE THAT SUMMARIZES THE PERFORMANCE OF THE CLASSIFIER.



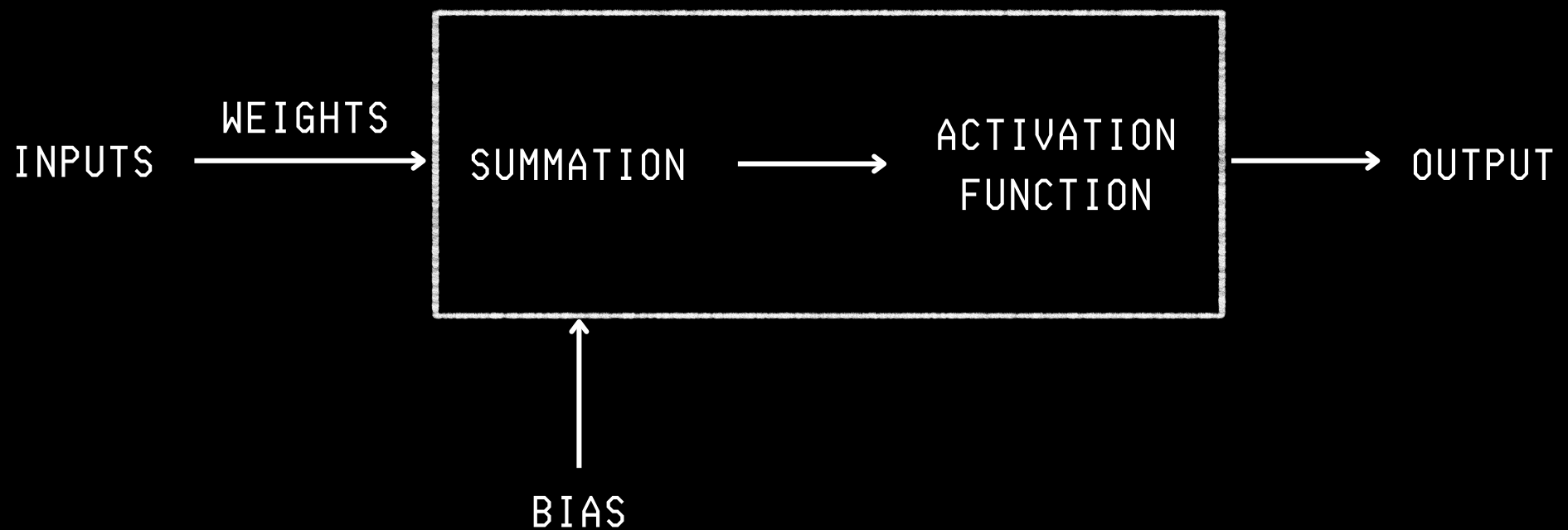
NAIVE BAYES

A PROBABILISTIC CLASSIFIER THAT APPLIES BAYES' THEOREM WITH THE ASSUMPTION OF FEATURE INDEPENDENCE, COMMONLY USED FOR TASKS LIKE TEXT CLASSIFICATION.

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

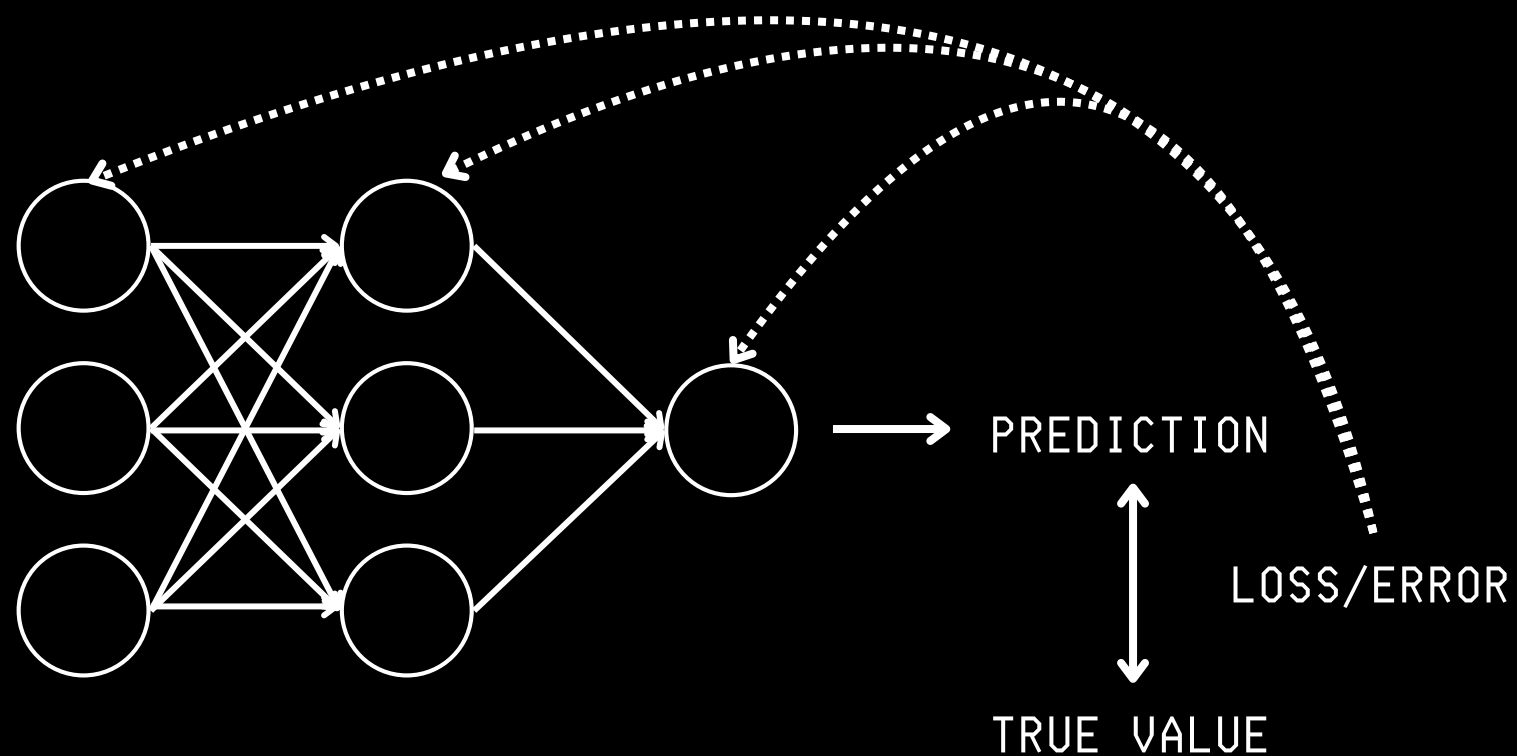
PERCEPTRON

THE BASIC UNIT OF A NEURAL NETWORK WHICH PERFORMS A LINEAR BINARY CLASSIFICATION BY APPLYING A WEIGHTED SUM OF INPUT FEATURES FOLLOWED BY AN ACTIVATION FUNCTION TO PRODUCE AN OUTPUT.



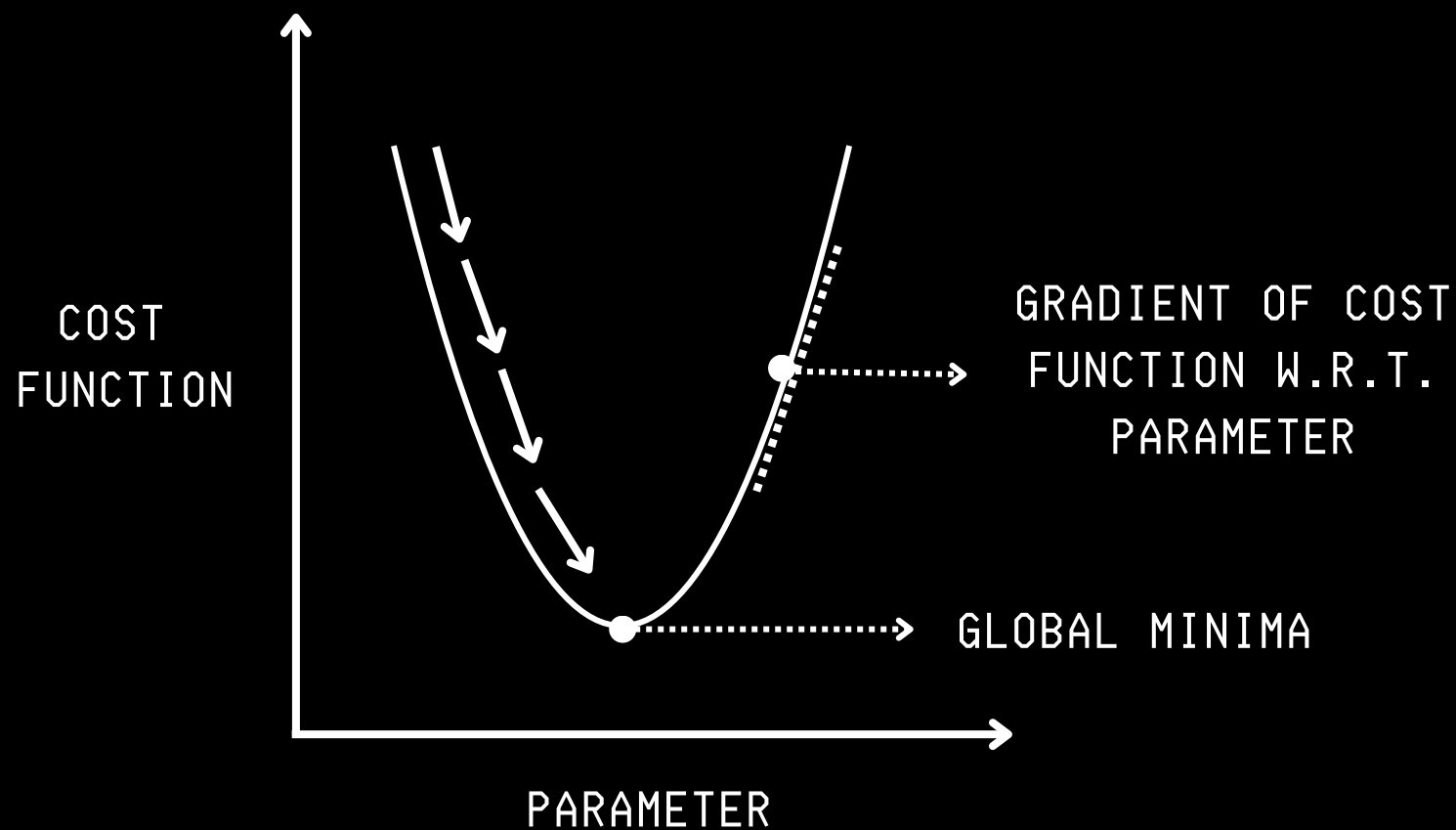
BACKPROPAGATION

AN ALGORITHM THAT COMPUTES THE GRADIENT OF THE LOSS FUNCTION WITH RESPECT TO EACH PARAMETER IN A NEURAL NETWORK AND UPDATES THEM TO MINIMIZE THE ERROR BY PROPAGATING THE ERRORS BACKWARDS FROM THE OUTPUT TO THE INPUT LAYER.



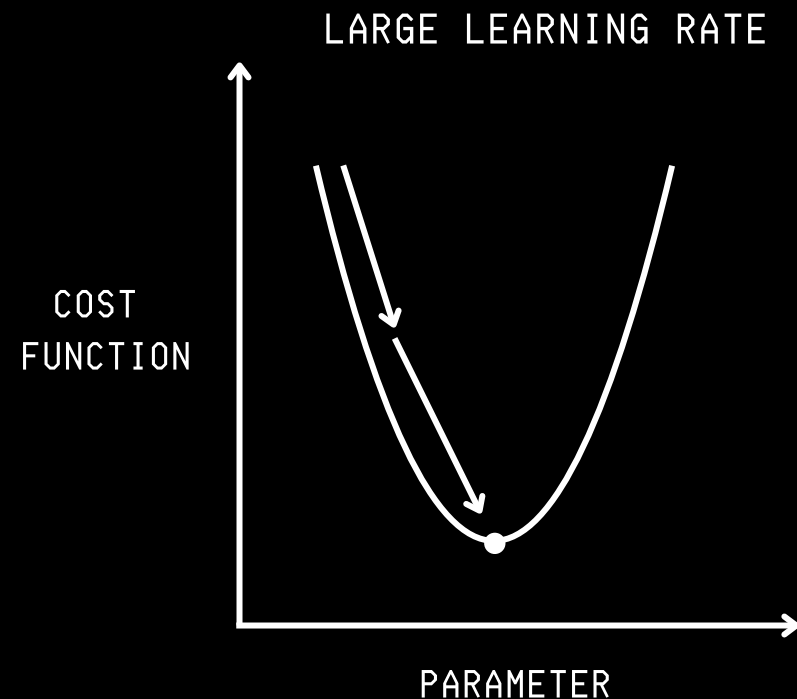
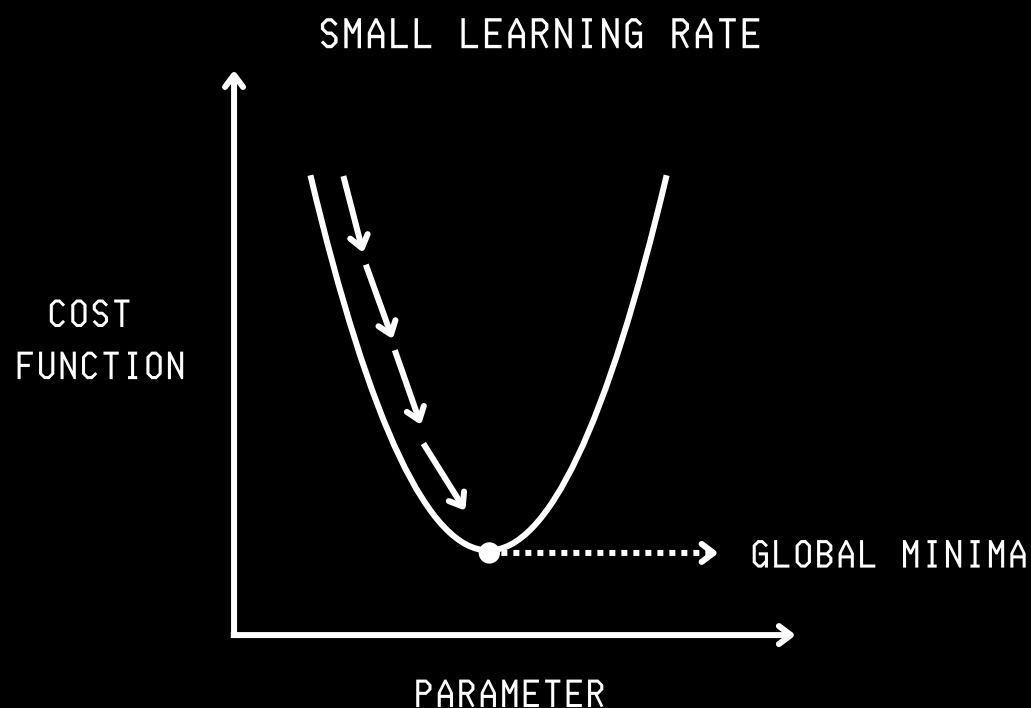
GRADIENT DESCENT

AN ITERATIVE OPTIMIZATION ALGORITHM THAT ADJUSTS THE MODEL PARAMETERS IN THE DIRECTION OPPOSITE TO THE GRADIENT OF THE LOSS FUNCTION WITH RESPECT TO THOSE PARAMETERS.



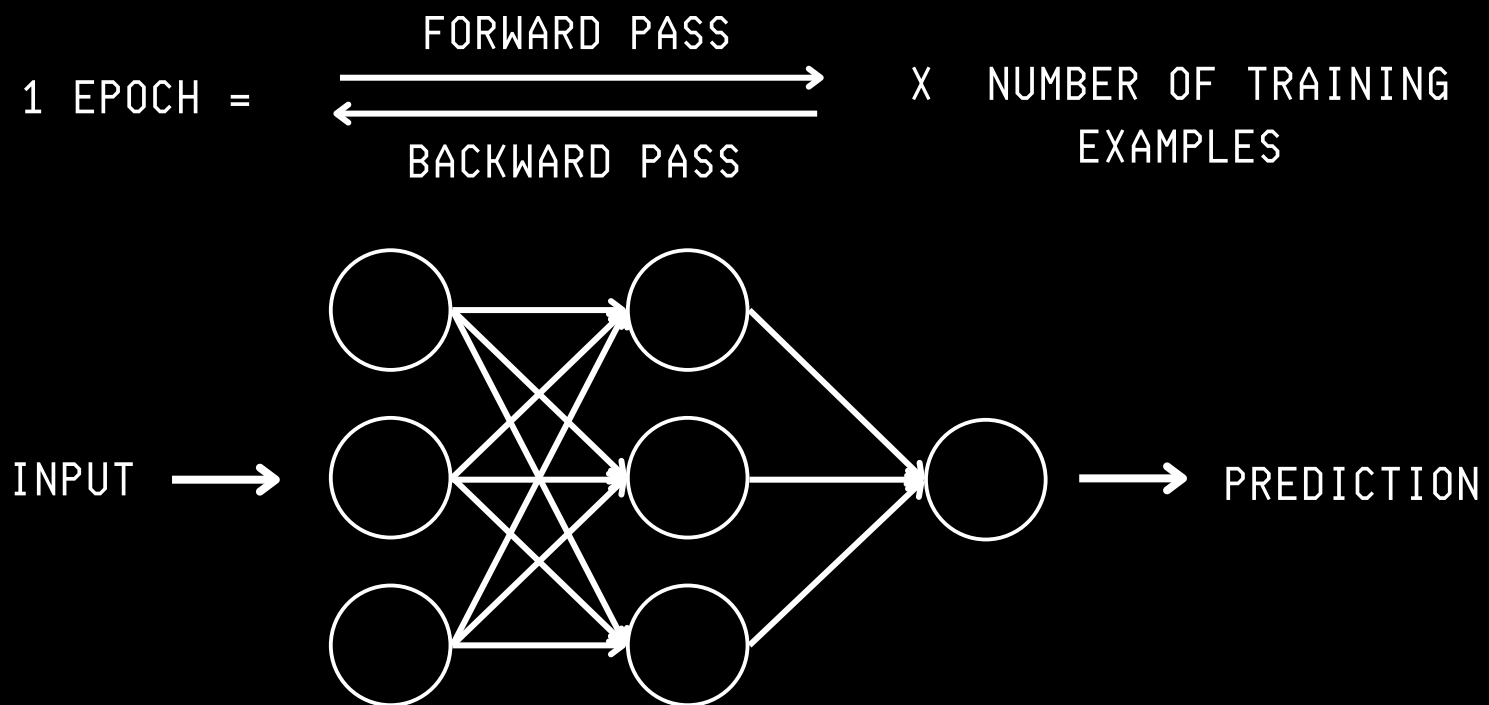
LEARNING RATE

A HYPERPARAMETER THAT DETERMINES THE STEP SIZE AT EACH ITERATION WHILE MOVING TOWARDS THE MINIMUM OF THE LOSS FUNCTION (GLOBAL MINIMA) IN OPTIMIZATION ALGORITHMS.



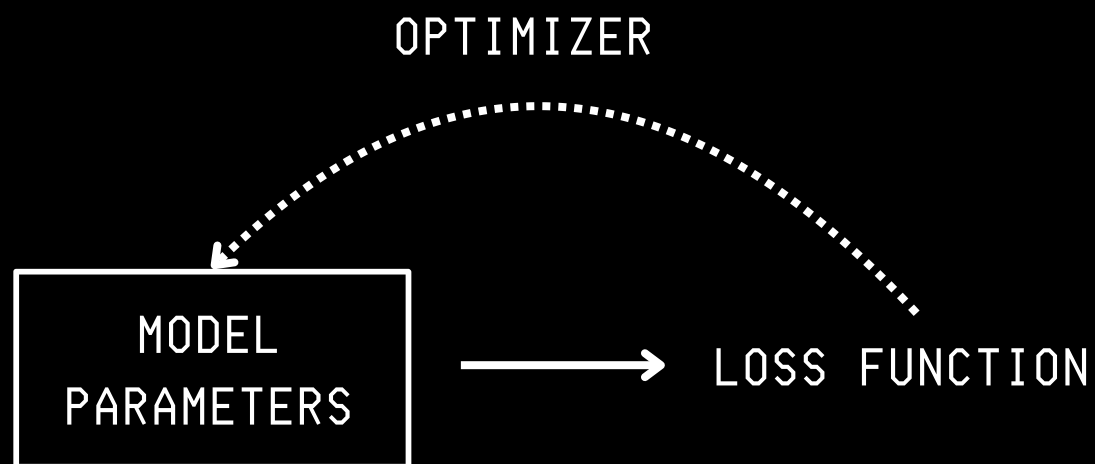
EPOCH

ONE COMPLETE CYCLE THROUGH THE ENTIRE TRAINING DATASET, INVOLVING ONE FORWARD PASS AND ONE BACKWARD PASS FOR EACH TRAINING EXAMPLE IN THE DATASET.



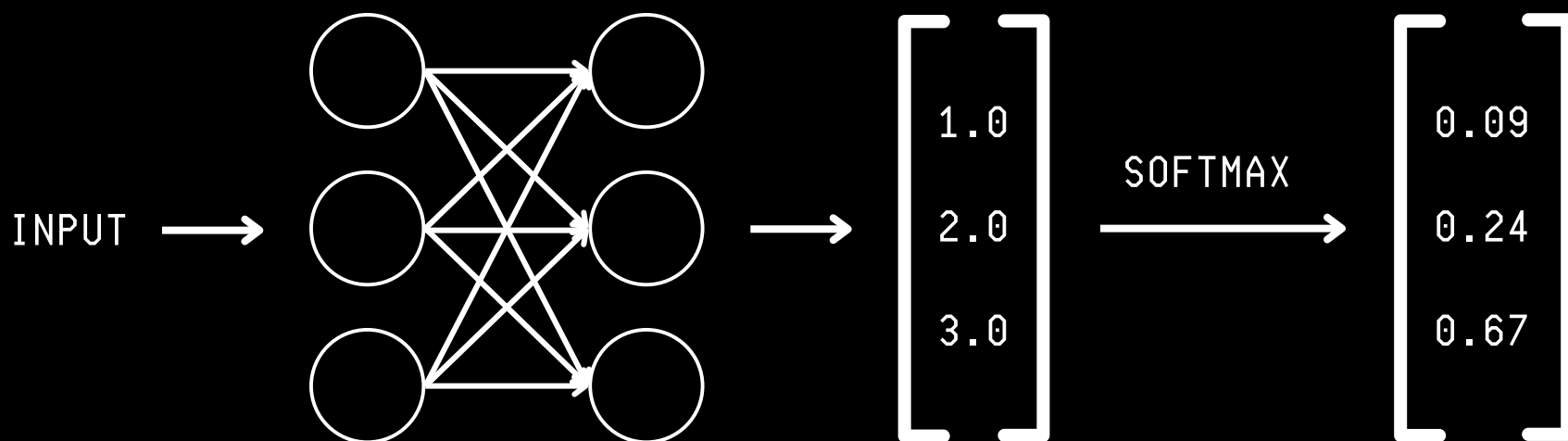
OPTIMIZER

AN ALGORITHM THAT ADJUSTS THE PARAMETERS OF AN AI MODEL TO MINIMIZE THE LOSS FUNCTION AND IMPROVE ITS PERFORMANCE ON A GIVEN TASK.



SOFTMAX

AN ACTIVATION FUNCTION THAT IS USED IN MULTI-CLASS CLASSIFICATION PROBLEMS THAT CONVERTS A VECTOR OF RAW SCORES INTO A PROBABILITY DISTRIBUTION BY EXPONENTIATING EACH SCORE AND NORMALIZING BY THE SUM OF THE EXPONENTIATED SCORES.



CROSS-ENTROPY

(ALSO CALLED LOGARITHMIC LOSS OR LOG LOSS)

A LOSS FUNCTION USED FOR CLASSIFICATION TASKS THAT MEASURES THE DIFFERENCE BETWEEN THE PREDICTED PROBABILITY DISTRIBUTION AND THE TRUE DISTRIBUTION, AIMING TO MINIMIZE THIS DIFFERENCE.

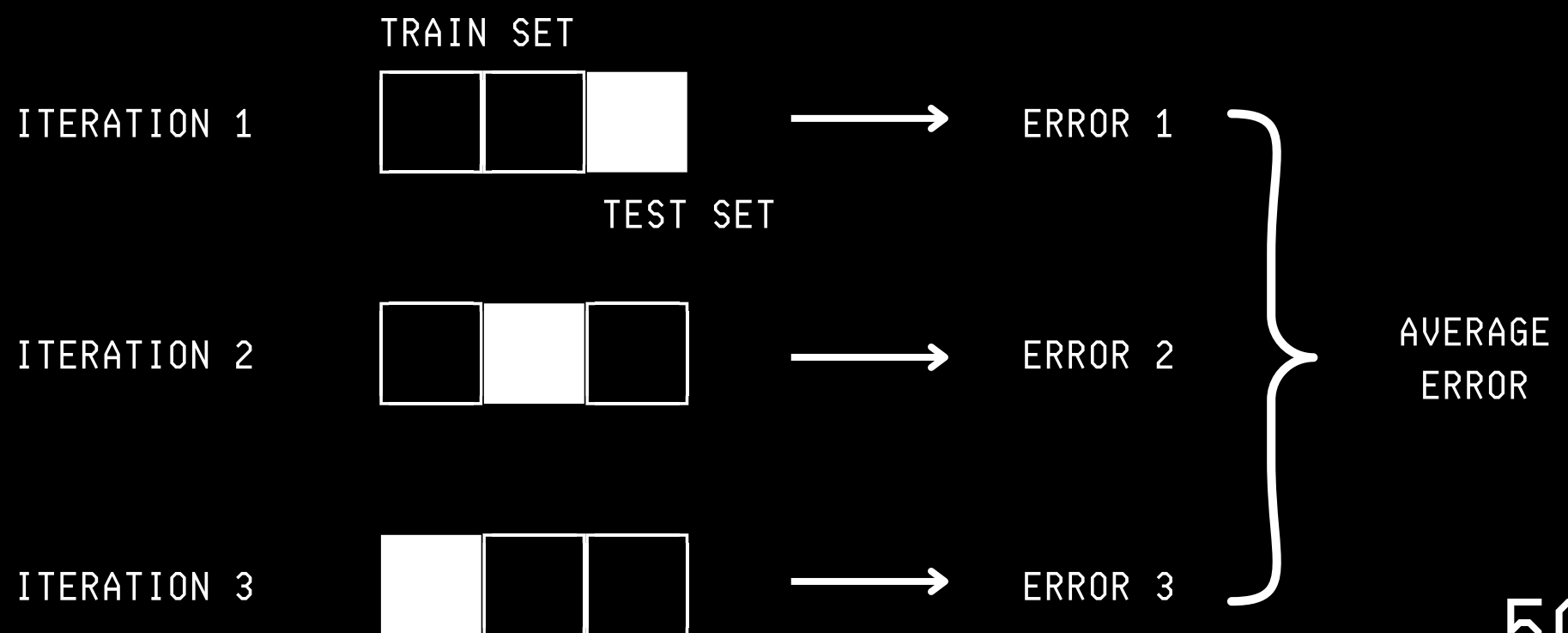
$$\text{BINARY CROSS-ENTROPY} = -(Y * \log Y^{\wedge}) + (1-Y) * \log(1-Y^{\wedge})$$

- Y: TRUE LABEL
- Y[^]: PREDICTED PROBABILITY

K-FOLD CROSS VALIDATION

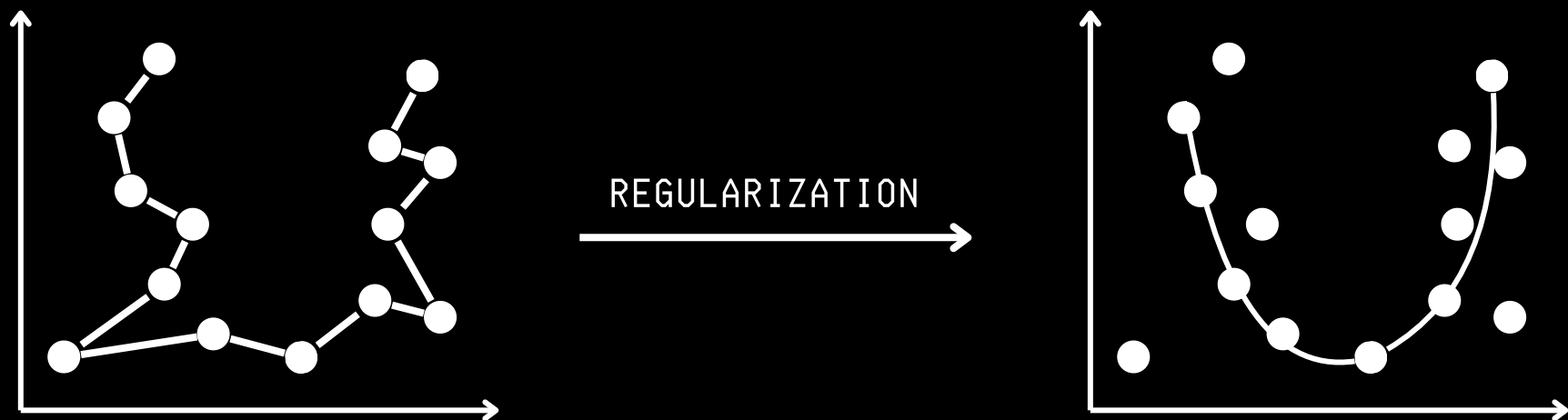
AN AI MODEL TRAINING TECHNIQUE WHERE THE DATASET IS DIVIDED INTO K EQUAL PARTS.

THE MODEL IS THEN TRAINED AND VALIDATED K TIMES, EACH TIME USING A DIFFERENT PART AS THE VALIDATION SET AND THE REST AS THE TRAINING SET.



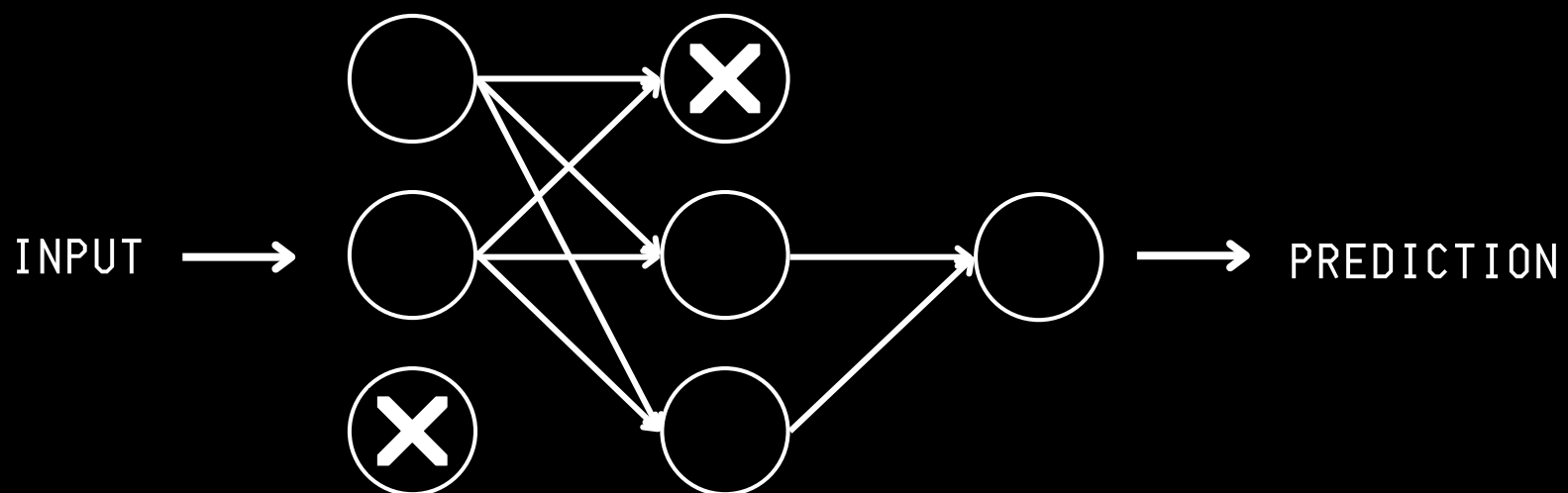
REGULARIZATION

A TECHNIQUE USED TO PREVENT OVERFITTING BY ADDING A PENALTY TERM TO THE LOSS FUNCTION, WHICH DISCOURAGES COMPLEX MODELS AND ENCOURAGES SIMPLER MODELS WITH SMALLER OR MORE EVENLY DISTRIBUTED PARAMETER VALUES.



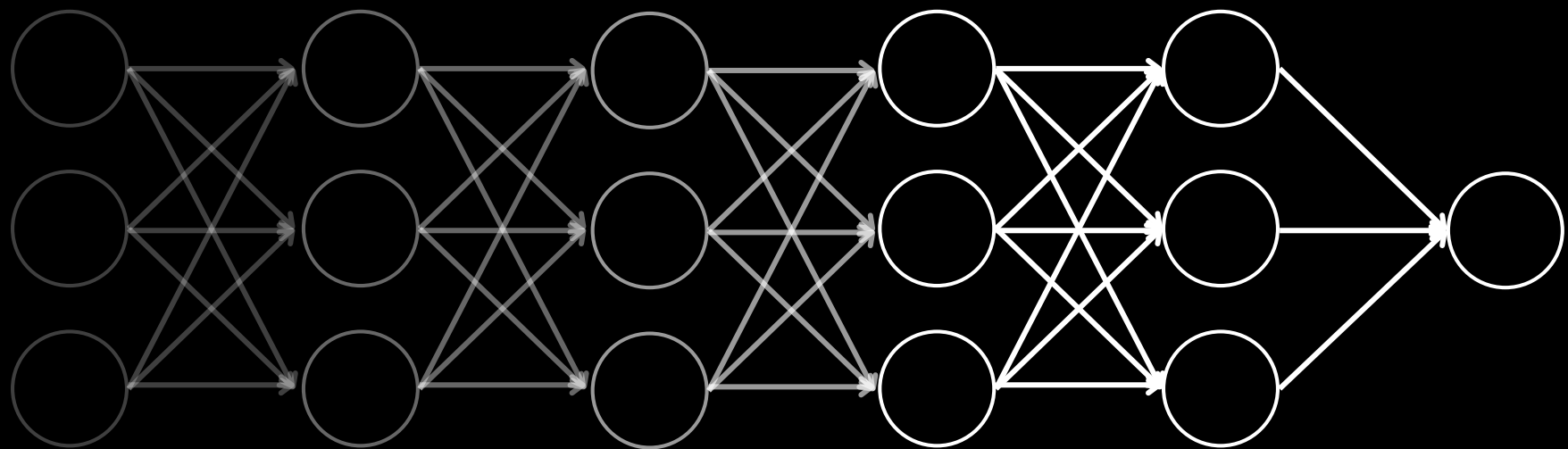
DROPOUT

A REGULARIZATION TECHNIQUE FOR NEURAL NETWORKS THAT RANDOMLY DEACTIVATES A FRACTION OF NEURONS DURING TRAINING TO PREVENT OVERFITTING.



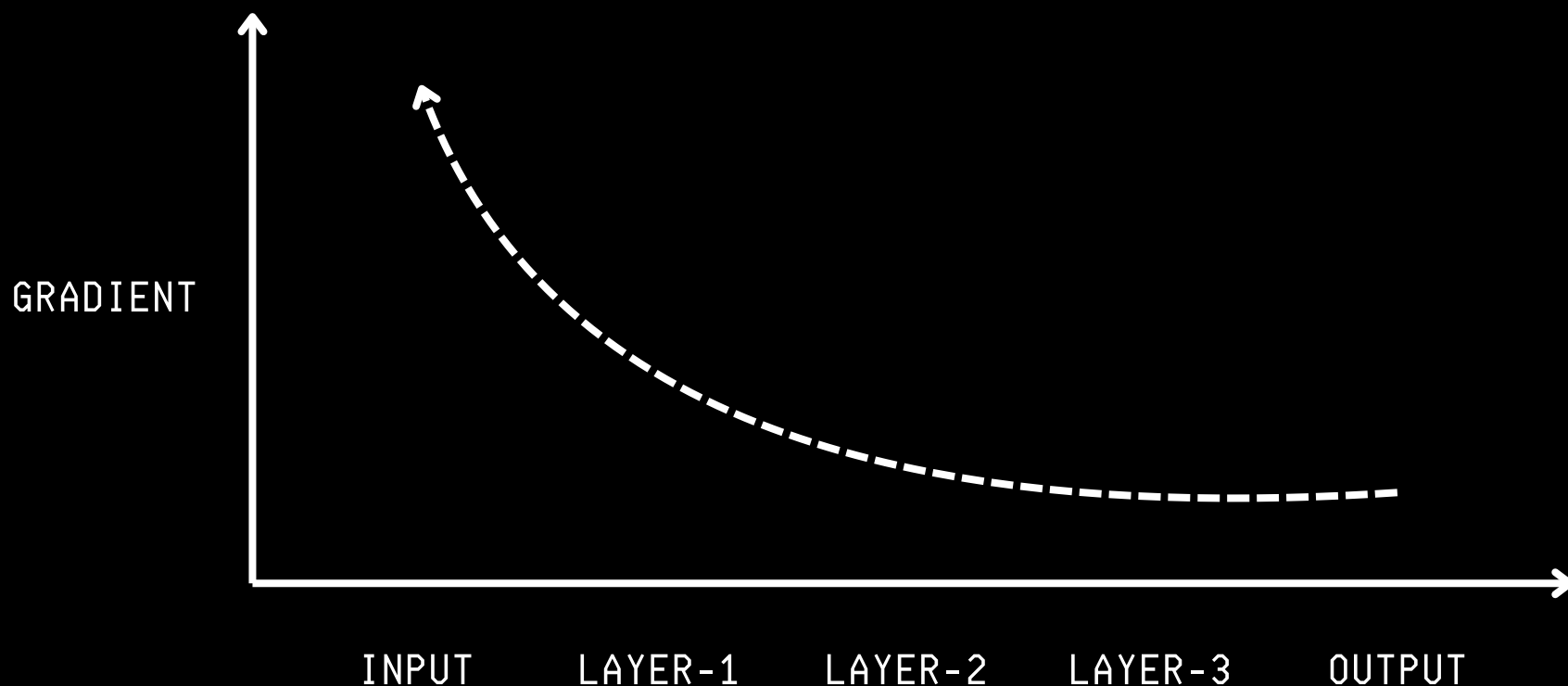
VANISHING GRADIENT

AN ISSUE IN TRAINING DEEP NEURAL NETWORKS WHERE GRADIENTS OF THE LOSS FUNCTION BECOME EXCEEDINGLY SMALL DURING BACKPROPAGATION, CAUSING THE EARLIER LAYERS OF THE NETWORK TO LEARN VERY SLOWLY OR NOT AT ALL.



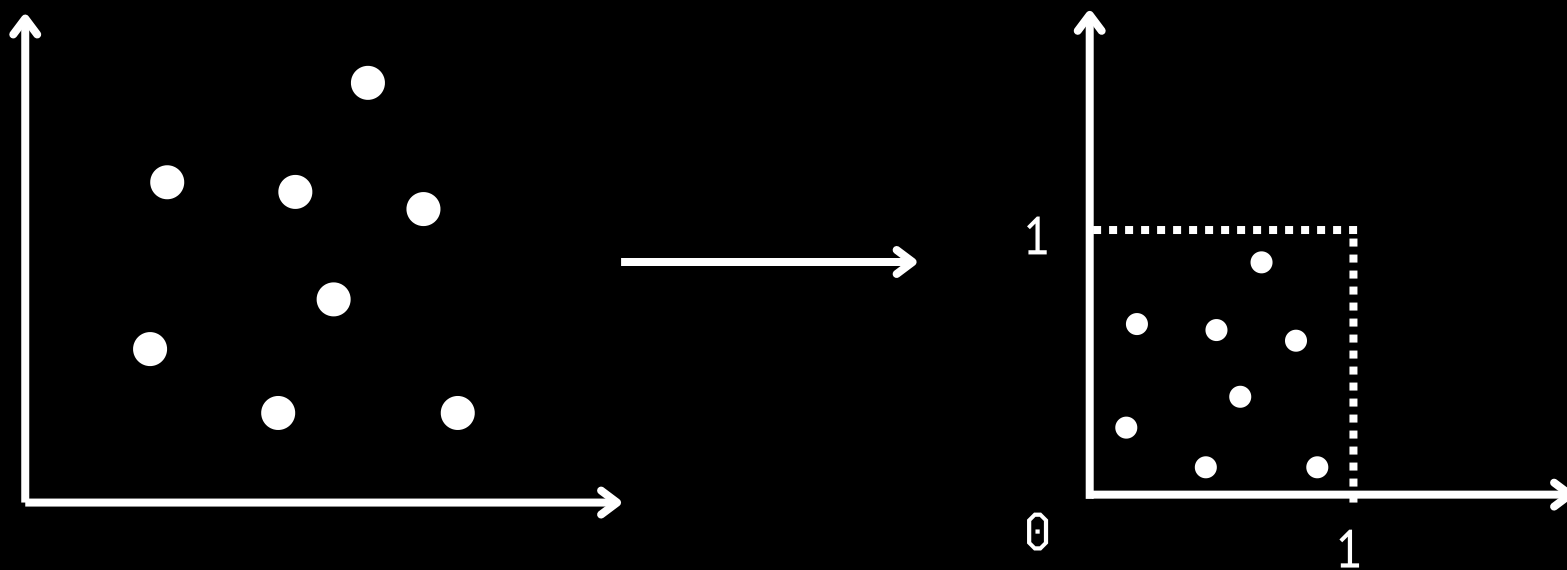
EXPLODING GRADIENT

AN ISSUE IN TRAINING DEEP NEURAL NETWORKS WHEN GRADIENTS OF THE LOSS FUNCTION BECOME EXCESSIVELY LARGE DURING BACKPROPAGATION, CAUSING THE MODEL PARAMETERS TO OSCILLATE WILDLY OR DIVERGE, MAKING TRAINING UNSTABLE.



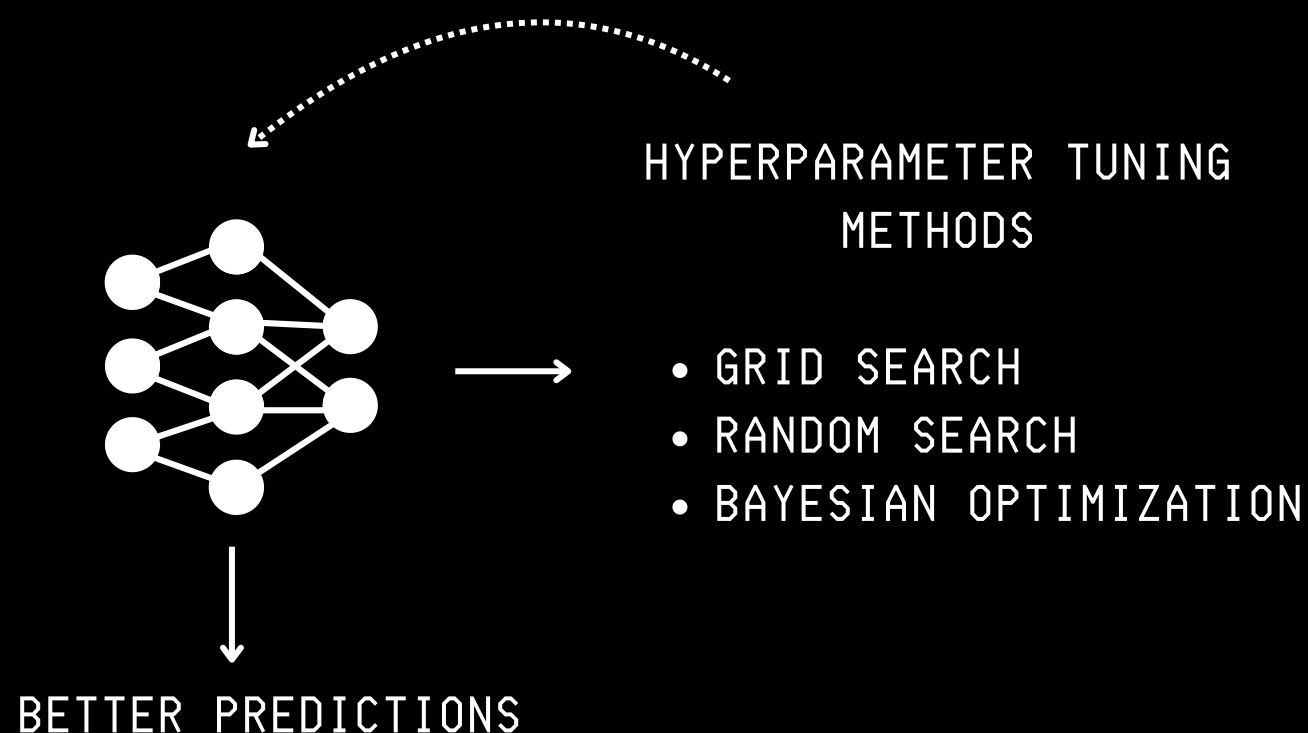
NORMALIZATION

A TECHNIQUE THAT SCALES AND ADJUSTS DATA VALUES TO A STANDARD RANGE, TYPICALLY BETWEEN 0 AND 1 OR -1 AND 1, TO IMPROVE THE PERFORMANCE AND CONVERGENCE SPEED OF MACHINE LEARNING ALGORITHMS BY ENSURING CONSISTENT DATA DISTRIBUTION.



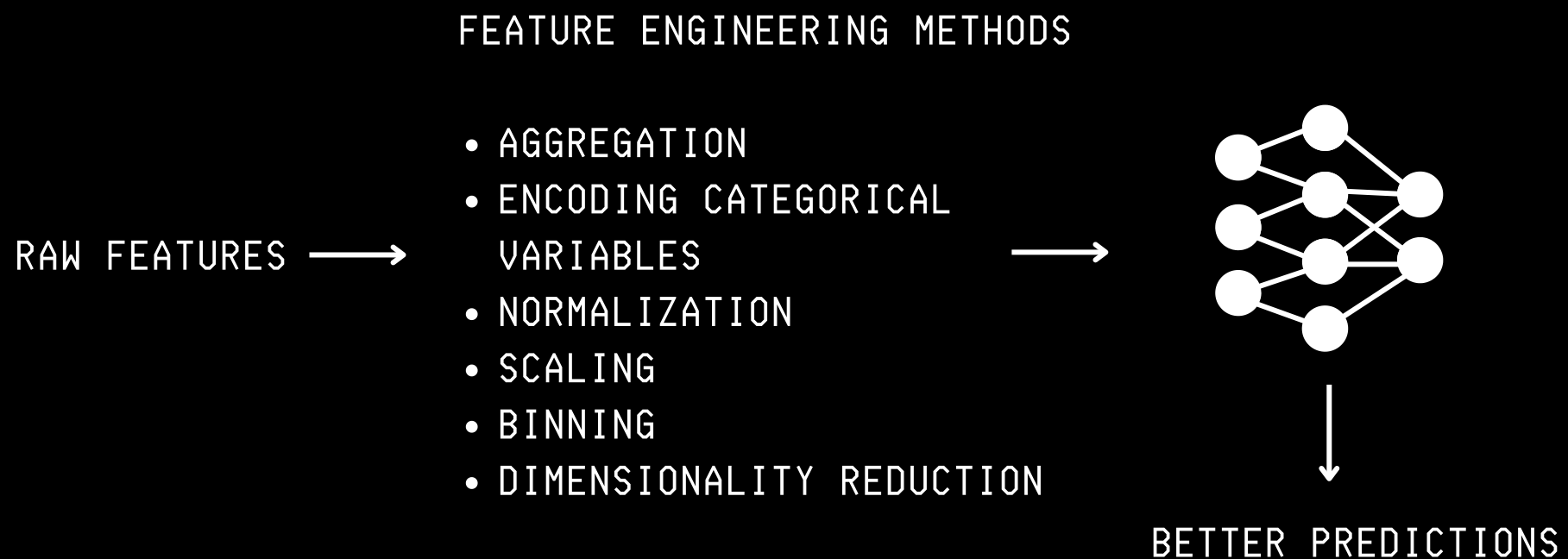
HYPERPARAMETER TUNING

THE PROCESS OF OPTIMIZING THE HYPERPARAMETERS OF A MACHINE LEARNING MODEL TO IMPROVE ITS PERFORMANCE BY SYSTEMATICALLY SEARCHING THROUGH A PREDEFINED SPACE OF HYPERPARAMETER VALUES



FEATURE ENGINEERING

THE PROCESS OF CREATING NEW FEATURES OR MODIFYING EXISTING ONES FROM RAW DATA TO IMPROVE THE PERFORMANCE OF MACHINE LEARNING MODELS BY ENHANCING THE QUALITY AND INFORMATIVENESS OF THE INPUT DATA.



DATA IMPUTATION

THE PROCESS OF FILLING IN MISSING VALUES IN A DATASET USING STATISTICAL OR MACHINE LEARNING TECHNIQUES TO MAINTAIN THE INTEGRITY OF THE DATA AND ALLOW FOR ACCURATE ANALYSIS AND MODELLING.

STUDENT	SCORE
ALICE	85
BOB	NaN
CHARLIE	75
DANIEL	90

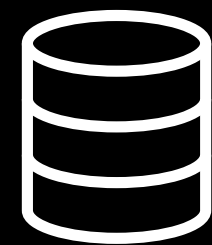
IMPUTATION
WITH THE
MEAN SCORE



STUDENT	SCORE
ALICE	85
BOB	83.33
CHARLIE	75
DANIEL	90

DATA AUGMENTATION

A TECHNIQUE USED TO INCREASE THE DIVERSITY OF A TRAINING DATASET BY APPLYING TRANSFORMATIONS SUCH AS ROTATIONS, TRANSLATIONS, FLIPS, AND NOISE TO EXISTING DATA, IMPROVING MODEL GENERALIZATION AND PERFORMANCE.



ORIGINAL
IMAGE
DATASET



- ROTATIONS
- FLIPPING
- CROPPING
- SCALING
- SHEARING

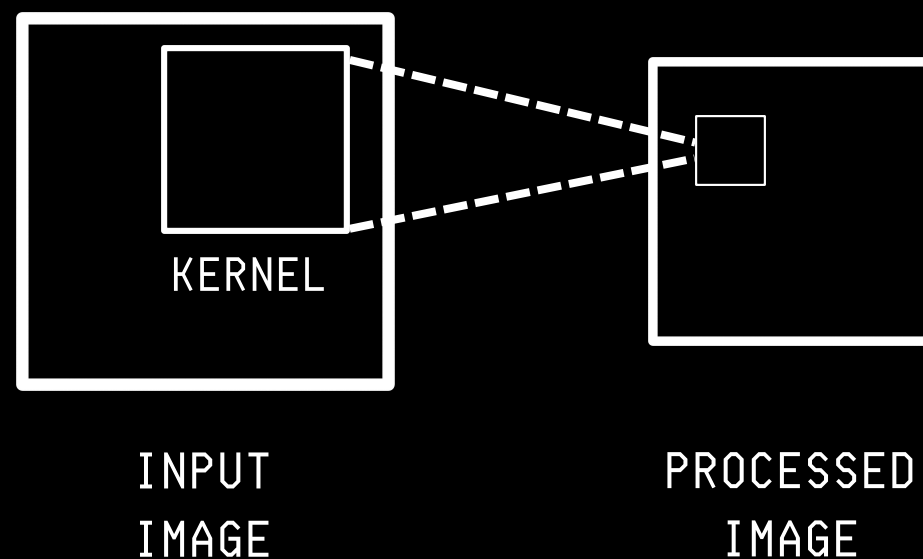


AUGMENTED
IMAGE
DATASET

CNN

CONVOLUTIONAL NEURAL NETWORK

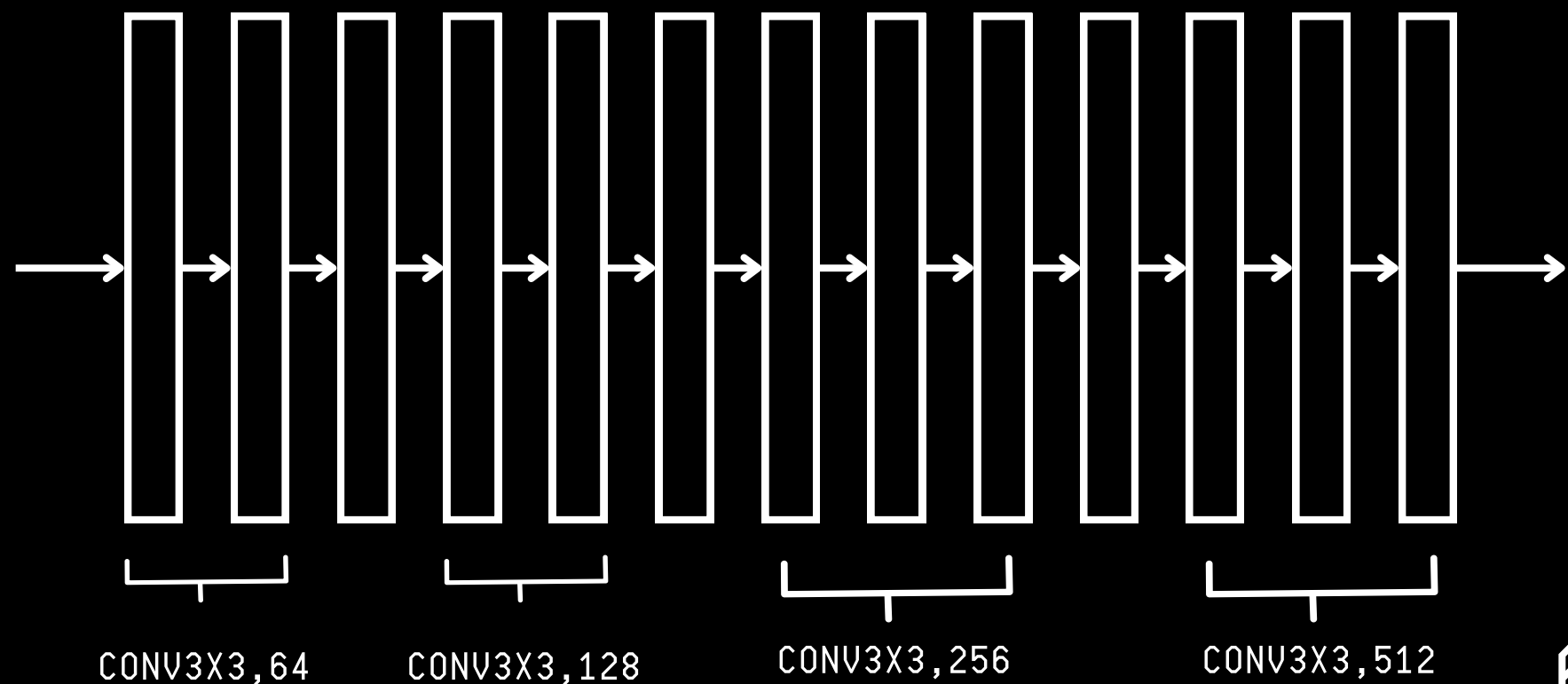
A DEEP LEARNING ALGORITHM SPECIFICALLY DESIGNED FOR PROCESSING STRUCTURED GRID DATA, SUCH AS IMAGES, USING CONVOLUTIONAL LAYERS THAT APPLY FILTERS TO CAPTURE SPATIAL HIERARCHIES AND PATTERNS.



VGG-NET

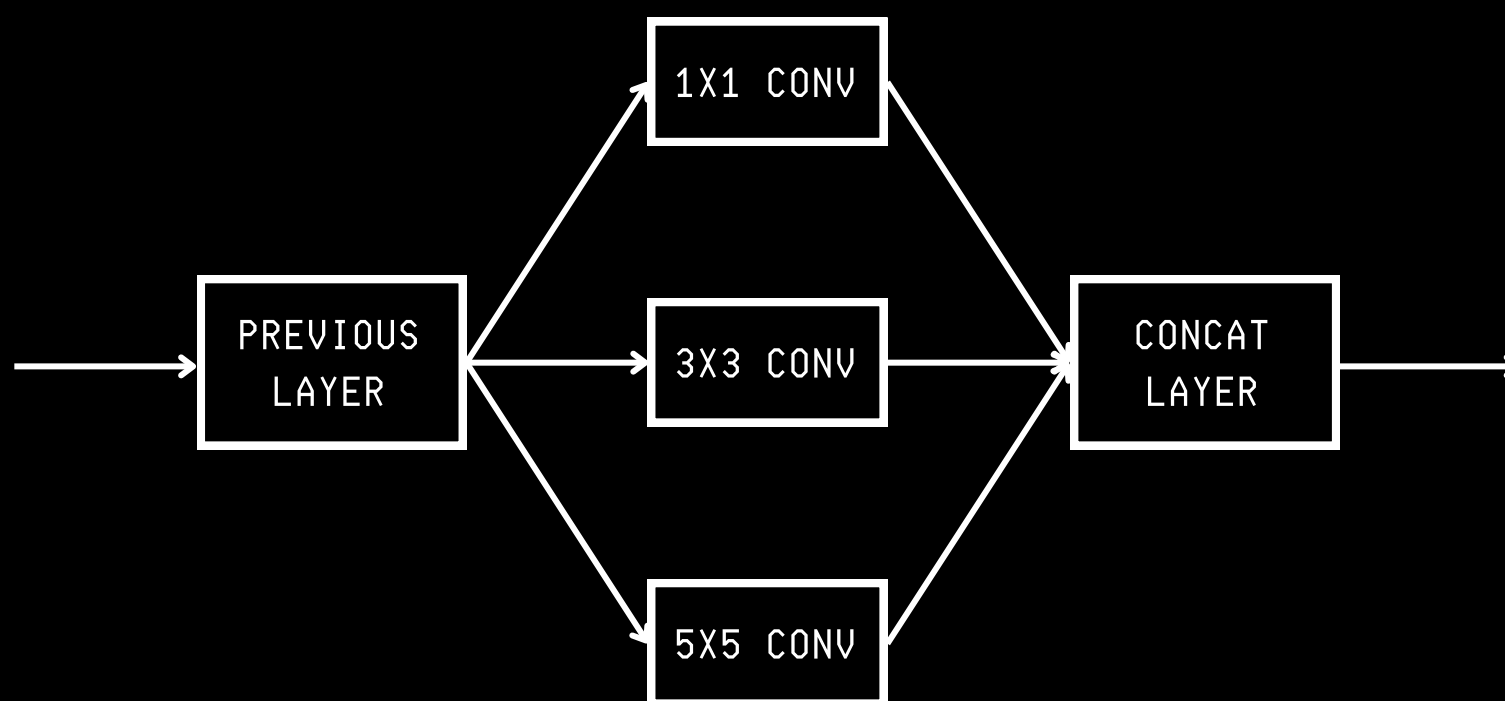
VISUAL GEOMETRY GROUP NETWORK

A DEEP CONVOLUTIONAL NEURAL NETWORK ARCHITECTURE KNOWN FOR ITS SIMPLICITY AND DEPTH, USING SMALL 3X3 FILTERS AND STACKING THEM IN INCREASING DEPTH TO ACHIEVE EXCELLENT PERFORMANCE ON IMAGE CLASSIFICATION TASKS.



INCEPTION NETWORK

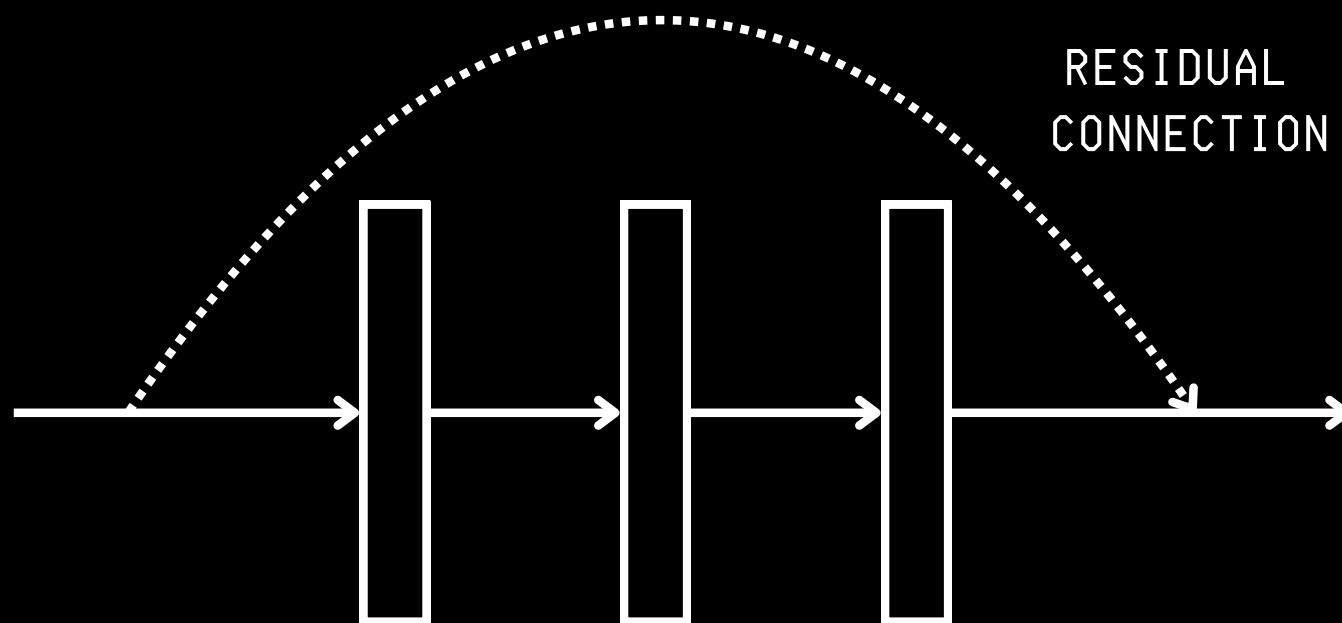
A DEEP CONVOLUTIONAL NEURAL NETWORK ARCHITECTURE FEATURING INCEPTION MODULES THAT APPLY MULTIPLE FILTER SIZES SIMULTANEOUSLY WITHIN THE SAME LAYER TO CAPTURE MULTI-SCALE FEATURES AND IMPROVE COMPUTATIONAL EFFICIENCY THROUGH DIMENSIONALITY REDUCTION.



RESNET

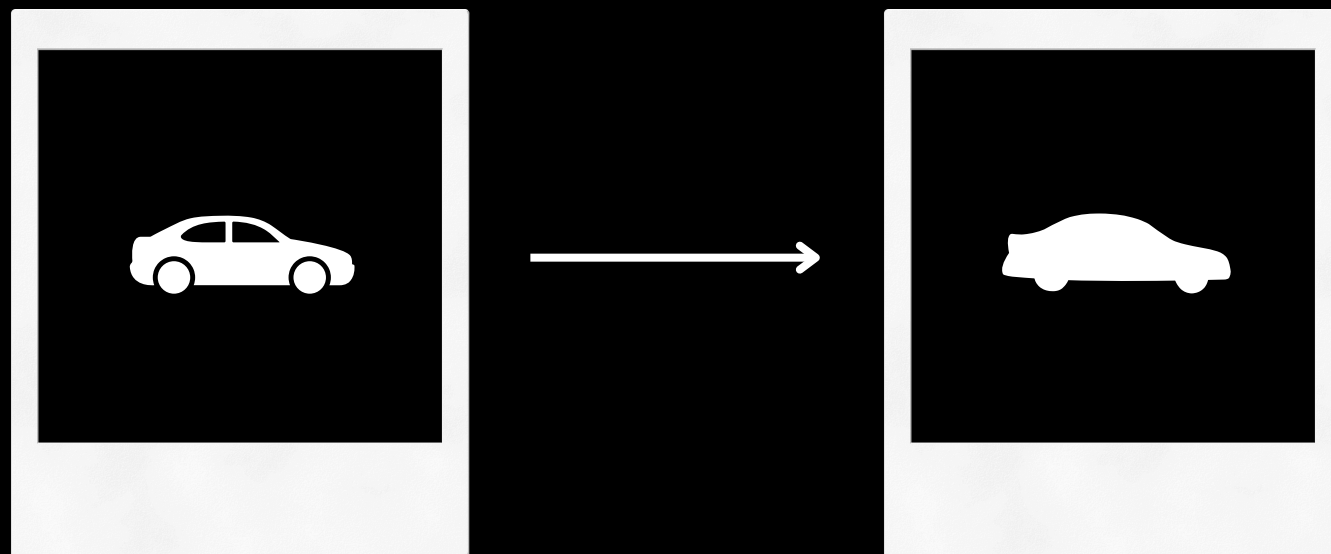
RESIDUAL NETWORK

A DEEP NEURAL NETWORK ARCHITECTURE THAT USES RESIDUAL CONNECTIONS (SKIP CONNECTIONS) TO ENABLE THE TRAINING OF VERY DEEP NETWORKS BY MITIGATING THE VANISHING GRADIENT PROBLEM.



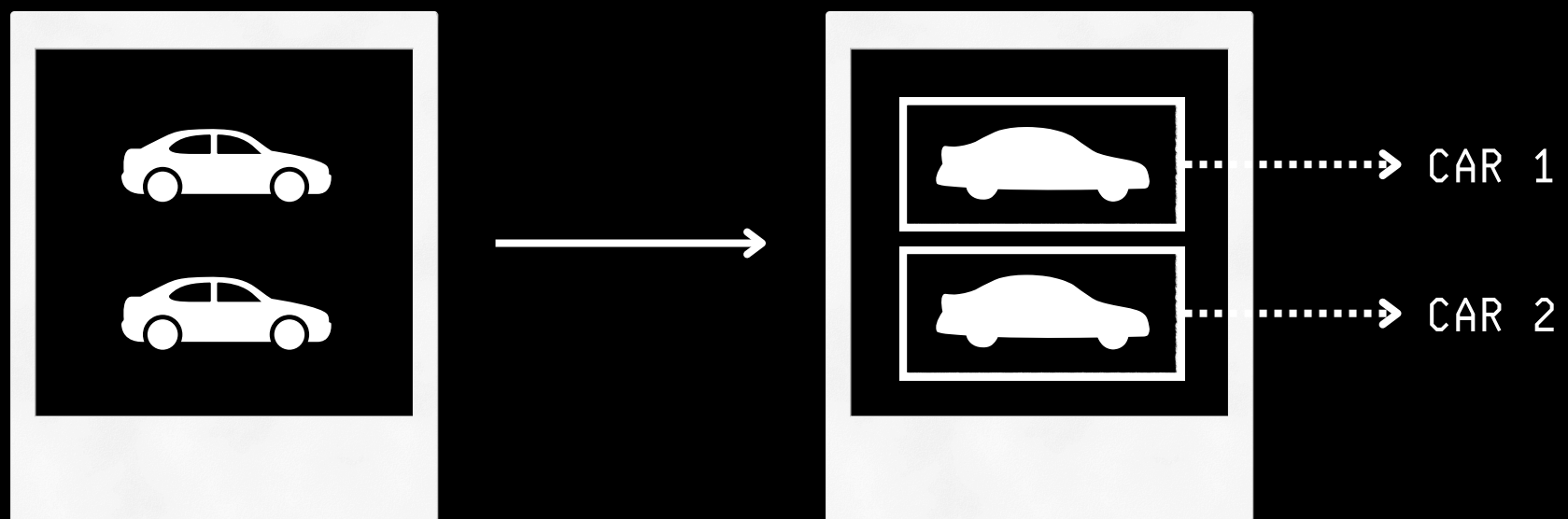
SEMANTIC SEGMENTATION

A COMPUTER VISION TASK THAT INVOLVES CLASSIFYING EACH PIXEL IN AN IMAGE INTO A SPECIFIC CATEGORY, EFFECTIVELY SEGMENTING THE IMAGE INTO MEANINGFUL REGIONS CORRESPONDING TO DIFFERENT OBJECTS OR PARTS.



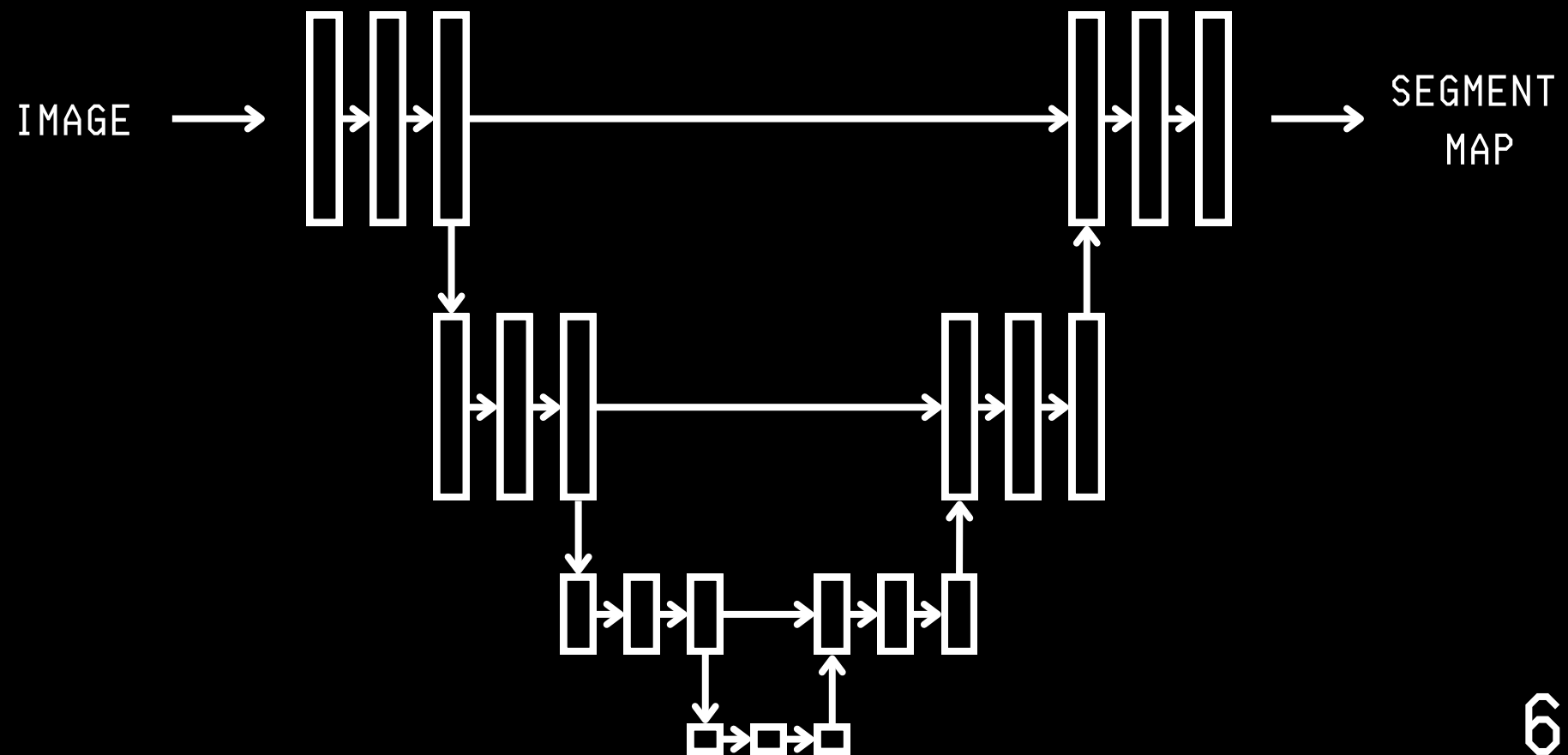
INSTANCE SEGMENTATION

A COMPUTER VISION TASK THAT INVOLVES IDENTIFYING AND SEGMENTING EACH DISTINCT OBJECT INSTANCE IN AN IMAGE, ASSIGNING EACH PIXEL NOT ONLY A CLASS LABEL BUT ALSO AN INSTANCE ID TO DISTINGUISH BETWEEN DIFFERENT OBJECTS OF THE SAME CLASS.



U-NET

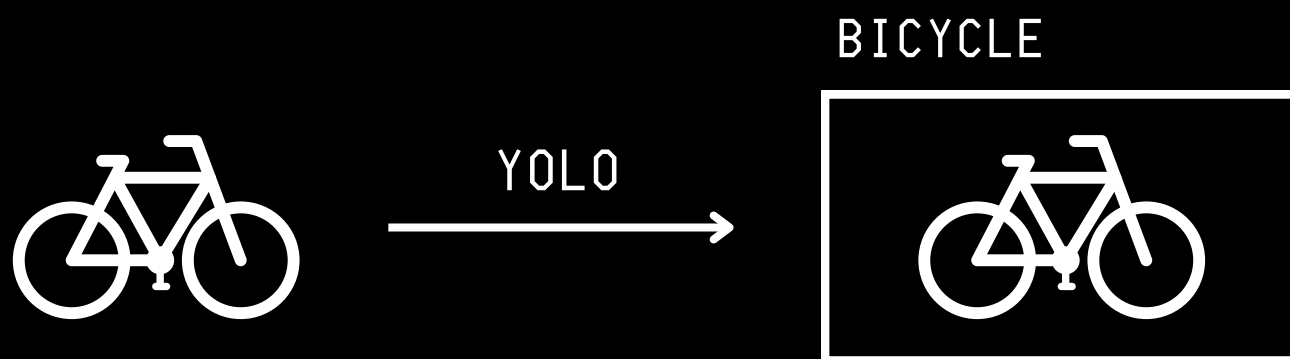
A TYPE OF CONVOLUTIONAL NEURAL NETWORK DESIGNED FOR IMAGE SEGMENTATION, CHARACTERIZED BY ITS U-SHAPED ARCHITECTURE THAT COMBINES ENCODER-DECODER PATHS TO PRODUCE HIGH-RESOLUTION SEGMENTATION MAPS.



YOLO

YOU ONLY LOOK ONCE

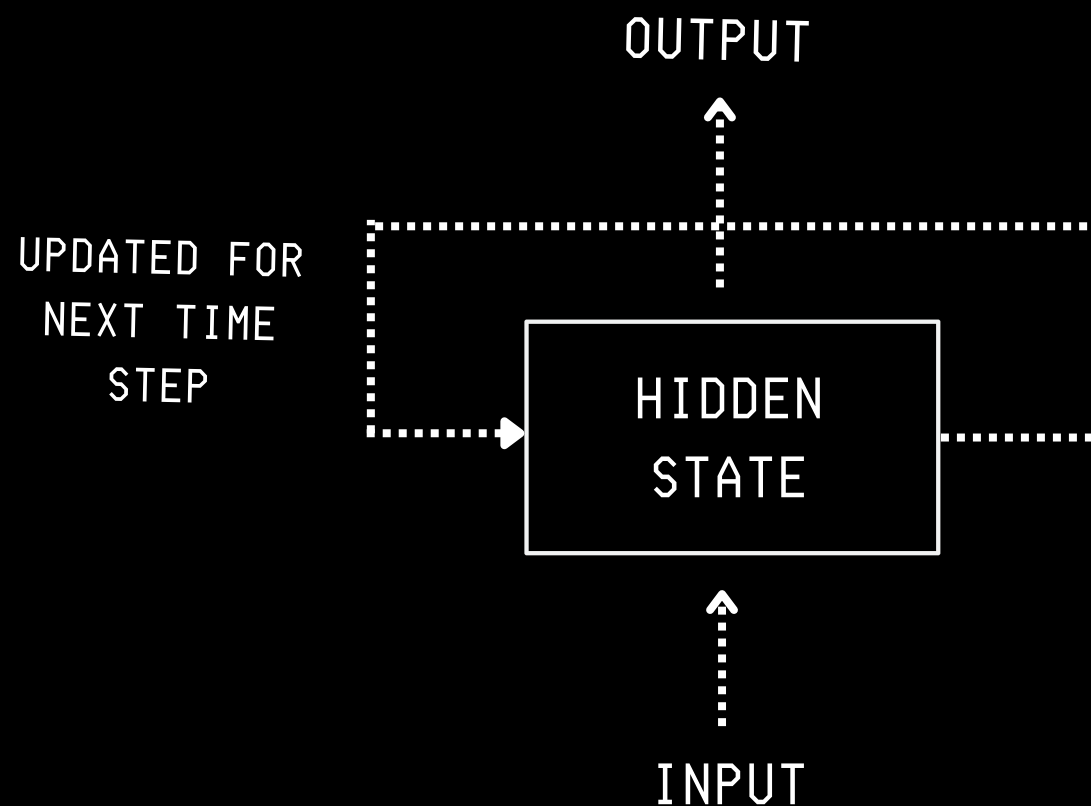
A REAL-TIME OBJECT DETECTION SYSTEM THAT FRAMES OBJECT DETECTION AS A SINGLE REGRESSION PROBLEM, DIRECTLY PREDICTING BOUNDING BOXES AND CLASS PROBABILITIES FROM FULL IMAGES IN ONE EVALUATION.



RNN

RECURRENT NEURAL NETWORK

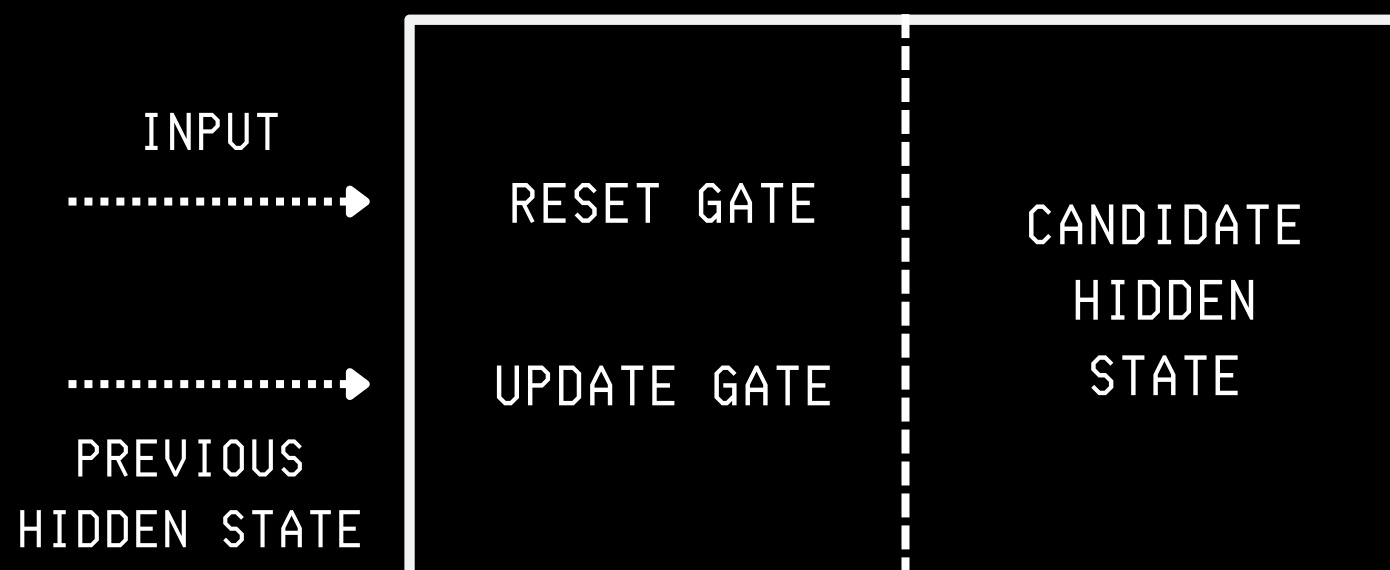
A NEURAL NETWORK ARCHITECTURE DESIGNED FOR SEQUENTIAL DATA THAT MAINTAINS A HIDDEN STATE TO CAPTURE INFORMATION FROM PREVIOUS STEPS, ALLOWING IT TO MODEL TEMPORAL DEPENDENCIES.



GRU

GATED RECURRENT UNIT

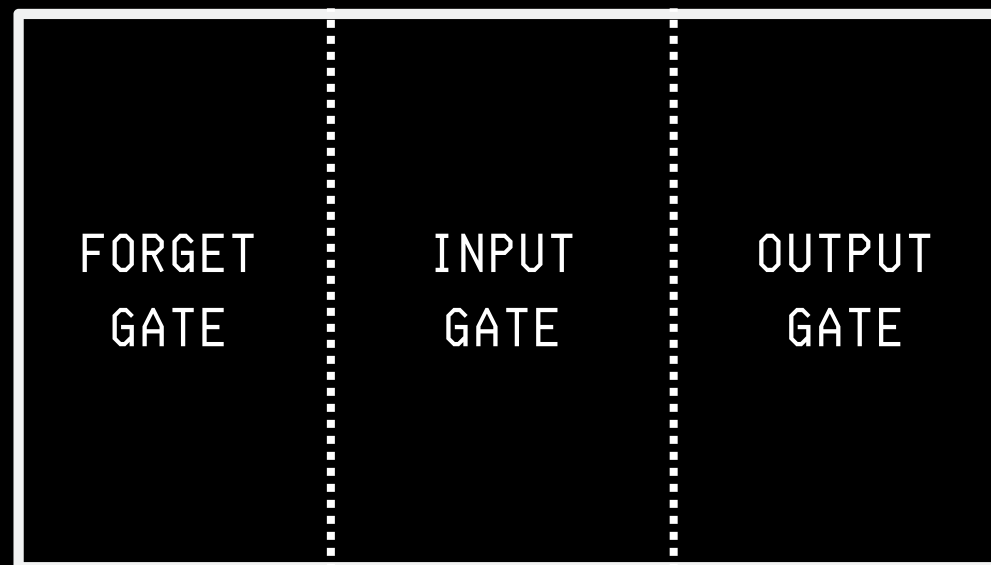
A TYPE OF RECURRENT NEURAL NETWORK (RNN) ARCHITECTURE DESIGNED TO CAPTURE DEPENDENCIES IN SEQUENCE DATA BY USING TWO GATES—AN UPDATE GATE AND A RESET GATE—TO CONTROL THE FLOW OF INFORMATION AND ADDRESS THE VANISHING GRADIENT PROBLEM.



LSTM

LONG SHORT-TERM MEMORY

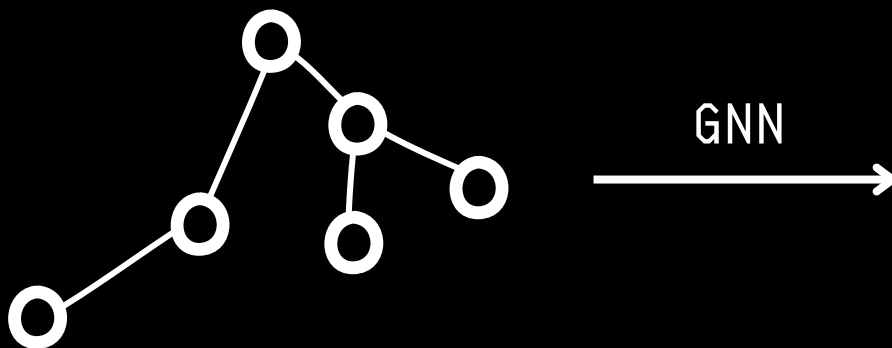
A TYPE OF RECURRENT NEURAL NETWORK (RNN) ARCHITECTURE DESIGNED TO EFFECTIVELY CAPTURE LONG-TERM DEPENDENCIES IN SEQUENCE DATA BY USING GATES TO CONTROL THE FLOW OF INFORMATION AND MITIGATE THE VANISHING GRADIENT PROBLEM.



GNN

GRAPH NEURAL NETWORK

A NEURAL NETWORK ARCHITECTURE DESIGNED TO PROCESS GRAPH-STRUCTURED DATA BY LEARNING NODE REPRESENTATIONS THROUGH MESSAGE PASSING AND AGGREGATION.

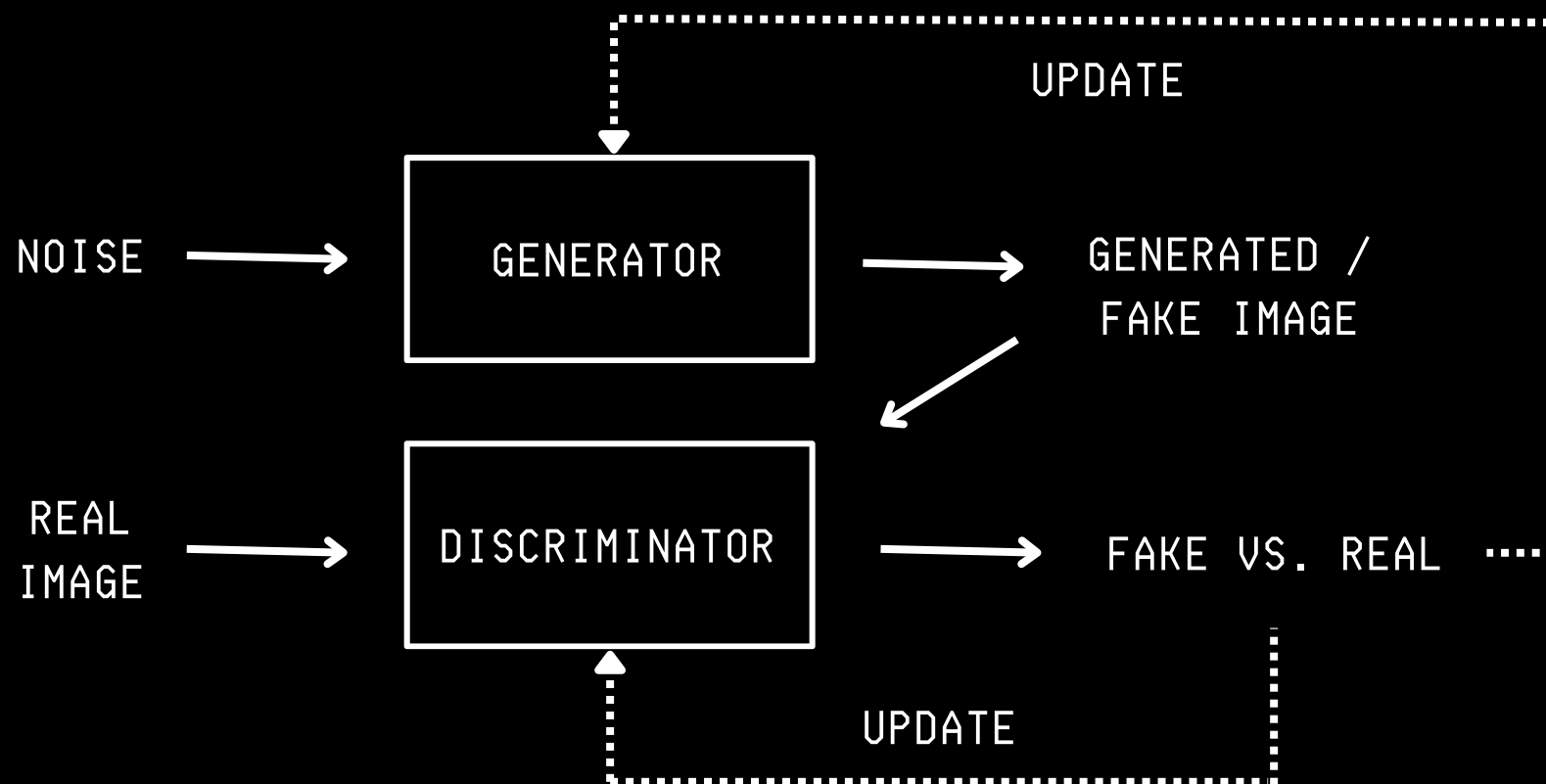


- GRAPH CLASSIFICATION
- NODE CLASSIFICATION
- LINK PREDICTION
- GRAPH GENERATION
- COMMUNITY DETECTION
- GRAPH EMBEDDING GENERATION

GAN

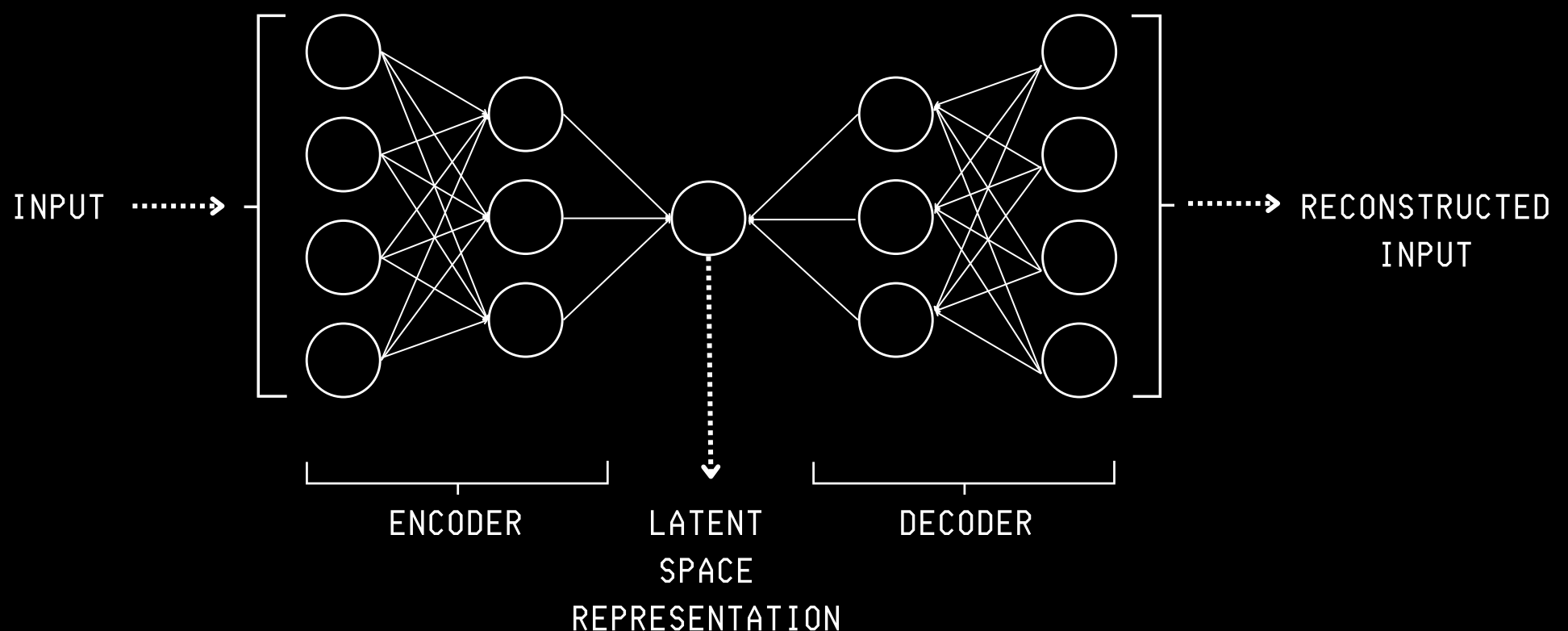
GENERATIVE ADVERSARIAL NETWORK

A NEURAL NETWORK ARCHITECTURE COMPOSED OF TWO
COMPETING SUB-NETWORKS—A GENERATOR AND A DISCRIMINATOR
—THAT LEARN TOGETHER TO PRODUCE REALISTIC SYNTHETIC
DATA FROM RANDOM NOISE.



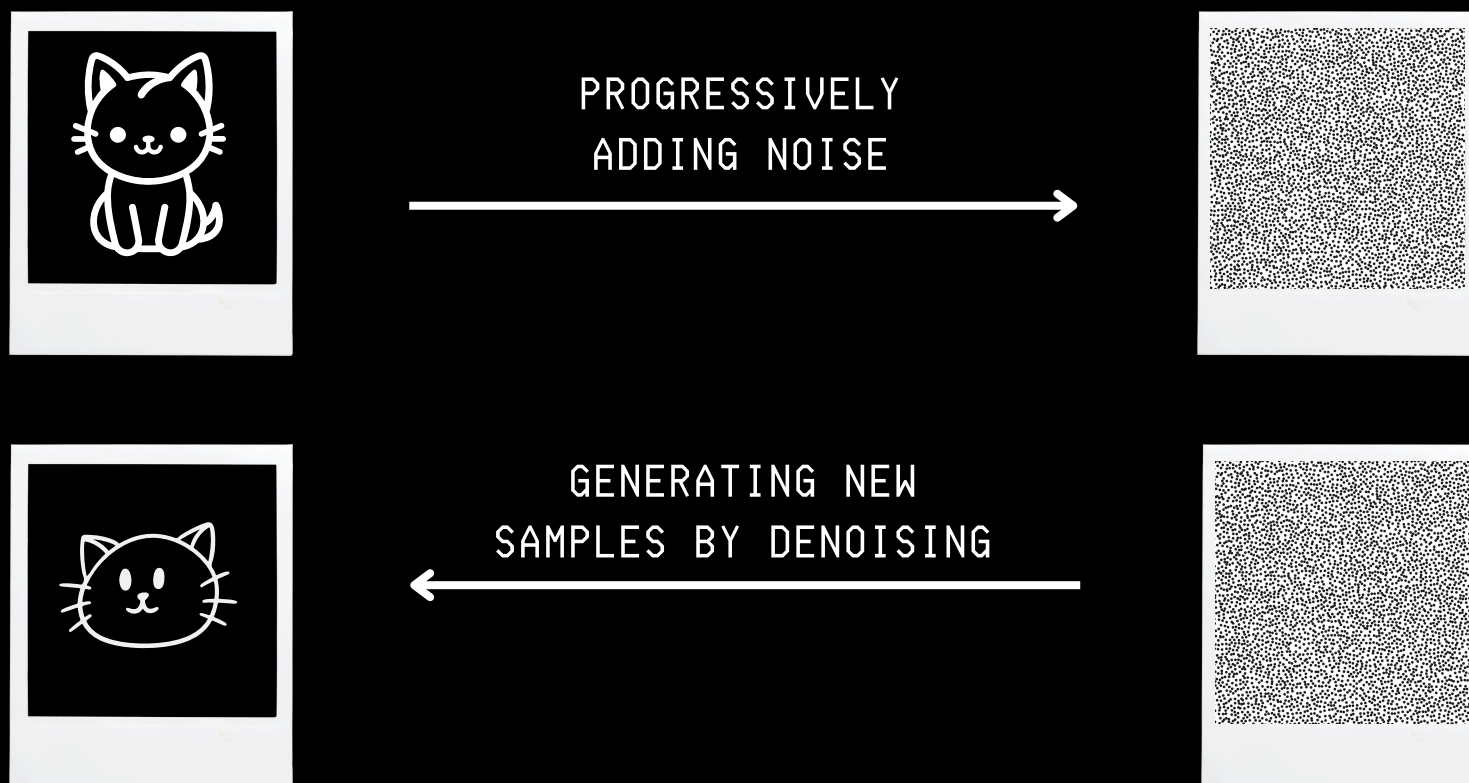
AUTOENCODER

A TYPE OF NEURAL NETWORKS DESIGNED TO LEARN EFFICIENT REPRESENTATIONS OF INPUT DATA BY ENCODING IT INTO A LOWER-DIMENSIONAL LATENT SPACE AND THEN RECONSTRUCTING THE ORIGINAL DATA FROM THIS COMPRESSED REPRESENTATION.



DIFFUSION MODEL

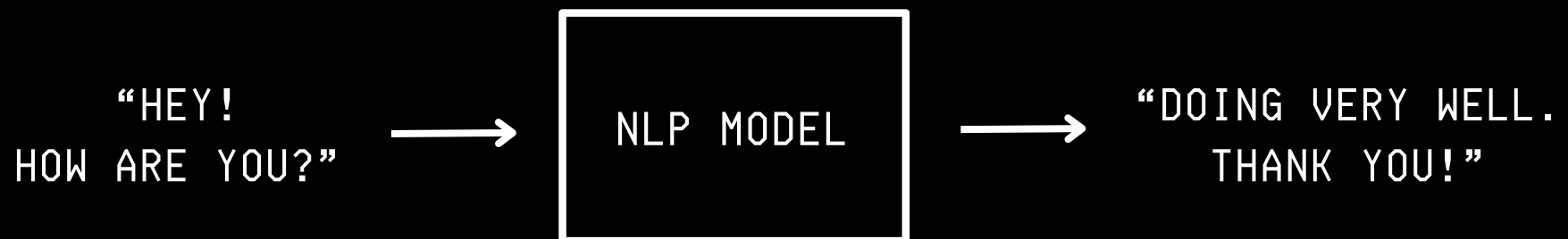
A GENERATIVE MODEL THAT LEARNS TO GENERATE DATA BY REVERSING A DIFFUSION PROCESS, PROGRESSIVELY DENOISING RANDOM NOISE TO PRODUCE REALISTIC SAMPLES.



NLP

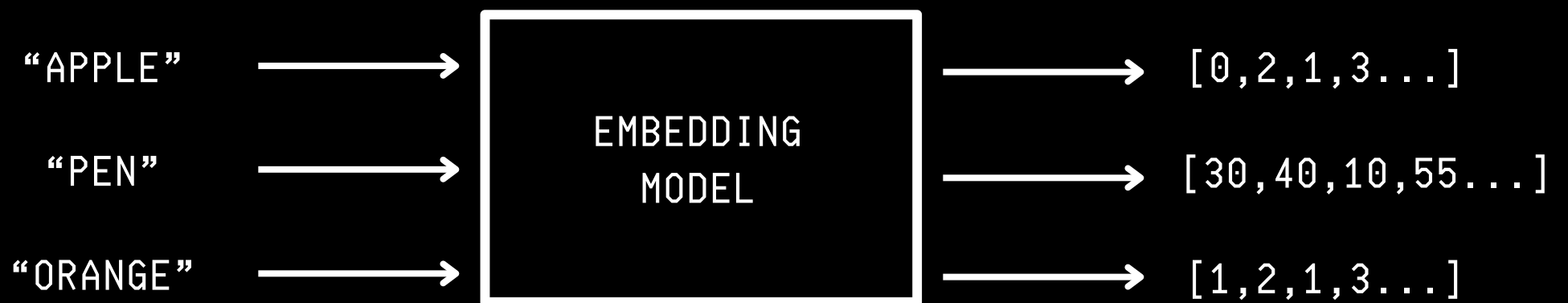
NATURAL LANGUAGE PROCESSING

A FIELD OF ARTIFICIAL INTELLIGENCE THAT FOCUSES ON THE INTERACTION BETWEEN COMPUTERS AND HUMANS THROUGH NATURAL LANGUAGE, AIMING TO ENABLE MACHINES TO UNDERSTAND, INTERPRET, GENERATE, AND RESPOND TO HUMAN LANGUAGE IN A VALUABLE WAY.



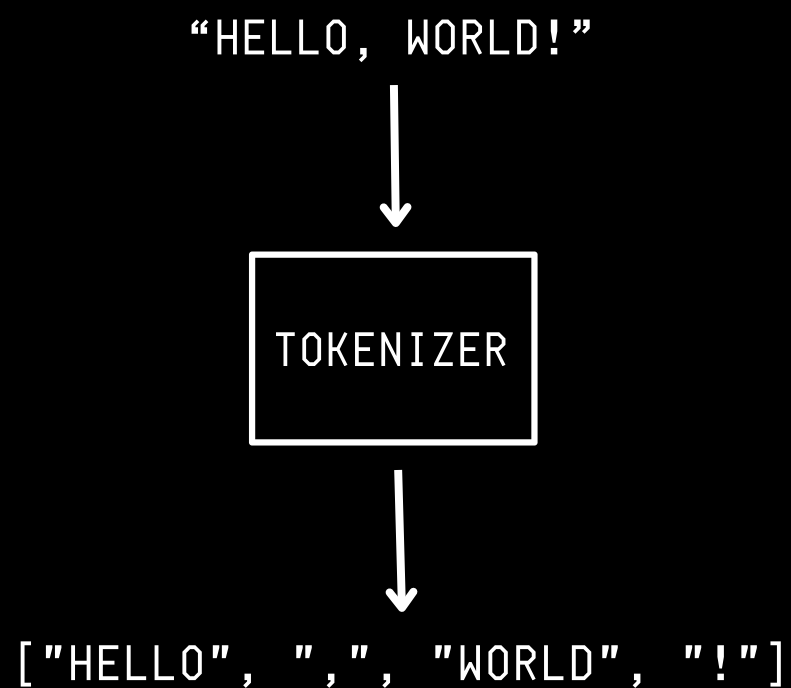
EMBEDDINGS

NUMERICAL REPRESENTATIONS OF TEXTUAL, CATEGORICAL, OR OTHER TYPES OF DATA IN A HIGH-DIMENSIONAL SPACE, WHICH CAPTURE THE CONTEXT OR SEMANTIC SIMILARITY BETWEEN DATA POINTS.



TOKENIZATION

THE PROCESS OF BREAKING DOWN TEXT INTO SMALLER UNITS (TOKENS), SUCH AS WORDS OR PHRASES, FOR EASIER PROCESSING AND ANALYSIS IN NATURAL LANGUAGE PROCESSING (NLP).



STOP WORDS

COMMONLY USED WORDS THAT ARE OFTEN FILTERED OUT IN NATURAL LANGUAGE PROCESSING (NLP) TO FOCUS ON MORE MEANINGFUL WORDS.

“THE”

“OF”

“FOR”

“IS”

“AND”

“WITH”

“AT”

“A”

“HAVE”

“WHICH”

“TO”

“BE”

“ON”

“IN”

“WAS”

STEMMING

THE PROCESS OF REDUCING WORDS TO THEIR ROOT FORM BY ALGORITHMICALLY CUTTING OFF PREFIXES AND SUFFIXES. IT FOCUSES ON REDUCING THE WORD TO ITS BASE FORM WITHOUT CONSIDERING THE CONTEXT.



LEMMATIZATION

THE PROCESS OF REDUCING WORDS TO THEIR BASE OR
DICTIONARY FORM (LEMMA) BASED ON THE WORD'S PART OF
SPEECH AND MEANING IN THE SENTENCE.



NER

NAMED ENTITY RECOGNITION

AN NLP TASK OF IDENTIFYING AND CLASSIFYING KEY INFORMATION (ENTITIES) IN TEXT INTO PREDEFINED CATEGORIES.

“ELON MUSK
FOUNDED
SPACEX.”



NER
CLASSIFIER

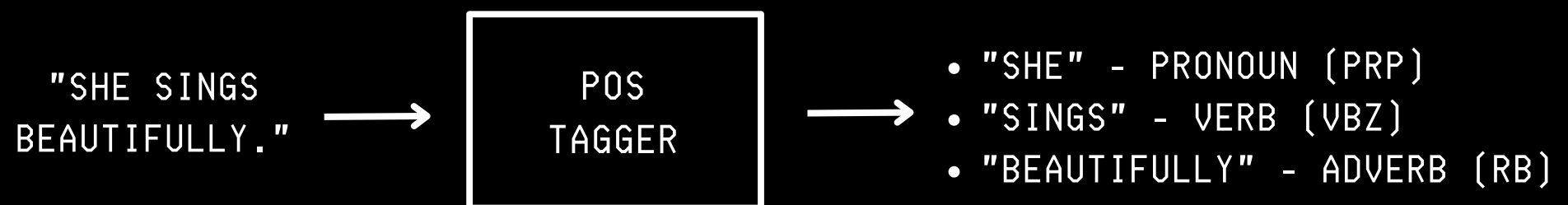


- ELON MUSK: PERSON
- SPACEX: ORGANIZATION

POS TAGGING

PART-OF-SPEECH TAGGING

THE PROCESS OF LABELING EACH WORD IN A SENTENCE WITH ITS APPROPRIATE PART OF SPEECH (E.G., NOUN, VERB, ADJECTIVE), BASED ON BOTH ITS DEFINITION AND ITS CONTEXT.

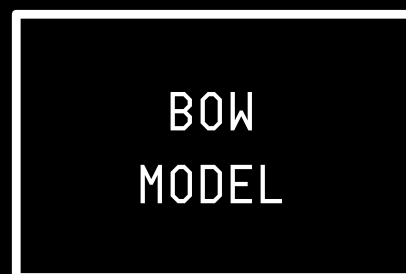


BOW MODEL

BAG OF WORDS MODEL

A MODEL THAT REPRESENTS TEXT AS AN UNORDERED COLLECTION OF WORDS, IGNORING SYNTAX AND WORD ORDER BUT MAINTAINING WORD FREQUENCY.

“THE CAT SAT
ON THE MAT.”



- THE: 2
- CAT: 1
- SAT: 1
- ON: 1
- MAT: 1

TF-IDF

TERM FREQUENCY-INVERSE DOCUMENT FREQUENCY

A STATISTICAL MEASURE THAT EVALUATES THE IMPORTANCE OF A WORD IN A DOCUMENT RELATIVE TO A CORPUS BY COMBINING TERM FREQUENCY AND INVERSE DOCUMENT FREQUENCY.

$$\text{TF-IDF (T, D)} = \text{TF (T,D)} \times \text{IDF (T)}$$

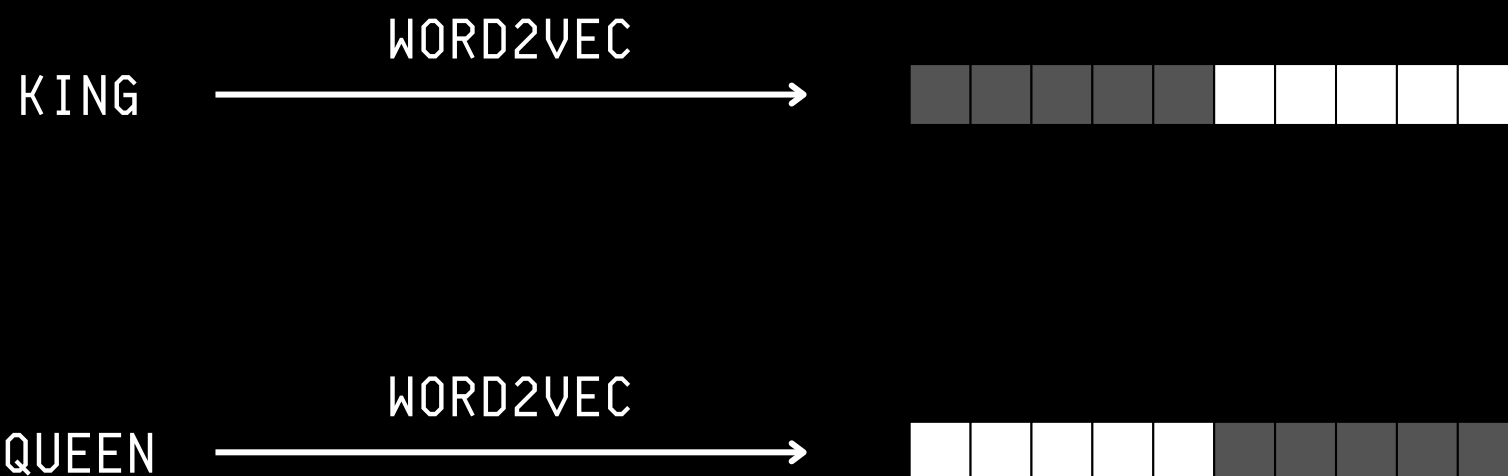
WHERE:

$$\text{TF (T,D)} = \frac{\text{NUMBER OF TIMES TERM T APPEARS IN DOCUMENT D}}{\text{TOTAL NUMBER OF TERMS IN DOCUMENT D}}$$

$$\text{IDF (T)} = \frac{\text{TOTAL NUMBER OF DOCUMENTS}}{\text{NUMBER OF DOCUMENTS CONTAINING THE TERM T}}$$

WORD2VEC

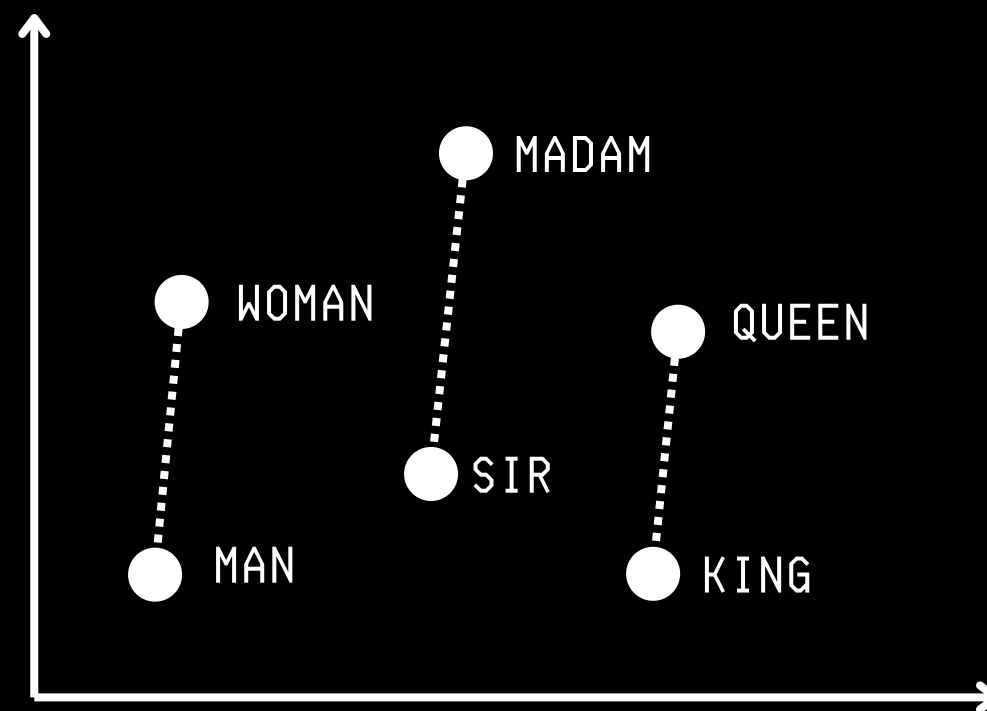
A NEURAL NETWORK-BASED MODEL THAT LEARNS VECTOR REPRESENTATIONS OF WORDS IN A CONTINUOUS VECTOR SPACE, CAPTURING SEMANTIC MEANINGS AND RELATIONSHIPS BY TRAINING ON LARGE TEXT CORPORA.



GLOVE

GLOBAL VECTORS FOR WORD REPRESENTATION

AN UNSUPERVISED LEARNING ALGORITHM THAT GENERATES WORD EMBEDDINGS BY FACTORIZING THE CO-OCCURRENCE MATRIX OF A CORPUS, CAPTURING BOTH GLOBAL STATISTICAL INFORMATION AND LOCAL CONTEXT.



ATTENTION

A METHOD IN NEURAL NETWORKS THAT ALLOWS THE MODEL TO SELECTIVELY FOCUS ON SPECIFIC PARTS OF THE INPUT SEQUENCE, ENHANCING ITS ABILITY TO CAPTURE RELEVANT CONTEXTUAL INFORMATION AND RELATIONSHIPS FOR IMPROVED PREDICTION ACCURACY.

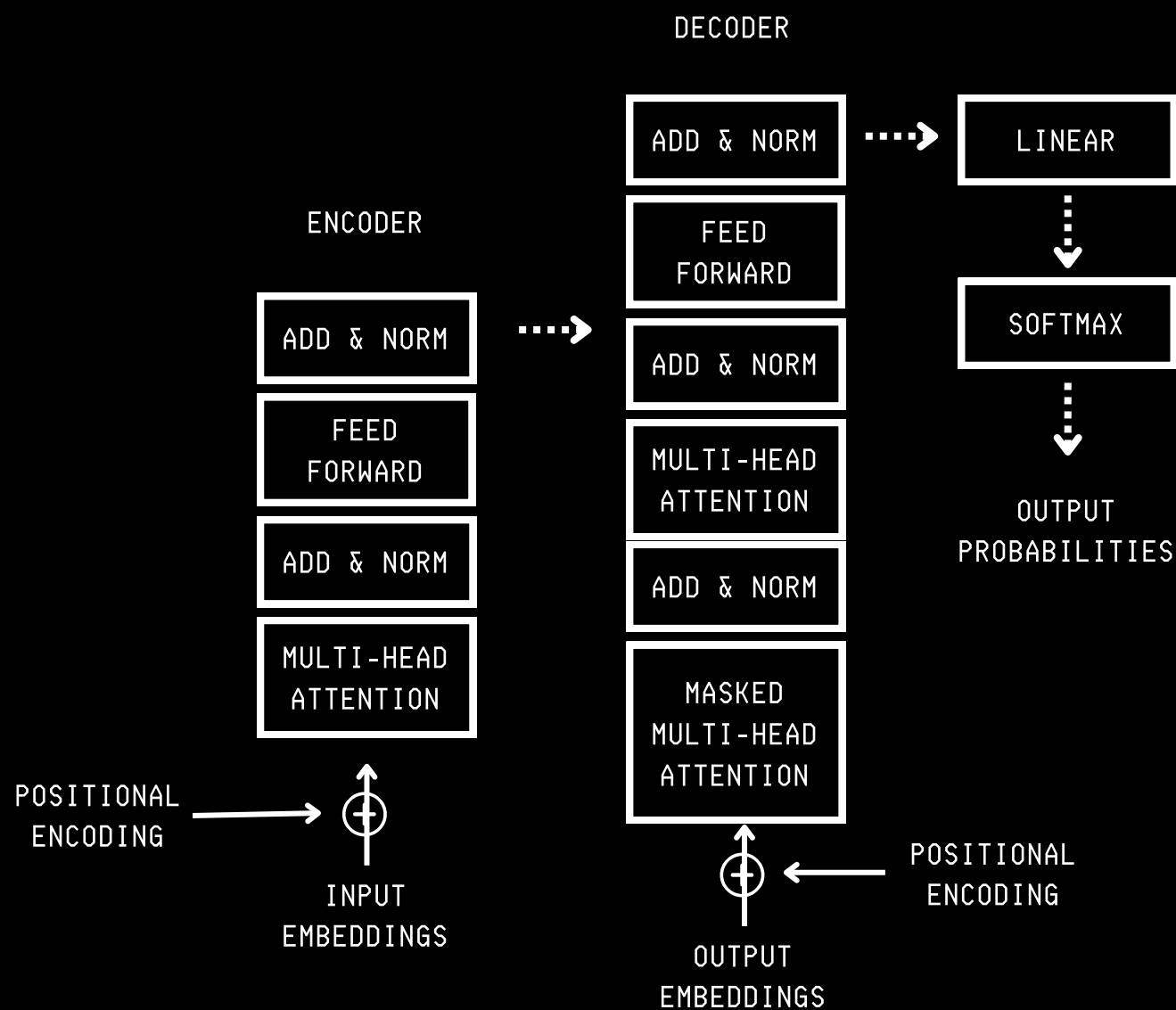
$$\text{ATTENTION}(Q, K, V) = \text{SOFTMAX} \left(Q * K^T / \sqrt{D} \right) * V$$

WHERE:

- Q -> QUERIES
- K -> KEYS
- V -> VALUES
- D -> DIMENSION OF KEYS

TRANSFORMER

A NEURAL NETWORK ARCHITECTURE DESIGNED FOR HANDLING SEQUENTIAL DATA, RELYING ON SELF-ATTENTION MECHANISMS TO CAPTURE DEPENDENCIES BETWEEN INPUT ELEMENTS WITHOUT REGARD TO THEIR DISTANCE IN THE SEQUENCE.



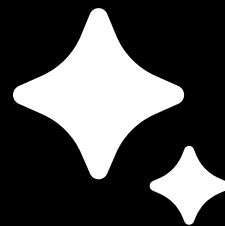
LLM

LARGE LANGUAGE MODEL

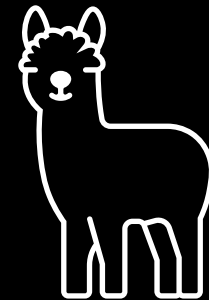
AN ADVANCED ARTIFICIAL INTELLIGENCE SYSTEM TRAINED ON VAST AMOUNTS OF TEXT DATA TO UNDERSTAND, GENERATE, AND INTERACT WITH HUMAN LANGUAGE IN A WAY THAT MIMICS HUMAN UNDERSTANDING AND RESPONSE.



GPT-4



BARD

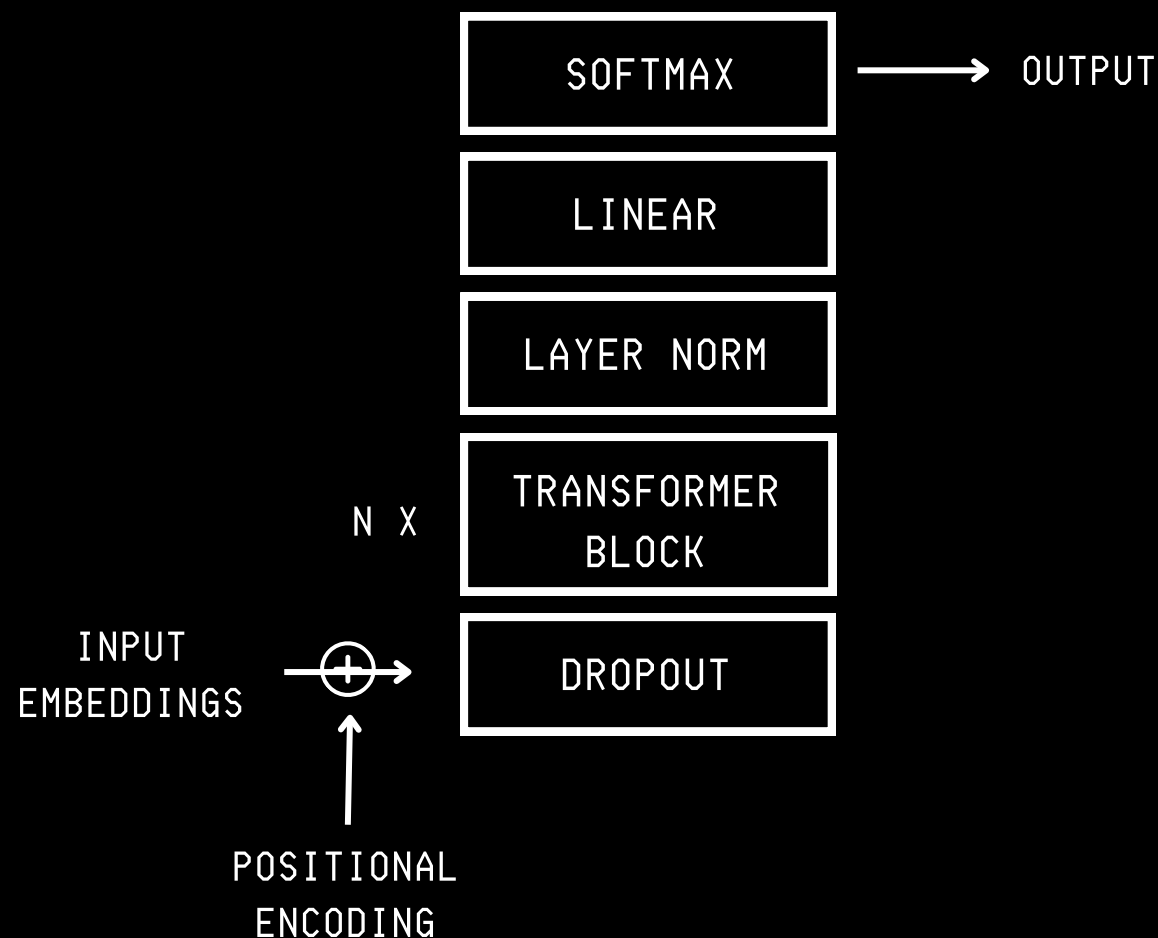


LLAMA

GPT

GENERATIVE PRE-TRAINED TRANSFORMER

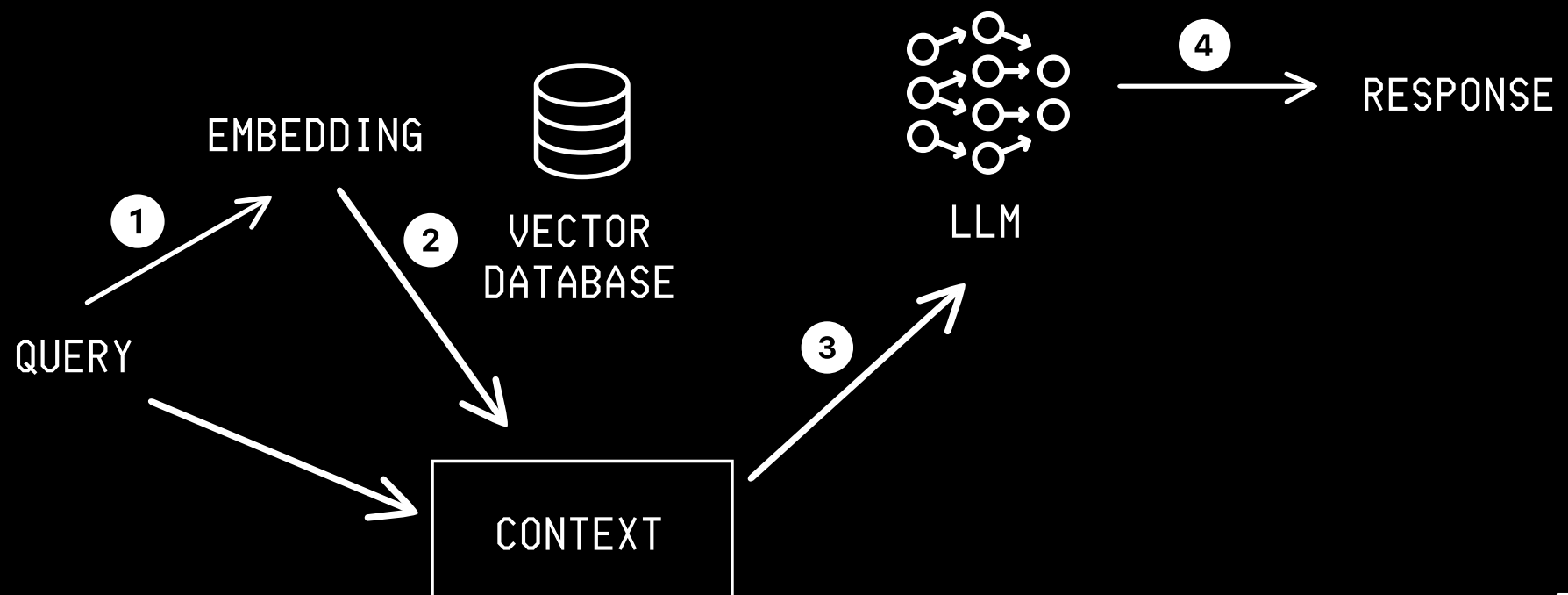
A LANGUAGE MODEL THAT USES THE TRANSFORMER ARCHITECTURE TO GENERATE HUMAN-LIKE TEXT BY PREDICTING THE NEXT WORD IN A SEQUENCE, PRE-TRAINED ON VAST TEXT DATA AND FINE-TUNED FOR SPECIFIC TASKS.



RAG

RETRIEVAL AUGMENTED GENERATION

A MACHINE LEARNING APPROACH THAT ENHANCES LANGUAGE MODEL RESPONSES BY DYNAMICALLY RETRIEVING AND INCORPORATING RELEVANT INFORMATION FROM A LARGE CORPUS OF DOCUMENTS DURING THE GENERATION PROCESS.

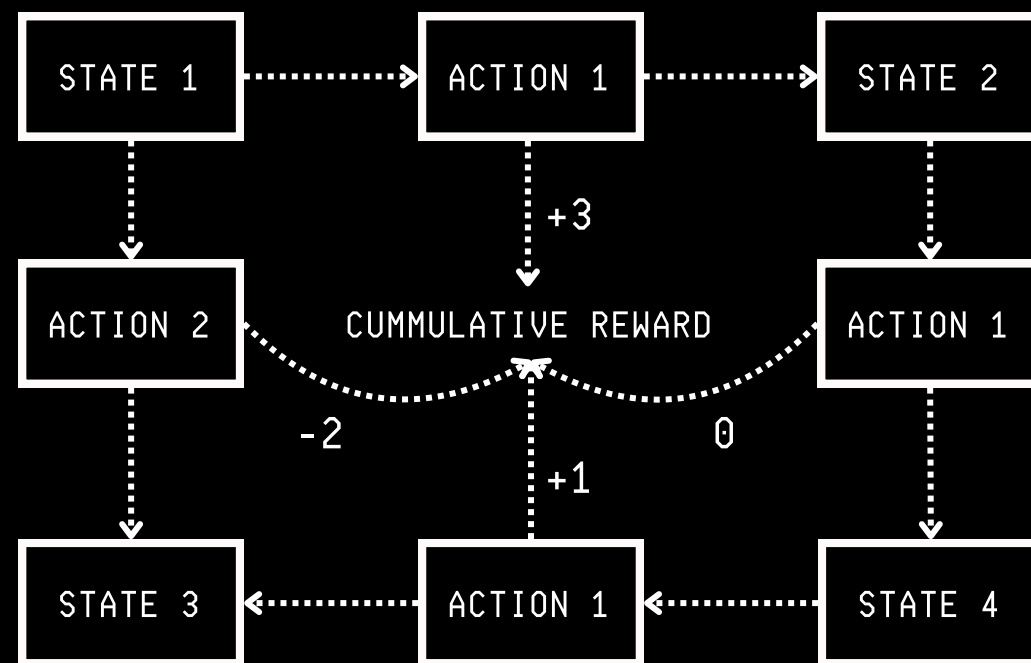


MDP

MARKOV DECISION PROCESS

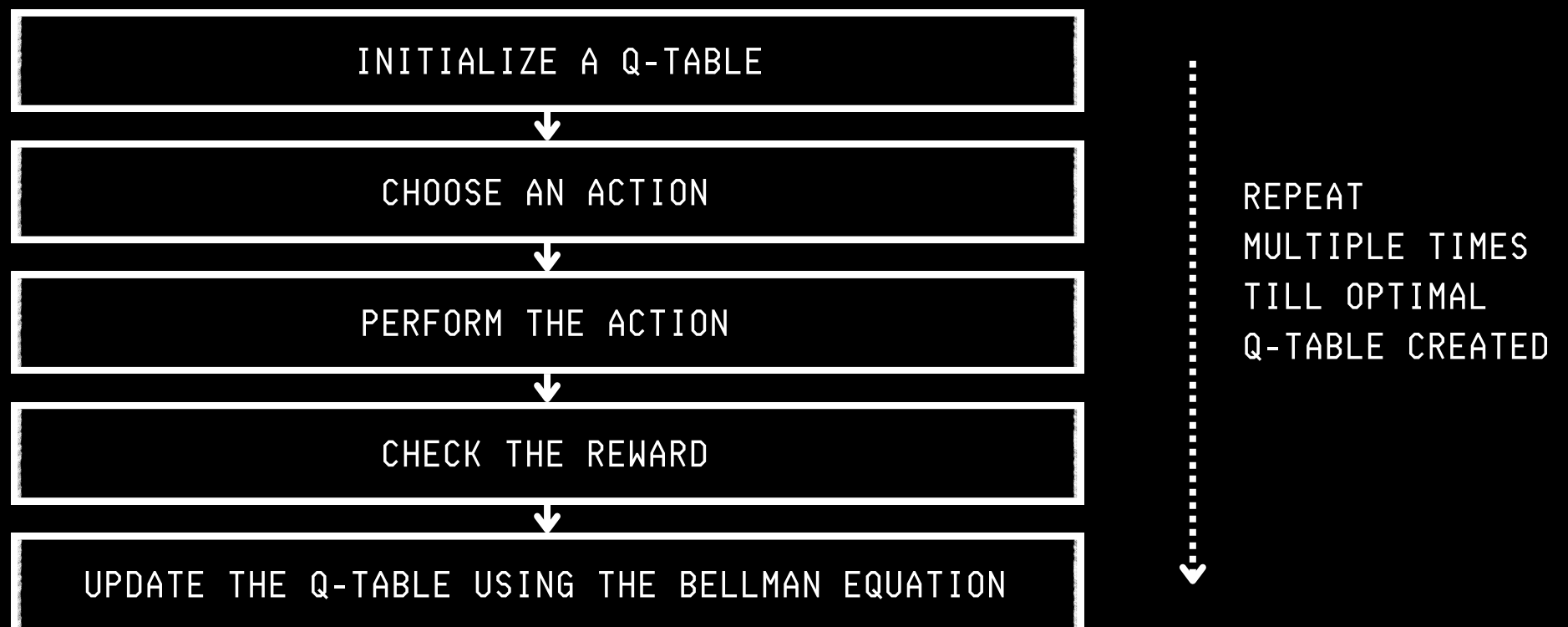
A MATHEMATICAL FRAMEWORK USED FOR MODELLING SEQUENTIAL DECISION-MAKING PROBLEMS WHERE OUTCOMES ARE PARTLY RANDOM AND PARTLY UNDER THE CONTROL OF A DECISION-MAKER.

THE OBJECTIVE OF AN MDP IS TYPICALLY TO FIND A POLICY THAT MAXIMIZES THE EXPECTED CUMULATIVE REWARD OVER TIME.



Q-LEARNING

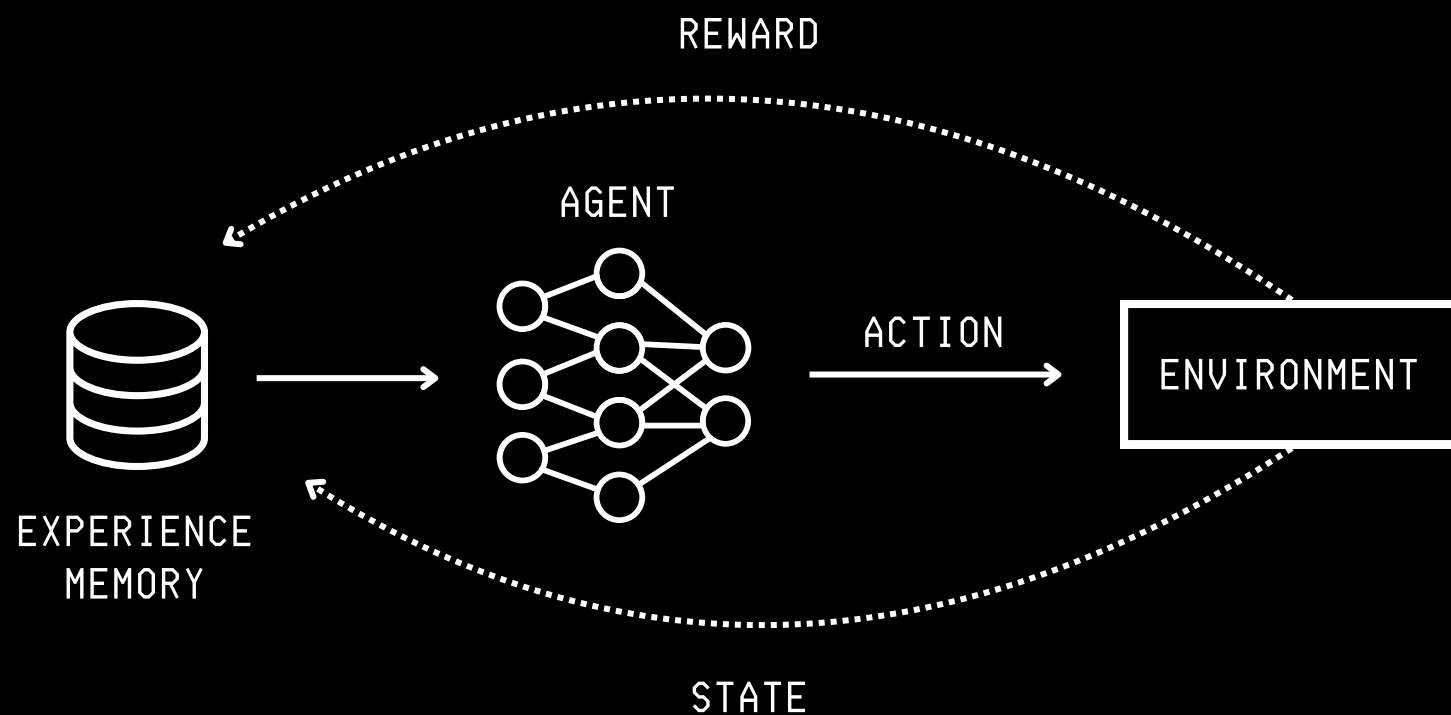
A REINFORCEMENT LEARNING ALGORITHM THAT AIMS TO LEARN THE OPTIMAL POLICY BY UPDATING Q-VALUES, WHICH ESTIMATE THE EXPECTED CUMULATIVE REWARD FOR TAKING A PARTICULAR ACTION IN A GIVEN STATE AND FOLLOWING THE OPTIMAL POLICY THEREAFTER.



DQN

DEEP Q-NETWORK

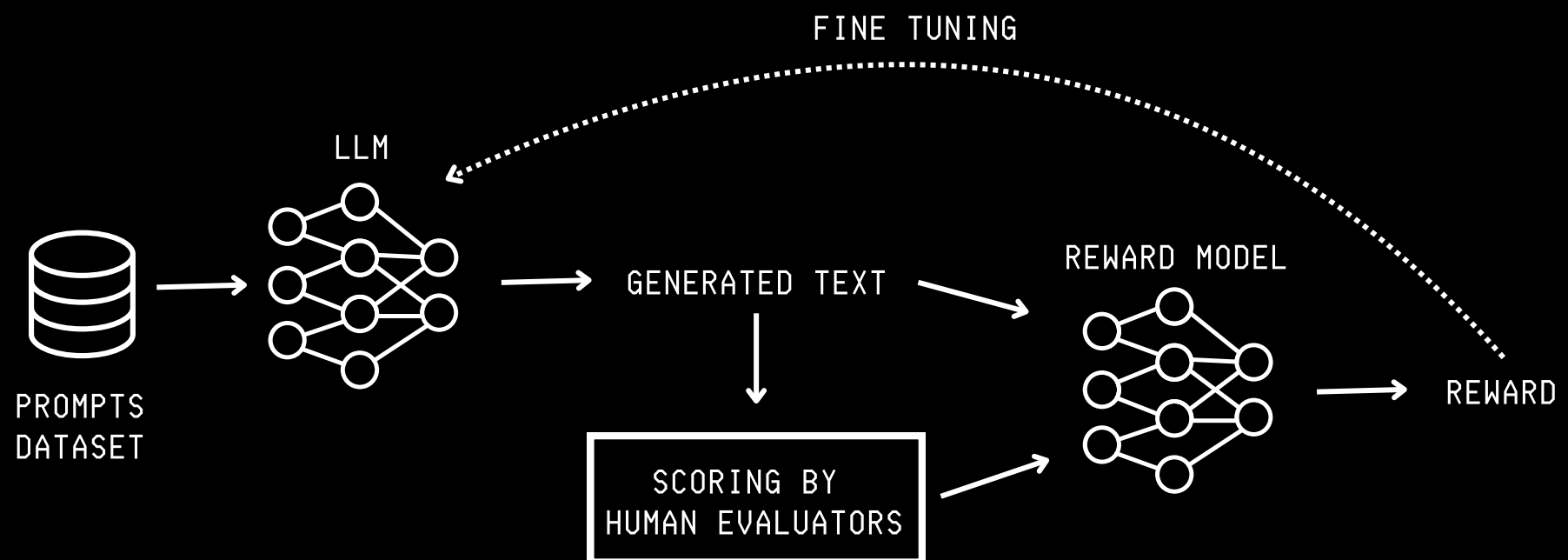
A REINFORCEMENT LEARNING ALGORITHM THAT USES DEEP NEURAL NETWORKS TO APPROXIMATE THE Q-VALUE FUNCTION, ENABLING AGENTS TO LEARN OPTIMAL POLICIES IN COMPLEX, HIGH-DIMENSIONAL ENVIRONMENTS.



RLHF

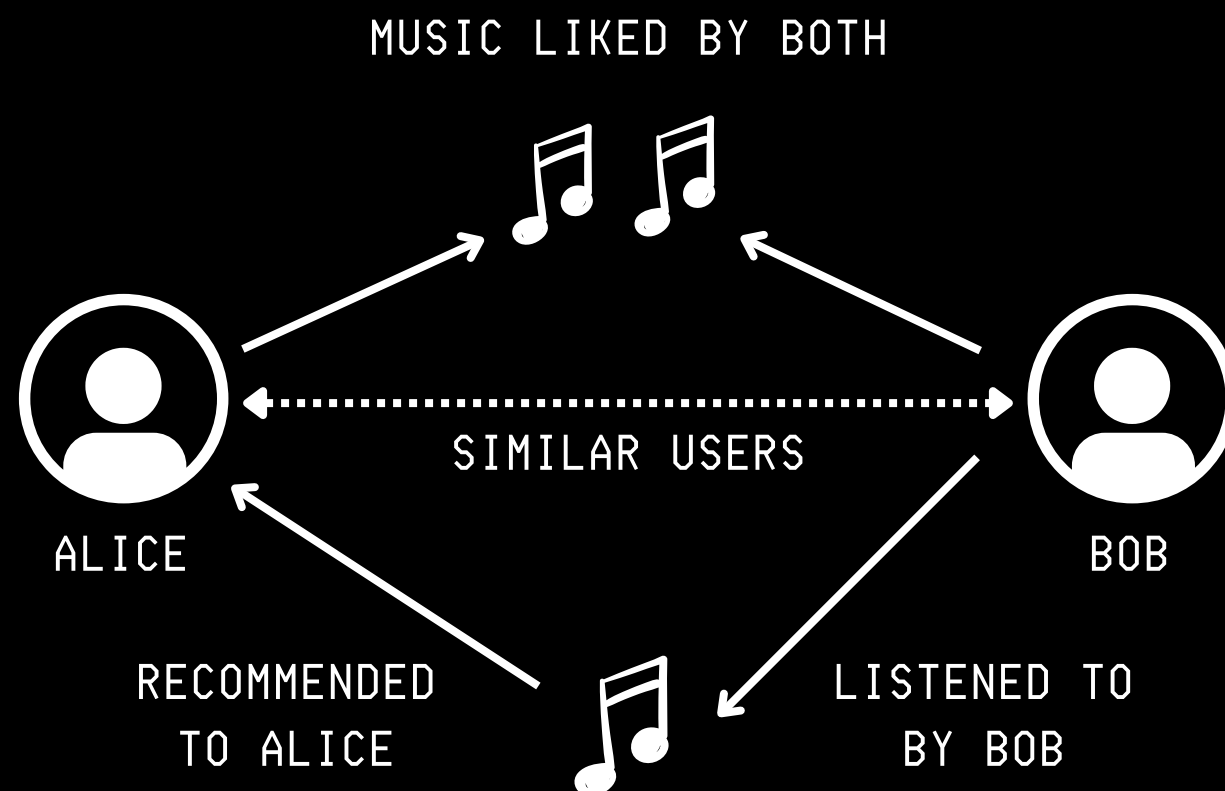
REINFORCEMENT LEARNING FROM HUMAN FEEDBACK

A METHOD WHERE AN AI MODEL IS TRAINED AND FINE-TUNED USING FEEDBACK FROM HUMAN EVALUATORS TO ALIGN ITS OUTPUTS WITH HUMAN PREFERENCES AND VALUES.



COLLABORATIVE FILTERING

A RECOMMENDATION SYSTEM TECHNIQUE THAT PREDICTS A USER'S INTERESTS BY AGGREGATING AND ANALYZING PATTERNS OF PREFERENCES AND BEHAVIOURS FROM A LARGE COMMUNITY OF USERS, IDENTIFYING SIMILAR USER GROUPS TO MAKE RECOMMENDATIONS.



TRANSFER LEARNING

A MACHINE LEARNING TECHNIQUE WHERE A PRE-TRAINED MODEL, DEVELOPED FOR ONE TASK, IS ADAPTED AND FINE-TUNED FOR A DIFFERENT BUT RELATED TASK, LEVERAGING THE KNOWLEDGE GAINED FROM THE INITIAL TRAINING TO IMPROVE PERFORMANCE ON THE NEW TASK.



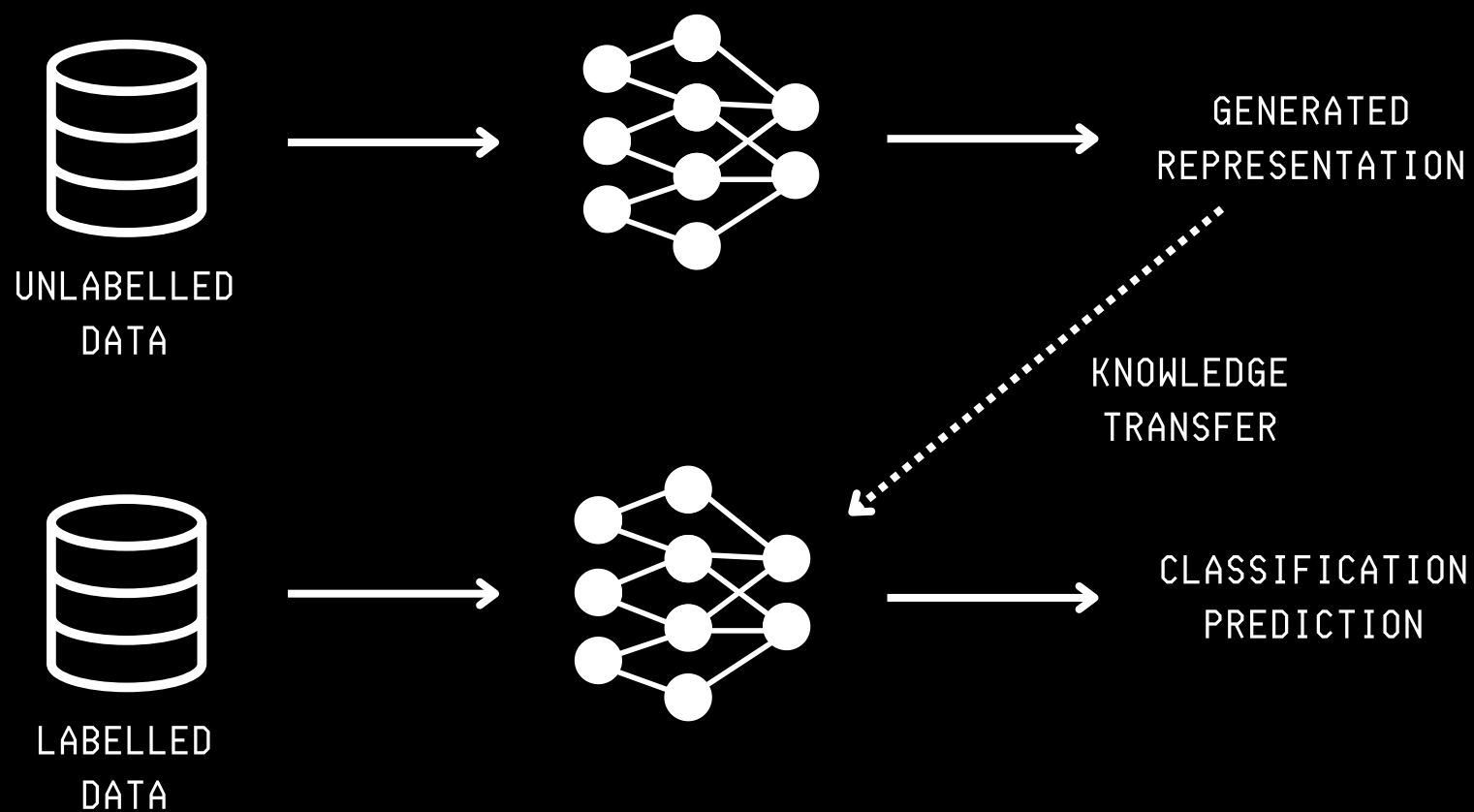
ZERO-SHOT LEARNING

A MACHINE LEARNING PARADIGM WHERE THE MODEL CAN CLASSIFY DATA FROM CLASSES IT HAS NEVER SEEN DURING TRAINING BY LEVERAGING KNOWLEDGE TRANSFER AND SEMANTIC RELATIONSHIPS.



SELF-SUPERVISED LEARNING

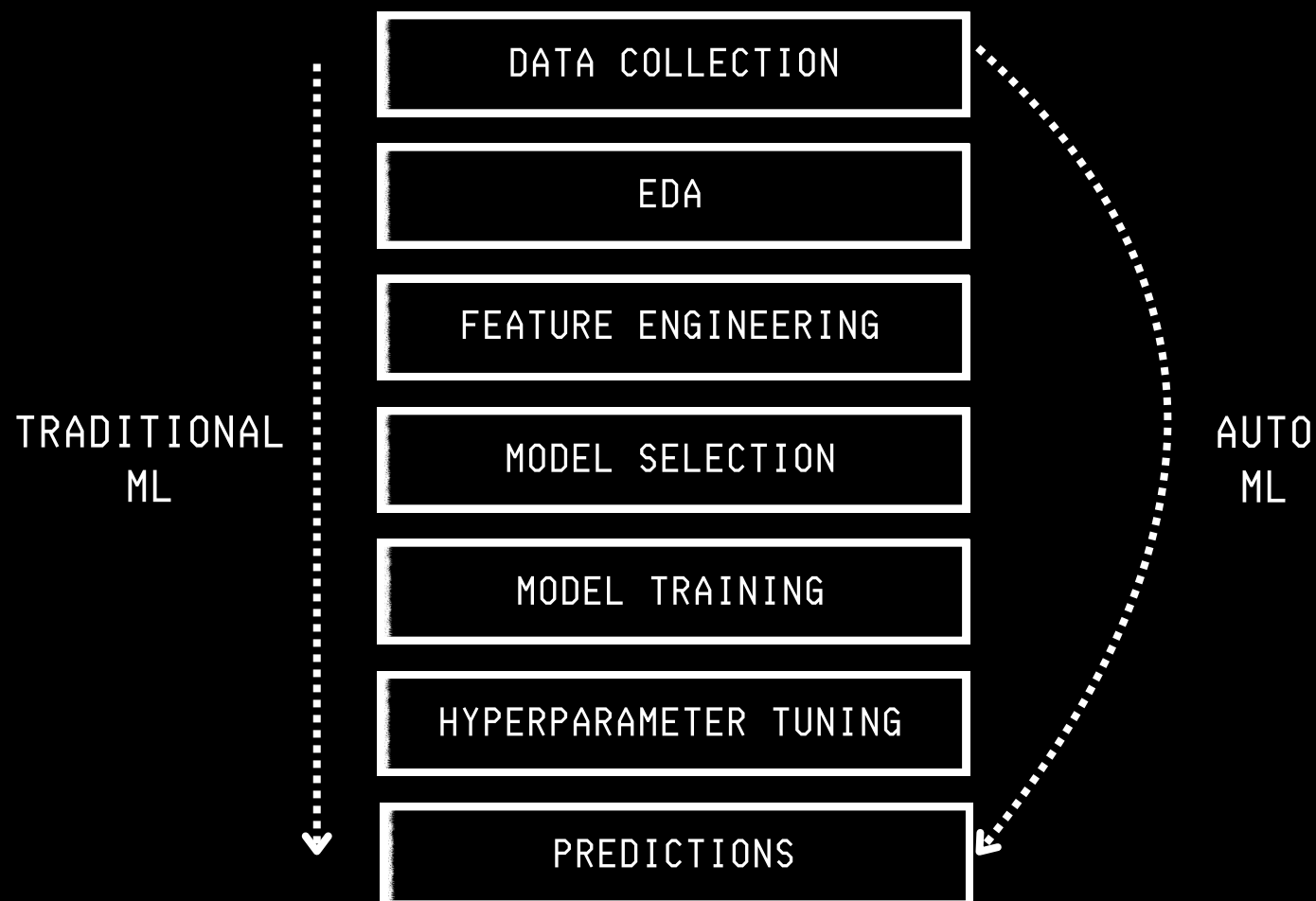
A TYPE OF MACHINE LEARNING WHERE THE MODEL LEARNS TO PREDICT PART OF ITS INPUT FROM OTHER PARTS, GENERATING ITS OWN LABELS FROM THE INPUT DATA, ALLOWING IT TO LEVERAGE LARGE AMOUNTS OF UNLABELED DATA FOR PRETRAINING AND IMPROVING PERFORMANCE ON DOWNSTREAM TASKS.

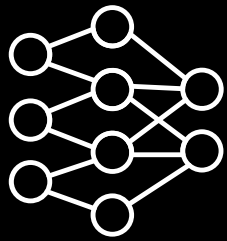


AUTO ML

AUTOMATED MACHINE LEARNING

THE PROCESS OF AUTOMATING THE END-TO-END PROCESS OF APPLYING MACHINE LEARNING, INCLUDING MODEL SELECTION, HYPERPARAMETER TUNING, FEATURE ENGINEERING, AND DEPLOYMENT.





STAY CONNECTED TO MY WORK THROUGH THE FOLLOWING LINKS:

1. SUBSTACK

[HTTPS://INTOAI.PUB](https://intoai.pub)

[HTTPS://WWW.INTOQUANTUM.PUB](https://www.intoquantum.pub)

2. MEDIUM

[HTTPS://MEDIUM.COM/@BAMANIA-ASHISH](https://medium.com/@bamania-ashish)

3. LINKEDIN

[HTTPS://WWW.LINKEDIN.COM/IN/ASHISHBAMANIA](https://www.linkedin.com/in/ashishbamania)

4. GUMROAD

[HTTPS://BAMANIAASHISH.GUMROAD.COM](https://bamaniaashish.gumroad.com)